

Tangent bundles. Lie algebras

SGA 3, Exposé II

English draft

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About this draft

This is an unofficial English translation of Exposé II, “Tangent bundles. Lie algebras” from *Schémas en groupes* (SGA 3). The source is the recomposition edited by P. Gille and P. Polo, PDF `Exp2-14oct24.pdf` (52 pages), from <https://webusers.imj-prg.fr/~patrick.polo/SGA3/>. The seminar was directed by M. Demazure and A. Grothendieck (IHÉS, 1962–1964).

Statement numbers follow that PDF. A clear typographical slip is corrected in the English and marked with a `% typo:` comment; such slips are listed in the README. A slip that would change the mathematics is left as printed and marked `% typo?:`. Scholarly proofreading remains outstanding.

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Contents

About this draft	iii
Exposé II. Tangent bundles. Lie algebras	1
1. The functors $\mathrm{Hom}_{Z/S}(X, Y)$	1
2. The schemes $I_S(\mathcal{M})$	3
3. The tangent bundle, the condition (E)	5
4. Tangent space to a group – Lie algebras	17
5. Computation of some Lie algebras	34
6. Miscellaneous remarks	39

Exposé II. Tangent bundles. Lie algebras

by *M. Demazure*

⁽⁰⁾We propose in this exposé to construct the analogue, in the theory of schemes, of the *tangent bundles* and *Lie algebras* of the classical theory. It will however be useful not to restrict oneself to schemes properly so called, but also to take an interest in certain functors on the category of schemes which are not necessarily *representable* (for example functors Hom , Norm , etc.). As was announced in the previous exposé (cf. I 1.1), we shall identify a scheme with the functor associated with it.

On the other hand, the constructions set out below go beyond the framework of the theory of schemes. They are equally valid, for example, in the theory of *analytic spaces* with nilpotent elements, modulo a few modifications of detail.

Before beginning this construction, we must set down a few general definitions which complete those of I 1.7.

1. The functors $\text{Hom}_{Z/S}(X, Y)$

Let us take up again the notations of I 1.1. One identifies the category $\mathcal{C}$ with a full subcategory of $\widehat{\mathcal{C}} = \text{Hom}(\mathcal{C}^\circ, (\text{Ens}))$ (in particular one omits the underlinings⁽¹⁾ which allowed us to distinguish graphically an object of $\widehat{\mathcal{C}}$ from an object of $\mathcal{C}$).

Let us consider the following situation: four objects of $\widehat{\mathcal{C}}$, denoted S, X, Y, Z , the first being in fact an object of $\mathcal{C}$, X and Y over Z , Z over S :

$$\begin{array}{ccc} X & & Y \\ & \searrow p_X & \swarrow p_Y \\ & Z & \\ & \downarrow & \\ & S. & \end{array}$$

Definition 1.1. One defines an object of $\widehat{\mathcal{C}}_S$, denoted $\text{Hom}_{Z/S}(X, Y)$, by:

$$(1) \quad \text{Hom}_{Z/S}(X, Y)(S') = \text{Hom}_{Z_{S'}}(X_{S'}, Y_{S'}) = \text{Hom}_Z(X \times_S S', Y),$$

for every object S' of $\mathcal{C}_S$. One sees at once that $\text{Hom}_{Z/S}(X, Y)$ is nothing other than the subobject of $\text{Hom}_S(X, Y)$ formed of the morphisms compatible with p_X and p_Y , that is, the kernel of the pair of morphisms

$$\text{Hom}_S(X, Y) \rightrightarrows \text{Hom}_S(X, Z)$$

defined, the first by composition with p_Y , the second as being the constant morphism whose “image” is p_X .

⁽⁰⁾**N.D.E.** Version of 14/10/2024

⁽¹⁾**N.D.E.** Rendered in this edition by bold characters **F, G, V, W**, cf. Exposé I. However, for functors such as Norm and Centr (cf. 5.2) the underlinings have been kept as in the original, and one has added them for the functor Lie, in order to distinguish the functor Lie(G/S) from the $\Gamma(S, \mathcal{O}_S)$ -Lie algebra $\text{Lie}(G/S) = \underline{\text{Lie}}(G/S)(S)$, used for example in Exposé VII_A.

⁽²⁾On the other hand, one sees as in I 1.7 that, for every object T of $\widehat{\mathcal{C}}$ over S , one has a natural bijection:

$$(2) \quad \mathrm{Hom}_S(T, \mathrm{Hom}_{Z/S}(X, Y)) \simeq \mathrm{Hom}_Z(X \times_S T, Y).$$

Moreover, by I 1.7.1, if E, F are objects of $\widehat{\mathcal{C}}$ over Z , one has:

$$\mathrm{Hom}_Z(E, \mathrm{Hom}_Z(F, Y)) \simeq \mathrm{Hom}_Z(E \times_Z F, Y) \simeq \mathrm{Hom}_Z(F, \mathrm{Hom}_Z(E, Y)).$$

Applying this to $E = X$ and $F = Z \times_S T$, one obtains natural bijections, for every object T of $\widehat{\mathcal{C}}_S$:

$$(3) \quad \mathrm{Hom}_S(T, \mathrm{Hom}_{Z/S}(X, Y)) \simeq \mathrm{Hom}_Z(X \times_S T, Y) \simeq \begin{cases} \mathrm{Hom}_Z(Z \times_S T, \mathrm{Hom}_Z(X, Y)) \\ \mathrm{Hom}_Z(X, \mathrm{Hom}_Z(Z \times_S T, Y)) \end{cases}.$$

Moreover, these bijections are functorial in T , so one obtains isomorphisms of S -functors:

$$(4) \quad \begin{array}{ccc} \mathrm{Hom}_S(T, \mathrm{Hom}_{Z/S}(X, Y)) & \xrightarrow{\sim} & \mathrm{Hom}_{Z/S}(X, \mathrm{Hom}_Z(Z \times_S T, Y)) \\ & \searrow \simeq \quad \swarrow \simeq & \\ & \mathrm{Hom}_{Z/S}(X \times_S T, Y) & \end{array}$$

Let us point out two particular cases of the definition. If $Z = S$, one has:

$$\mathrm{Hom}_{S/S}(X, Y) = \mathrm{Hom}_S(X, Y).$$

On the other hand, when $X = Z$, one sets

$$(5) \quad \prod_{Z/S} Y = \mathrm{Hom}_{Z/S}(Z, Y),$$

one therefore has by definition

$$\left(\prod_{Z/S} Y \right)(S') = \mathrm{Hom}_Z(Z \times_S S', Y) \simeq \Gamma(Y_{S'}/Z_{S'}).$$

The functor $\prod_{Z/S}: \widehat{\mathcal{C}}_Z \rightarrow \widehat{\mathcal{C}}_S$ is right adjoint to the functor of base change from S to Z : for every S -functor U one has

$$\mathrm{Hom}_S\left(U, \prod_{Z/S} Y\right) = \mathrm{Hom}_Z(U \times_S Z, Y).$$

(If $\mathcal{C} = (\mathrm{Sch})$ and if Z is an S -scheme, the functor $\prod_{Z/S}$ is called “*Weil restriction of scalars*”).⁽³⁾ Let us also note that one has an isomorphism:

$$(6) \quad \mathrm{Hom}_{Z/S}(X, Y) \simeq \mathrm{Hom}_{X/S}(X, Y \times_Z X) = \prod_{X/S} (Y \times_Z X),$$

which gives in particular, for $Z = S$, an isomorphism:

$$(7) \quad \mathrm{Hom}_S(X, Y) \simeq \prod_{X/S} Y_X.$$

Remark 1.2. The functor $Y \mapsto \mathrm{Hom}_{Z/S}(X, Y)$ commutes with the product in the following sense: one has a functorial isomorphism

$$(*) \quad \mathrm{Hom}_{Z/S}(X, Y \times_Z Y') \simeq \mathrm{Hom}_{Z/S}(X, Y) \times_S \mathrm{Hom}_{Z/S}(X, Y').$$

It follows that if Y is a Z -group, resp. a Z -ring, etc., then $\mathrm{Hom}_{Z/S}(X, Y)$ is an S -group, resp. an S -ring, etc.

⁽²⁾**N.D.E.** Points (2), (3), (4) have been spelled out; in particular, (4) will be used in 3.11.

⁽³⁾**N.D.E.** The two preceding sentences have been added.

Remark 1.3. ⁽⁴⁾Moreover, let $\pi: M \rightarrow Y$ be a Y -functor in $\mathbf{O}_Y$ -modules (cf. I, 4.3.3.1). Set $H = \mathrm{Hom}_{Z/S}(X, Y)$. Then $\mathrm{Hom}_{Z/S}(X, M)$ is endowed with a natural structure of $\mathbf{O}_H$ -module; more precisely, for every H' over H , $\mathrm{Hom}_H(H', \mathrm{Hom}_{Z/S}(X, M))$ is endowed with a natural structure of $\mathbf{O}(H' \times_S X)$ -module.

Indeed, let us denote by $m: M \times_Y M \rightarrow M$ and $\lambda: \mathbf{O}_Y \times_Y M \rightarrow M$ the morphisms defining the structures of Y -group (abelian) and of $\mathbf{O}_Y$ -module. Let H' be an S -scheme over $H = \mathrm{Hom}_{Z/S}(X, Y)$, that is, one is given a Z -morphism $f: X \times_S H' \rightarrow Y$, which therefore makes $X \times_S H'$ a Y -object. Then

$$\mathrm{Hom}_H(H', \mathrm{Hom}_{Z/S}(X, M))$$

is the set of Z -morphisms $\phi: X \times_S H' \rightarrow M$ such that $\pi \circ \phi = f$, that is, of Y -morphisms $X \times_S H' \rightarrow M$.

If ϕ, ψ are two such morphisms, one defines $\phi + \psi$ as the composite Y -morphism

$$X \times_S H' \xrightarrow{\phi \times \psi} M \times_Y M \xrightarrow{m} M$$

and one checks that this endows $\mathrm{Hom}_{Z/S}(X, M)$ with the structure of an abelian group over $H = \mathrm{Hom}_{Z/S}(X, Y)$.⁽⁵⁾

Likewise, if a is an element of $\mathbf{O}(X \times_S H')$, i.e. an S -morphism $a: X \times_S H' \rightarrow \mathbf{O}_S$, one defines $a\phi$ as the composite $\lambda \circ (a \times \phi)$, where $a \times \phi$ denotes the Y -morphism from $X \times_S H'$ to $\mathbf{O}_Y \times_Y M \simeq \mathbf{O}_S \times_S M$ with components a and ϕ ; one checks that this endows $\mathrm{Hom}_H(H', \mathrm{Hom}_{Z/S}(X, M))$ with the structure of an $\mathbf{O}(X \times_S H')$ -module, functorial in the H -object H' .

2. The schemes $I_S(\mathcal{M})$

Definition 2.1. Let S be a scheme and $\mathcal{M}$ a quasi-coherent $\mathcal{O}_S$ -module. One denotes by $\mathcal{D}_{\mathcal{O}_S}(\mathcal{M})$ the quasi-coherent $\mathcal{O}_S$ -algebra $\mathcal{O}_S \oplus \mathcal{M}$ (where $\mathcal{M}$ is regarded as a square-zero ideal). One denotes by $I_S(\mathcal{M})$ the S -scheme $\mathrm{Spec} \mathcal{D}_{\mathcal{O}_S}(\mathcal{M})$.⁽⁶⁾ In particular one denotes $\mathcal{D}_{\mathcal{O}_S} = \mathcal{D}_{\mathcal{O}_S}(\mathcal{O}_S)$, $I_S = I_S(\mathcal{O}_S)$ and one calls them respectively the *algebra of dual numbers over S* and the *scheme of dual numbers over S* .

Then $\mathcal{M} \mapsto I_S(\mathcal{M})$ is a contravariant functor from the category of quasi-coherent $\mathcal{O}_S$ -modules to that of S -schemes. In particular the morphisms $0 \rightarrow \mathcal{M}$ and $\mathcal{M} \rightarrow 0$ define respectively the structural morphism $\rho: I_S(\mathcal{M}) \rightarrow I_S(0) = S$ and a section $\varepsilon_{\mathcal{M}}$ of the latter, which one calls the *zero section*.⁽⁷⁾

2.1.1. — ⁽⁸⁾As $\mathcal{M} \rightarrow I_S(\mathcal{M})$ is a contravariant functor, every $a \in \mathrm{End}_{\mathcal{O}_S}(\mathcal{M})$ defines an S -endomorphism a^* of $I_S(\mathcal{M})$, and one has $1^* = \mathrm{id}$, $(ab)^* = b^* \circ a^*$, $0^* = \varepsilon_{\mathcal{M}} \circ \rho$ and $a^* \circ \varepsilon_{\mathcal{M}} = \varepsilon_{\mathcal{M}}$. Consequently, the S -scheme $I_S(\mathcal{M})$ is endowed with a right action of the multiplicative monoid of $\mathrm{End}_{\mathcal{O}_S}(\mathcal{M})$, which commutes with the S -morphisms $I_S(\mathcal{M}) \rightarrow I_S(\mathcal{M}')$ coming from morphisms $\mathcal{M}' \rightarrow \mathcal{M}$; in particular, the operators a^* preserve the zero section of $I_S(\mathcal{M})$.

⁽⁴⁾**N.D.E.** This remark has been added; it is useful for Corollary 3.11.1.

⁽⁵⁾**N.D.E.** Of course, if M is a Y -group (not necessarily abelian) and if one sets $\phi \cdot \psi = m \circ (\phi \times \psi)$, this makes $\mathrm{Hom}_{Z/S}(X, M)$ a group over $\mathrm{Hom}_{Z/S}(X, Y)$.

⁽⁶⁾**N.D.E.** Note that $I_S(\mathcal{M})$ has the same underlying space as S .

⁽⁷⁾**N.D.E.** Compare with 3.1.1.

⁽⁸⁾**N.D.E.** The original has been set out in detail in what follows, and the numbering 2.1.1 to 2.1.3 has been added.

For every $a \in \text{End}_{\mathcal{O}_S}(\mathcal{M})$ and $f: S' \rightarrow S$ and $m \in I_S(\mathcal{M})(S')$, one will denote $m \cdot a = a^*(m)$; then $m \cdot 1 = m$, $(m \cdot a) \cdot b = m \cdot (ab)$, $m \cdot 0 = \varepsilon_{\mathcal{M}}(\rho(m))$ and, if $m = \varepsilon_{\mathcal{M}}(f)$, then $m \cdot a = m$.⁽⁹⁾

Remark 2.1.2. The formation of the $I_S(\mathcal{M})$ commutes with extension of the base: one has canonical isomorphisms

$$I_S(\mathcal{M})_{S'} \simeq I_{S'}(\mathcal{M} \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}).$$

For simplicity, one will denote $I_{S'}(\mathcal{M}) = I_S(\mathcal{M})_{S'}$; more generally, if X is an S -functor (not necessarily representable), one will denote $I_X(\mathcal{M}) = I_S(\mathcal{M}) \times_S X$.

2.1.3. — ⁽¹⁰⁾By what precedes, the multiplicative monoid of $\mathbf{O}(S')$ operates on the S' -scheme $I_{S'}(\mathcal{M})$, functorially in $\mathcal{M}$, i.e. the S -scheme $I_S(\mathcal{M})$ is endowed with the structure of an object with monoid of operators $\mathbf{O}_S$, this structure being functorial in $\mathcal{M}$.

One therefore has a morphism of S -schemes

$$\lambda: I_S(\mathcal{M}) \times_S \mathbf{O}_S \rightarrow I_S(\mathcal{M}),$$

satisfying obvious conditions. For every S -functor X , one obtains by base change a morphism of X -functors:

$$\lambda_X: I_X(\mathcal{M}) \times_S \mathbf{O}_S \rightarrow I_X(\mathcal{M})$$

which makes the S -functor $I_X(\mathcal{M})$ an object with monoid of operators $\mathbf{O}(X)$: every element a of $\mathbf{O}(X) = \text{Hom}_S(X, \mathbf{O}_S)$ defines an X -endomorphism

$$a^* = \lambda_X \circ (\text{id}_{I_X(\mathcal{M})} \times (a \circ \text{pr}_X))$$

of $I_X(\mathcal{M})$; explicitly, if $x \in X(S')$ and $m \in I_S(\mathcal{M})(S') = I_{S'}(\mathcal{M})(S')$, then $a(x) = a \circ x$ belongs to $\mathbf{O}(S')$ and one has:

$$(m, x) \cdot a = (m \cdot a(x), x).$$

This operation is functorial in $\mathcal{M}$ and preserves the zero section $\varepsilon_{\mathcal{M}}: X \rightarrow I_X(\mathcal{M})$, i.e. $a^* \circ \varepsilon_{\mathcal{M}} = \varepsilon_{\mathcal{M}}$ for every $a \in \mathbf{O}(X)$.

Moreover, this operation is “functorial in X ” in the following sense: if $\pi: Y \rightarrow X$ is a morphism of S -functors and $u: \mathbf{O}(X) \rightarrow \mathbf{O}(Y)$ the corresponding morphism of rings (i.e. $u(a) = a \circ \pi \in \mathbf{O}(Y)$ for every $a \in \mathbf{O}(X) = \text{Hom}_S(X, \mathbf{O}_S)$), then the following diagram is commutative:

$$\begin{array}{ccc} I_X(\mathcal{M}) & \xrightarrow{a^*} & I_X(\mathcal{M}) \\ \pi \downarrow & & \downarrow \pi \\ I_Y(\mathcal{M}) & \xrightarrow{u(a)^*} & I_Y(\mathcal{M}). \end{array}$$

2.2. — Let now $\mathcal{M}$ and $\mathcal{N}$ be two quasi-coherent $\mathcal{O}_S$ -modules. The commutative diagram

$$\begin{array}{ccc} & \mathcal{M} \oplus \mathcal{N} & \\ \swarrow & & \searrow \\ \mathcal{M} & & \mathcal{N} \\ \searrow & & \swarrow \\ & 0 & \end{array}$$

⁽⁹⁾**N.D.E.** In the sequel, one will be concerned mainly with the case where $a \in \mathbf{O}(S)$, acting by homotheties on $\mathcal{M}$. For example, if $\mathcal{M}$ is a free $\mathcal{O}_S$ -module of rank r , then, for every $S' \rightarrow S$, $\text{Hom}_S(S', I_S(\mathcal{M}))$ is identified with the set of r -tuples $(\varepsilon_1, \dots, \varepsilon_r) \in \mathbf{O}(S')$ such that $\varepsilon_i \varepsilon_j = 0$ for all i, j , and one has $(\varepsilon_1, \dots, \varepsilon_r) \cdot a = (a\varepsilon_1, \dots, a\varepsilon_r)$ for every $a \in \mathbf{O}(S)$.

⁽¹⁰⁾**N.D.E.** This paragraph has been added; it will be useful in 3.4.2 and in 4.6.2.

defines a commutative diagram of S -schemes

$$(*) \quad \begin{array}{ccccc} & & \mathrm{I}_S(\mathcal{M} \oplus \mathcal{N}) & & \\ & \nearrow & \uparrow \varepsilon_{\mathcal{M} \oplus \mathcal{N}} & \nwarrow & \\ \mathrm{I}_S(\mathcal{M}) & & & & \mathrm{I}_S(\mathcal{N}) \\ & \nwarrow \varepsilon_{\mathcal{M}} & \downarrow & \nearrow \varepsilon_{\mathcal{N}} & \\ & & S & & \end{array}$$

Proposition 2.2. *For every S -scheme X , the diagram of functors over S obtained by applying the functor $\mathrm{Hom}_S(-, X)$ to the diagram $(*)$ is cartesian:*

$$\begin{array}{ccccc} & & \mathrm{Hom}_S(\mathrm{I}_S(\mathcal{M} \oplus \mathcal{N}), X) & & \\ & \swarrow & & \searrow & \\ \mathrm{Hom}_S(\mathrm{I}_S(\mathcal{M}), X) & & & & \mathrm{Hom}_S(\mathrm{I}_S(\mathcal{N}), X) \\ & \searrow & \downarrow & \swarrow & \\ & & \mathrm{Hom}_S(S, X) = X & & \end{array}$$

One must check that for every $S' \rightarrow S$, the diagram of sets obtained by taking the value of the functors on S' is cartesian. As the formation of $\mathrm{I}_S(\mathcal{P})$ commutes with extension of the base in the sense made explicit above, it suffices to do this for $S' = S$, hence to check that the following diagram of sets is cartesian:

$$\begin{array}{ccccc} & & X(\mathrm{I}_S(\mathcal{M} \oplus \mathcal{N})) & & \\ & \swarrow & \downarrow X(\varepsilon_{\mathcal{M} \oplus \mathcal{N}}) & \searrow & \\ X(\mathrm{I}_S(\mathcal{M})) & & & & X(\mathrm{I}_S(\mathcal{N})) \\ & \searrow X(\varepsilon_{\mathcal{M}}) & \downarrow & \swarrow X(\varepsilon_{\mathcal{N}}) & \\ & & X(S) & & \end{array}$$

Now, if $x \in X(S)$, it follows from SGA 1, III 5.1⁽¹¹⁾, that $X(\varepsilon_{\mathcal{M}})^{-1}(x)$ is isomorphic, functorially in $\mathcal{M}$, to

$$\mathrm{Hom}_{\mathcal{O}_S}(x^*(\Omega_{X/S}^1), \mathcal{M}),$$

where $\Omega_{X/S}^1$ denotes the sheaf of relative differentials of X with respect to S . Now this latter functor (in $\mathcal{M}$) evidently transforms a direct sum of $\mathcal{O}_S$ -modules into the product of the corresponding sets, whence the result.

Corollary 2.2.1. *Let X be an S -scheme and $\mathcal{M}$ a free $\mathcal{O}_S$ -module of finite type. The functor $\mathrm{Hom}_S(\mathrm{I}_S(\mathcal{M}), X)$ is isomorphic (as a functor over X) to a finite product (over X) of copies of $\mathrm{Hom}_S(\mathrm{I}_S, X)$.*

Nota 2.2.2. It follows from the proof of the proposition that $\mathrm{Hom}_S(\mathrm{I}_S, X)$ is isomorphic as an X -functor to $\mathbf{V}(\Omega_{X/S}^1)$ (I 4.6.1), hence representable by the vector fibration⁽¹²⁾ $\mathbf{V}(\Omega_{X/S}^1)$.

3. The tangent bundle, the condition (E)

In this section, unless the contrary is indicated, the letters $\mathcal{M}$, $\mathcal{M}'$, $\mathcal{N}$, etc., will always denote free $\mathcal{O}_S$ -modules of finite type (that is, isomorphic to a finite direct sum of copies of $\mathcal{O}_S$).

⁽¹¹⁾**N.D.E.** See also the addition 0.1.8 in Exp. III.

⁽¹²⁾**N.D.E.** Cf. N.D.E. (33) of Exposé I.

We shall systematically use the identifications justified in Exposé I; thus we shall say “functor over S ” to designate indiscriminately a functor equipped with a morphism to S ($= \mathbf{h}_S$) or a functor on the category of objects over S . One will likewise say “group functor over S ”, etc.

Definition 3.1. Let S be a scheme and $\mathcal{M}$ a free $\mathcal{O}_S$ -module of finite type. Let X be a functor over S . One calls the tangent bundle to X over S relative to the $\mathcal{O}_S$ -module $\mathcal{M}$, and one denotes by $T_{X/S}(\mathcal{M})$, the S -functor

$$T_{X/S}(\mathcal{M}) = \text{Hom}_S(I_S(\mathcal{M}), X).$$

In particular, one calls the tangent bundle to X over S , and one denotes by $T_{X/S}$, the functor

$$T_{X/S} = T_{X/S}(\mathcal{O}_S) = \text{Hom}_S(I_S, X).$$

(13)

Then $\mathcal{M} \mapsto T_{X/S}(\mathcal{M})$ is a covariant functor from the category of free $\mathcal{O}_S$ -modules of finite type to the category of S -functors. In particular the morphisms⁽¹⁴⁾ $\mathcal{M} \rightarrow 0$ and $0 \rightarrow \mathcal{M}$ define respectively an S -morphism

$$\pi_{\mathcal{M}}: T_{X/S}(\mathcal{M}) \rightarrow T_{X/S}(0) \simeq X$$

and a section τ_0 of the latter, called the zero section (or null section).

Remark 3.1.1. ⁽¹⁵⁾ One will note that the projection $\pi_{\mathcal{M}}: T_{X/S}(\mathcal{M}) \rightarrow X$ is induced by the zero section $\varepsilon_{\mathcal{M}}: S \rightarrow I_S(\mathcal{M})$, while the null section $\tau_0: X \rightarrow T_{X/S}(\mathcal{M})$ is induced by the structural morphism $\rho: I_S(\mathcal{M}) \rightarrow S$; i.e. for every point $t \in T_{X/S}(\mathcal{M})(S')$ (resp. $x \in X(S')$), corresponding to an S -morphism $f: S' \times_S I_S(\mathcal{M}) \rightarrow X$ (resp. $g: S' \rightarrow X$), one has $\pi(t) = f \circ (\text{id}_{S'} \times \varepsilon_{\mathcal{M}})$ (resp. $\tau_0(x) = g \circ (\text{id}_{S'} \times \rho)$).

It follows from 3.1 that $\mathcal{M} \mapsto T_{X/S}(\mathcal{M})$ is a covariant functor from the category of free $\mathcal{O}_S$ -modules of finite type to that of functors over X . In particular $\mathbf{O}(S)$ is a monoid of operators of the X -functor $T_{X/S}(\mathcal{M})$ which respects “the functoriality in $\mathcal{M}$ ”.

Scholium 3.1.2. ⁽¹⁵⁾ What precedes means, in particular, the following things. For every S -morphism $\phi: X' \rightarrow X$, set

$$\Sigma(X', \mathcal{M}) = \text{Hom}_X(X', T_{X/S}(\mathcal{M})).$$

One has an action of the multiplicative monoid $\text{End}_{\mathcal{O}_S}(\mathcal{M})$ on $\Sigma(X', \mathcal{M})$, denoted $(\lambda, x) \mapsto \lambda * x$, such that $\lambda * (\mu * x) = (\lambda\mu) * x$, $1 * x = x$, and $0 * x = \tau_0 \circ \phi$, where τ_0 is the null section $X \rightarrow T_{X/S}(\mathcal{M})$. One likewise has an action of $\text{End}_{\mathcal{O}_S}(\mathcal{M} \oplus \mathcal{M})$ on $\Sigma(X', \mathcal{M} \oplus \mathcal{M})$.⁽¹⁶⁾

Moreover, let $m: \mathcal{M} \oplus \mathcal{M} \rightarrow \mathcal{M}$ (resp. $\delta: \mathcal{M} \rightarrow \mathcal{M} \oplus \mathcal{M}$) be the addition (resp. the diagonal map) of $\mathcal{M}$, and denote by $m_{X'}: \Sigma(X', \mathcal{M} \oplus \mathcal{M}) \rightarrow \Sigma(X', \mathcal{M})$ and $\delta_{X'}: \Sigma(X', \mathcal{M}) \rightarrow \Sigma(X', \mathcal{M} \oplus \mathcal{M})$ the morphisms induced by m and δ . For $\lambda, \mu \in \mathbf{O}(S)$, denote by ℓ_{λ} (resp. $\ell_{\lambda, \mu}$) multiplication by λ in $\mathcal{M}$ (resp. by (λ, μ) in $\mathcal{M} \oplus \mathcal{M}$). Since $m \circ \ell_{\lambda, \lambda} = \ell_{\lambda} \circ m$ and $m \circ \ell_{\lambda, \mu} \circ \delta = \ell_{\lambda + \mu}$, one has, for every $z \in \Sigma(X', \mathcal{M} \oplus \mathcal{M})$ and $x \in \Sigma(X', \mathcal{M})$:

$$(\dagger) \quad \lambda * m(z) = m((\lambda, \lambda) * z), \quad m((\lambda, \mu) * \delta(x)) = (\lambda + \mu) * x.$$

Definition 3.2. Let $u \in X(S) = \text{Hom}_S(S, X) = \Gamma(X/S)$. One calls the tangent space to X over S at the point u relative to $\mathcal{M}$, and one denotes by $L_{X/S}^u(\mathcal{M})$, the S -functor obtained

⁽¹³⁾**N.D.E.** When $S = \text{Spec}(k)$ and X is a k -scheme, one has $T_{X/S}(S') = \text{Hom}_S(S' \otimes_k k[\varepsilon], X) = X(S' \otimes_k k[\varepsilon])$; one thus recovers one of the usual definitions of the tangent bundle.

⁽¹⁴⁾**N.D.E.** “ $0 \rightarrow \mathcal{M}$ and $\mathcal{M} \rightarrow 0$ ” has been corrected to: “ $\mathcal{M} \rightarrow 0$ and $0 \rightarrow \mathcal{M}$ ”.

⁽¹⁵⁾**N.D.E.** Paragraphs 3.1.1 and 3.1.2 have been added.

⁽¹⁶⁾**N.D.E.** In fact, one is interested only in the action of $\mathbf{O}(S)$ (resp. $\mathbf{O}(S) \times \mathbf{O}(S)$) on $\Sigma(X', \mathcal{M})$ (resp. $\Sigma(X', \mathcal{M} \oplus \mathcal{M})$), cf. below and the proof of 3.6.

from the X -functor $T_{X/S}(\mathcal{M})$ by inverse image under the morphism $u: S \rightarrow X$:

$$\begin{array}{ccc} L_{X/S}^u(\mathcal{M}) & \longrightarrow & T_{X/S}(\mathcal{M}) \\ \downarrow & & \downarrow \pi \\ S & \xrightarrow{u} & X. \end{array}$$

In particular $L_{X/S}^u(\mathcal{O}_S)$ is denoted by $L_{X/S}^u$. It is the tangent space to X over S at the point u .

Remark 3.2.1. ⁽¹⁷⁾ It follows from 3.1.1 that, for every $t: S' \rightarrow S$, $L_{X/S}^u(\mathcal{M})(S')$ is the set of S -morphisms $f: I_{S'}(\mathcal{M}) \rightarrow X$ such that $f \circ \varepsilon_{\mathcal{M}} = u \circ t$, where $\varepsilon_{\mathcal{M}}: S' \rightarrow I_{S'}(\mathcal{M})$ is the zero section.

Let us immediately record the

Proposition 3.3. *If X is representable by an S -scheme denoted $\tilde{X}$, then $T_{X/S}(\mathcal{M})$ and $L_{X/S}^u(\mathcal{M})$ are representable. In particular $T_{X/S}$ and $L_{X/S}^u$ are representable by vector fibrations over $\tilde{X}$ and over S which are respectively $\mathbf{V}(\Omega_{\tilde{X}/S}^1)$ and $\mathbf{V}(u^*(\Omega_{\tilde{X}/S}^1))$.*

It obviously suffices to prove the proposition for $T_{X/S}(\mathcal{M})$, the analogous results for $L_{X/S}^u(\mathcal{M})$ being deduced from it by inverse image. By Corollary 2.2.1, it even suffices to do so for $T_{X/S}$, and in this case the proposition is nothing other than the remark indicated in 2.2.2.

Remark 3.3.1. From this proposition there follows a particularly simple description of the vector fibration representing $L_{X/S}^u$: the image of the section u of X over S is locally closed⁽¹⁸⁾, hence defined by a quasi-coherent ideal $\mathfrak{m}$ of a scheme induced on an open of X . The quotient $\mathfrak{m}/\mathfrak{m}^2$ may be regarded as a quasi-coherent module over S . It is this module which defines the vector fibration sought.

For example, let X be an algebraic scheme over a field k and u a point of X rational over k . Let $\mathfrak{m}$ be the maximal ideal of the local ring $\mathcal{O}_{X,u}$ and let t be the k -vector space dual of $\mathfrak{m}/\mathfrak{m}^2$; it is the Zariski tangent space of $\mathcal{O}_{X,u}$ at the point u . Then, with the notations of I 4.6.5.1, one has:

$$L_{X/k}^u = \mathbf{V}(\mathfrak{m}/\mathfrak{m}^2) = \mathbf{W}(t).$$

This parenthesis being closed, let us return to the general situation. Let us first remark that $L_{X/S}^u(\mathcal{M})$ is a covariant functor from the category of free $\mathcal{O}_S$ -modules of finite type to that of functors over S . In particular $\mathbf{O}(S)$ is a set of operators of the S -functor $L_{X/S}^u(\mathcal{M})$ which respects the functoriality in $\mathcal{M}$.⁽¹⁹⁾

Proposition 3.4. *The formation of $T_{X/S}(\mathcal{M})$ and $L_{X/S}(\mathcal{M})$ commutes with extension of the base: for every S -scheme S' , one has isomorphisms functorial in $\mathcal{M}$*

$$\begin{aligned} T_{X_{S'}/S'}(\mathcal{M} \otimes \mathcal{O}_{S'}) &\xrightarrow{\sim} T_{X/S}(\mathcal{M})_{S'}, \\ L_{X_{S'}/S'}^{u'}(\mathcal{M} \otimes \mathcal{O}_{S'}) &\xrightarrow{\sim} L_{X/S}^u(\mathcal{M})_{S'}, \quad \text{where } u' = u_{S'}. \end{aligned}$$

This follows immediately from the fact that the Hom commute with extension of the base.

Corollary 3.4.1. *The X -functor $T_{X/S}(\mathcal{M})$ (resp. the S -functor $L_{X/S}^u(\mathcal{M})$) is endowed naturally with a structure of object with operators $\mathbf{O}_X$ (resp. $\mathbf{O}_S$), this structure being functorial in $\mathcal{M}$.*

⁽¹⁷⁾**N.D.E.** This remark has been added.

⁽¹⁸⁾**N.D.E.** Cf. EGA I, 5.3.11.

⁽¹⁹⁾**N.D.E.** Cf. 3.1.2.

Let us first show this for $L_{X/S}^u(\mathcal{M})$. For each S' over S , $\mathbf{O}(S')$ operates on $\mathcal{M} \otimes \mathcal{O}_{S'}$, hence on $L_{X_{S'}/S'}^u(\mathcal{M} \otimes \mathcal{O}_{S'}) = L_{X/S}^u(\mathcal{M})_{S'}$; now one checks that this operation is functorial in S' . It therefore endows $L_{X/S}^u(\mathcal{M})$, as announced, with a structure of functor with operators $\mathbf{O}_S$.

For $T_{X/S}(\mathcal{M})$ it is a little more complicated. For each X' over X , set $T_{X/S}(\mathcal{M})_{X'} = T_{X/S}(\mathcal{M}) \times_X X'$; one must endow $T_{X/S}(\mathcal{M})_{X'}(X') = \text{Hom}_X(X', T_{X/S}(\mathcal{M}))$ with a structure of set with monoid of operators $\mathbf{O}(X')$, in a manner functorial in X' . For this one constructs the following diagram, where $X_{X'}$ denotes $X \times_S X'$ and f' the section of $X_{X'}$ over X' defined by $f: X' \rightarrow X$.

$$\begin{array}{ccc}
 & T_{X_{X'}/X'}(\mathcal{M}) & \\
 \swarrow & & \searrow \\
 T_{X/S}(\mathcal{M}) & & T_{X/S}(\mathcal{M})_{X'} \\
 & & \\
 & X_{X'} & \\
 \swarrow & \xleftarrow{f'} & \searrow \\
 X & \xleftarrow{f} & X' \\
 \searrow & & \swarrow \\
 & S &
 \end{array}$$

⁽²⁰⁾ This diagram, together with 3.2.1, shows that $T_{X/S}(\mathcal{M})_{X'}(X')$ is identified with

$$L_{X_{X'}/X'}^{f'}(\mathcal{M})(X') = \{X'\text{-morphisms } \psi: I_{X'}(\mathcal{M}) \rightarrow X_{X'} \text{ such that } \psi \circ \varepsilon_{\mathcal{M}} = f'\},$$

on which every $a \in \mathbf{O}(X')$ operates via its action on $I_{X'}(\mathcal{M})$, i.e. with the notations of 2.1.1, one has: $a\psi = \psi \circ a^*$, i.e. for every $X'' \rightarrow X'$ and $x \in I_{X'}(\mathcal{M})(X'')$, $(a\psi)(x) = \psi(x \cdot a)$. One then checks easily that this construction is functorial in X' .

The isomorphisms of Proposition 3.4 are then by construction isomorphisms for the structures of $\mathbf{O}_{X_{S'}}$ -objects, resp. $\mathbf{O}_{S'}$ -objects.

Remark 3.4.2. ⁽²¹⁾ The operation of $\mathbf{O}_X$ on $T_{X/S}(\mathcal{M})$ may be seen, more simply, as follows. For every $f: X' \rightarrow X$, one has

$$\text{Hom}_X(X', T_{X/S}(\mathcal{M})) = \{\phi \in \text{Hom}_S(I_{X'}(\mathcal{M}), X) \mid \phi \circ \varepsilon_{\mathcal{M}} = f\},$$

and one has seen in 2.1.3 that $I_{X'}(\mathcal{M})$, considered as an S -functor, is endowed with an operation of the monoid $\mathbf{O}(X')$ which preserves the zero section $\varepsilon_{\mathcal{M}}: X' \rightarrow I_{X'}(\mathcal{M})$. Consequently, if one denotes by a^* the X' -endomorphism of $I_{X'}(\mathcal{M})$ defined by $a \in \mathbf{O}(X')$, one has: $a\phi = \phi \circ a^*$, i.e. for every $S' \rightarrow S$ and $(m, x') \in \text{Hom}_S(S', I_S(\mathcal{M}) \times_S X')$,

$$(a\phi)(m, x') = \phi(m \cdot a(x'), x')$$

(note that $a^* \circ \varepsilon_{\mathcal{M}} = \varepsilon_{\mathcal{M}}$, whence $(a\phi) \circ \varepsilon_{\mathcal{M}} = \phi \circ \varepsilon_{\mathcal{M}} = f$).

Likewise, the operation of $\mathbf{O}_S$ on $L_{X/S}^u(\mathcal{M})$ may be described as follows. For every $t: S' \rightarrow S$, $L_{X/S}^u(\mathcal{M})(S')$ is the set of S -morphisms $\phi: I_{S'}(\mathcal{M}) \rightarrow X$ such that $\phi \circ \varepsilon_{\mathcal{M}} = u \circ t$; for such a ϕ and $a \in \mathbf{O}(S')$, one has: $a\phi = \phi \circ a^*$.

Remark 3.4.3. ⁽²¹⁾ The observations of 3.1.2 hold equally for the operation of $\mathbf{O}_S$ on $L_{X/S}^u(\mathcal{M})$ and for that of $\mathbf{O}_X$ on $T_{X/S}(\mathcal{M})$.

Definition 3.5. Let S be a scheme and X an S -functor. One says that X satisfies the condition (E) relative to S if, for every S' over S and all free $\mathcal{O}_{S'}$ -modules of finite type $\mathcal{M}$

⁽²⁰⁾**N.D.E.** The original has been set out in detail in what follows.

⁽²¹⁾**N.D.E.** Remarks 3.4.2 and 3.4.3 have been added.

and $\mathcal{N}$, the diagram of sets

$$\begin{array}{ccc} & X(\mathrm{I}_{S'}(\mathcal{M} \oplus \mathcal{N})) & \\ \swarrow & & \searrow \\ X(\mathrm{I}_{S'}(\mathcal{M})) & & X(\mathrm{I}_{S'}(\mathcal{N})) \\ \searrow & & \swarrow \\ & X(S') & \end{array}$$

obtained by applying X to the diagram $(*)$ defined in 2.1, is cartesian.⁽²²⁾

3.5.1. — It comes to the same to say that the functor $\mathcal{M} \mapsto \mathrm{T}_{X/S}(\mathcal{M})$ takes direct sums of free $\mathcal{O}_S$ -modules of finite type to products of X -functors;⁽²³⁾ in this case, the same holds for the functor $\mathcal{M} \mapsto \mathrm{L}_{X/S}^u(\mathcal{M}) = S \times_X \mathrm{T}_{X/S}(\mathcal{M})$, for every $u \in \Gamma(X/S)$.

Proposition 2.2 shows that *every representable functor satisfies the condition (E)*.

Abbreviation: instead of saying “ X satisfies the condition (E) with respect to S ”, one will sometimes say “ X/S satisfies the condition (E)”.

If X/S satisfies the condition (E), the functor $\mathcal{M} \mapsto \mathrm{T}_{X/S}(\mathcal{M})$ commutes with products, hence takes groups to groups. In particular $\mathrm{T}_{X/S}(\mathcal{M})$ is a commutative X -group. For the same reason, $\mathrm{L}_{X/S}^u(\mathcal{M})$ is a commutative S -group.

Proposition 3.6. *If X/S satisfies (E), the structure of abelian group on $\mathrm{T}_{X/S}(\mathcal{M})$ (resp. $\mathrm{L}_{X/S}^u(\mathcal{M})$) and the operation of $\mathbf{O}_X$ (resp. $\mathbf{O}_S$) endow $\mathrm{T}_{X/S}(\mathcal{M})$ (resp. $\mathrm{L}_{X/S}^u(\mathcal{M})$) with a structure of $\mathbf{O}_X$ -module (resp. $\mathbf{O}_S$ -module).*

The operation of $\mathbf{O}_X$ (resp. $\mathbf{O}_S$) is functorial in $\mathcal{M}$; it therefore respects the structure of abelian group which is deduced by functoriality from that of $\mathcal{M}$.⁽²⁴⁾ Indeed, let us take up again the notations of 3.1.2. The structure of (abelian) X -group on $\mathrm{T}_{X/S}(\mathcal{M})$ is defined by the composite:

$$\mathrm{T}_{X/S}(\mathcal{M}) \times_X \mathrm{T}_{X/S}(\mathcal{M}) \simeq \mathrm{T}_{X/S}(\mathcal{M} \oplus \mathcal{M}) \xrightarrow{m} \mathrm{T}_{X/S}(\mathcal{M}),$$

and on the other hand the morphism

$$\mathrm{T}_{X/S}(\mathcal{M}) \xrightarrow{\delta} \mathrm{T}_{X/S}(\mathcal{M} \oplus \mathcal{M}) \simeq \mathrm{T}_{X/S}(\mathcal{M}) \times_X \mathrm{T}_{X/S}(\mathcal{M})$$

is the diagonal morphism. Taking Remark 3.4.3 into account, one deduces from the equalities $(\dagger)$ of 3.1.2 that

$$\lambda(x + y) = \lambda x + \lambda y, \quad (\lambda + \mu)x = \lambda x + \mu x,$$

for every $f: X' \rightarrow X$, $x, y \in \mathrm{Hom}_X(X', \mathrm{T}_{X/S}(\mathcal{M}))$ and $\lambda, \mu \in \mathbf{O}(X')$.

Remark 3.6.1. If X is representable, in which case, on the one hand it satisfies (E), and on the other hand $\mathrm{T}_{X/S}$ and $\mathrm{L}_{X/S}^u$ are representable by vector fibrations, the preceding laws are the same as those which are deduced from the structures of vector fibration (cf. I 4.6).⁽²⁵⁾

Proposition 3.4. bis. *If X/S satisfies (E), then $X_{S'}/S'$ satisfies (E) and the isomorphisms of 3.4 respect the structures of $\mathbf{O}_{X_{S'}}$ -modules, resp. of $\mathbf{O}_{S'}$ -modules.*

Without comment.

⁽²²⁾**N.D.E.** For examples of S -functors and S -groups not satisfying (E), see 6.2 and 6.3.

⁽²³⁾**N.D.E.** What follows has been added.

⁽²⁴⁾**N.D.E.** What follows has been set out in detail.

⁽²⁵⁾**N.D.E.** That is, for every S -morphism $f: X' \rightarrow X$, the action of $\mathbf{O}(X')$ on $\mathrm{Hom}_X(X', \mathrm{T}_{X/S}(\mathcal{M}))$ corresponds, via the identification $\mathrm{Hom}_X(X', \mathrm{T}_{X/S}(\mathcal{M})) \simeq \mathrm{Hom}_{\mathcal{O}_{X'}}(f^*(\Omega_{X/S}^1), \mathcal{M} \otimes_{\mathcal{O}_S} \mathcal{O}_{X'})$, to the natural action of $\mathbf{O}(X')$ on the term on the right; this follows from the proof of SGA 1, III 5.1 (see also the addition 0.1.8 in Exposé III).

Proposition 3.7. *The functors $T_{X/S}(\mathcal{M})$ and $L_{X/S}^u(\mathcal{M})$ are functorial in X , i.e. if $f: X \rightarrow X'$ is an S -morphism, one has commutative diagrams:*

$$\begin{array}{ccc} T_{X/S}(\mathcal{M}) & \xrightarrow{T(f)} & T_{X'/S}(\mathcal{M}) \\ \downarrow & & \downarrow \\ X & \xrightarrow{\quad} & X', \end{array} \quad \begin{array}{ccc} L_{X/S}^u(\mathcal{M}) & \xrightarrow{L(f)} & L_{X'/S}^{fou}(\mathcal{M}) \\ & \searrow & \swarrow \\ & S. & \end{array}$$

⁽²⁶⁾ Moreover, if f is a monomorphism, the same holds of $T(f)$ and $L(f)$.

The existence of $T(f)$ and $L(f)$, as well as the last assertion, are deduced immediately from the definitions. The commutativity of the diagrams then results from the functoriality of these morphisms with respect to $\mathcal{M}$ and from the fact that $T_{X/S}(0) = X$.

Remark 3.7.1. ⁽²⁷⁾ Suppose X and X' representable, and let r be the rank of the free $\mathcal{O}_S$ -module $\mathcal{M}$. Then, by 2.2.2, $T_{X/S}(\mathcal{M})$ is isomorphic to the product over X of r copies of $V(\Omega_{X/S}^1)$, and likewise for $T_{X'/S}(\mathcal{M})$. Consequently, the square above is cartesian when f is an open immersion, more generally when $f^*(\Omega_{X'/S}^1) = \Omega_{X/S}^1$, for example if f is étale; under these conditions, one has an isomorphism of S -functors

$$L_{X/S}^u(\mathcal{M}) \xrightarrow{\sim} L_{X'/S}^{fou}(\mathcal{M}).$$

In general, the cartesian square of 3.7 defines a morphism of X -functors:

$$\begin{array}{ccc} T_{X/S}(\mathcal{M}) & \xrightarrow{\quad} & T_{X'/S}(\mathcal{M}) \times_{X'} X \\ & \searrow & \swarrow \\ & X. & \end{array}$$

Proposition 3.7. bis. *Let $f: X \rightarrow X'$ be an S -morphism; if X and X' satisfy (E) with respect to S , then*

$$T_{X/S}(\mathcal{M}) \xrightarrow{T(f)} T_{X'/S}(\mathcal{M})_X \quad \text{resp.} \quad L_{X/S}^u(\mathcal{M}) \xrightarrow{L(f)} L_{X'/S}^{fou}(\mathcal{M})$$

is a morphism of $\mathcal{O}_X$ -modules (resp. of $\mathcal{O}_S$ -modules).

This follows from Proposition 3.7 by functoriality in $\mathcal{M}$.

Proposition 3.8. *Let X and Y be two functors over S . One has isomorphisms, functorial in $\mathcal{M}$:*

$$(3.8.1) \quad T_{X/S}(\mathcal{M}) \times_S T_{Y/S}(\mathcal{M}) \xrightarrow{\sim} T_{(X \times_S Y)/S}(\mathcal{M}),$$

$$(3.8.2) \quad L_{X/S}^u(\mathcal{M}) \times_S L_{Y/S}^v(\mathcal{M}) \xrightarrow{\sim} L_{(X \times_S Y)/S}^{(u,v)}(\mathcal{M}).$$

⁽²⁸⁾ The first isomorphism follows from 1.2 (*), and the second is deduced from it by the base change $(u, v): S \rightarrow X \times_S Y$.

Remark 3.8.0. Let us observe that (3.8.1) may also be interpreted as an isomorphism of $X \times_S Y$ -functors

$$(T_{X/S}(\mathcal{M}) \times_X (X \times_S Y)) \times_{X \times_S Y} (T_{Y/S}(\mathcal{M}) \times_Y (X \times_S Y)) \xrightarrow{\sim} T_{(X \times_S Y)/S}(\mathcal{M}).$$

⁽²⁶⁾**N.D.E.** The sentence which follows has been added.

⁽²⁷⁾**N.D.E.** The hypothesis that X and X' are representable has been added, and what follows has been set out in detail.

⁽²⁸⁾**N.D.E.** What follows has been spelled out.

Corollary 3.8.1. *If X/S is equipped with an algebraic structure defined by finite cartesian products, then $T_{X/S}(\mathcal{M})$ is equipped with a structure of the same species, and the projection $T_{X/S}(\mathcal{M}) \rightarrow X$ is a morphism of this species of structure.*

Proposition 3.8. bis. *If X/S and Y/S satisfy (E), then $(X \times_S Y)/S$ satisfies (E), and (3.8.1) (resp. (3.8.2)) is an isomorphism of $\mathcal{O}_{X \times_S Y}$ -modules (resp. of $\mathcal{O}_S$ -modules).*

PROOF. ⁽²⁹⁾ Suppose that X/S and Y/S satisfy (E). Then, by 3.5.1 and (3.8.1), the same holds for $(X \times_S Y)/S$. Let us show that (3.8.1) is an isomorphism of $\mathcal{O}_{X \times_S Y}$ -modules.

Let $(x, y): Z \rightarrow X \times_S Y$ be an S -morphism; taking 3.4.2 into account, it suffices to see that the map

$$\begin{aligned} & \{\phi \in \text{Hom}_S(I_Z(\mathcal{M}), X) \mid \phi \circ \varepsilon_{\mathcal{M}} = x\} \times \{\psi \in \text{Hom}_S(I_Z(\mathcal{M}), Y) \mid \psi \circ \varepsilon_{\mathcal{M}} = y\} \\ & \rightarrow \{\theta \in \text{Hom}_S(I_Z(\mathcal{M}), X \times_S Y) \mid \theta \circ \varepsilon_{\mathcal{M}} = (x, y)\} \end{aligned}$$

which to (ϕ, ψ) associates $\phi \times \psi$ is a morphism of $\mathcal{O}(Z)$ -modules. But this is clear, for if $a \in \mathcal{O}(Z)$, then $a \cdot (\phi, \psi) = (\phi \circ a^*, \psi \circ a^*)$ is sent to

$$(\phi \circ a^*) \times (\psi \circ a^*) = (\phi \times \psi) \circ a^* = a \cdot (\phi \times \psi).$$

Likewise, using 3.2.1, one shows that (3.8.2) is an isomorphism of $\mathcal{O}_S$ -modules. $\square$

3.9.0. — ⁽³⁰⁾ If X is an S -group and if $e: S \rightarrow X$ denotes its unit section, one denotes:

$$\text{Lie}(X/S, \mathcal{M}) = L_{X/S}^e(\mathcal{M}),$$

i.e. $\text{Lie}(X/S, \mathcal{M})$ is defined by the cartesian square:

$$\begin{array}{ccc} \text{Lie}(X/S, \mathcal{M}) & \xrightarrow{i} & T_{X/S}(\mathcal{M}) \\ \downarrow & & \downarrow p \\ S & \xrightarrow{e} & X. \end{array}$$

By Corollary 3.8.1, the projection $p: T_{X/S}(\mathcal{M}) \rightarrow X$ is a morphism of S -groups, and it follows that $\text{Lie}(X/S, \mathcal{M})$ is equipped with a structure of S -group, and is isomorphic via i to the kernel of p .

If, moreover, X/S satisfies condition (E), one will see in Proposition 3.9 that the structure of S -group on $\text{Lie}(X/S, \mathcal{M})$, induced by that of X , *coincides with the structure of abelian group induced by the functoriality in $\mathcal{M}$* (cf. 3.5.1). In fact, this result is valid under the weaker hypothesis that X be an S -functor in monoids or, more generally, an S -functor in H -sets (cf. the definition below).

Definitions 3.9.0.1. a) Let us introduce the following terminology:⁽³¹⁾ an H -set is a set X equipped with a composition law with two-sided unit, denoted e_X or simply e . If $f: X \rightarrow Y$ is a morphism of H -sets, its kernel $\text{Ker } f$ is $f^{-1}(e_Y)$; it is a sub- H -set of X .

b) An H -object in a category $\mathcal{C}$ is defined in the usual way: it is thus an object X of $\mathcal{C}$, equipped with a morphism $X \times X \rightarrow X$ such that there exists a section of X (over the final object) possessing the properties of a two-sided unit. Every $\mathcal{C}$ -monoid, in particular every $\mathcal{C}$ -group, is thus a $\mathcal{C}$ - H -object. In particular, an H -object of the category of functors over the scheme S will be called an S - H -functor.

c) If X is an S - H -functor (for example, an S -group), and if $e: S \rightarrow X$ denotes the unit section of X , one denotes:

$$\text{Lie}(X/S, \mathcal{M}) = L_{X/S}^e(\mathcal{M}) \quad \text{and} \quad \text{Lie}(X/S) = \text{Lie}(X/S, \mathcal{O}_S).$$

⁽²⁹⁾**N.D.E.** This proof has been added.

⁽³⁰⁾**N.D.E.** This paragraph has been added, in order to explain the introduction of the notion of S - H -functor.

⁽³¹⁾**N.D.E.** This is inspired by the notion of H -space in topology.

Let us then make explicit the following particular case of 3.8.1.⁽³²⁾

Corollary 3.9.0.2. *If X is an S - H -functor (resp. an S -group), then $T_{X/S}(\mathcal{M})$ and $\text{Lie}(X/S, \mathcal{M})$ are also S - H -functors (resp. S -groups), and one has morphisms of S - H -functors (resp. of S -groups):*

$$\text{Lie}(X/S, \mathcal{M}) \xrightarrow{i} T_{X/S}(\mathcal{M}) \xrightleftharpoons[s]{p} X,$$

where i is an isomorphism of $\text{Lie}(X/S)(\mathcal{M})$ onto $\text{Ker } p$ and s is a section of p .

Proposition 3.9. *Let X be an S - H -functor satisfying (E) with respect to S . The structure of S - H -functor on $\text{Lie}(X/S, \mathcal{M})$ coming from that of X coincides with the structure of S -group defined in 3.5.1.*

It follows from what has been said above that $\text{Lie}(X/S, \mathcal{M})$ is an H -object in the category of $\mathcal{O}_S$ -modules. The proposition will then follow from the following lemma:⁽³³⁾

Lemma 3.10. *Let $\mathcal{C}$ be a category. Let G be an H -object in the category of $\mathcal{C}$ - H -objects; G is thus a $\mathcal{C}$ - H -object (whose composition law we shall denote by $f: G \times G \rightarrow G$) equipped with a morphism of $\mathcal{C}$ - H -objects $h: G \times G \rightarrow G$.⁽³⁴⁾ Then $f = h$ and f is commutative.*

By taking the values of the functors on a variable argument, one reduces in the usual way to checking the lemma when $\mathcal{C}$ is the category of sets. One thus has a set G and two maps $f, h: G \times G \rightarrow G$ such that

$$h(f(x, y), f(z, t)) = f(h(x, z), h(y, t)).$$

On the other hand one has two elements of G , namely e and u , with $f(e, x) = f(x, e) = x$, $h(u, x) = h(x, u) = x$. One sees first that

$$h(f(u, y), f(x, u)) = f(x, y) = h(f(x, u), f(u, y)).$$

In particular, for $y = e$, resp. $x = e$, one obtains, respectively,

$$x = f(x, e) = h(f(u, e), f(x, u)) = h(u, f(x, u)) = f(x, u),$$

$$y = f(e, y) = h(f(e, u), f(u, y)) = h(u, f(u, y)) = f(u, y),$$

whence, substituting back into the original equality,

$$h(y, x) = f(x, y) = h(x, y).$$

This proves the lemma, and also Proposition 3.9. One then deduces from 3.9 the following corollaries.

Corollary 3.9.1. *If X is an S - H -functor satisfying (E) with respect to S , every element of $X(I_S(\mathcal{M}))$ which projects onto the unit element of $X(S)$ is invertible.*

Corollary 3.9.2. *If X is an S -monoid satisfying (E) with respect to S , an element of $X(I_S(\mathcal{M}))$ is invertible if and only if its image in $X(S)$ is invertible.*

Corollary 3.9.3. *If X is an S -group satisfying (E) with respect to S , the two S -group laws on $\text{Lie}(X/S, \mathcal{M})$ coincide.*

Corollary 3.9.4. ⁽³⁵⁾ *Let G be an S -group satisfying (E) with respect to S . For $n \in \mathbf{Z}$, let $n_G: G \rightarrow G$ be the morphism of S -functors defined by $g \mapsto g^n$. Then the derived morphism $L(n_G): \text{Lie}(G/S) \rightarrow \text{Lie}(G/S)$ is the “multiplication by n ”, i.e. the map which to every $x \in \text{Lie}(G/S)(S')$ associates nx .*

⁽³²⁾**N.D.E.** Corollary 3.9.0.2 has been added; it will be useful in 4.1.

⁽³³⁾**N.D.E.** Inspired by the standard proof that the fundamental group of an H -space is abelian.

⁽³⁴⁾**N.D.E.** $G \times G$ is equipped with the law $(G \times G) \times (G \times G) \rightarrow G \times G, ((x, z), (y, t)) \mapsto (f(x, y), f(z, t))$.

⁽³⁵⁾**N.D.E.** This corollary has been added; it amplifies Remark 4.1.1.2 further on.

Let us first observe that n_G is not in general a morphism of groups, but that it preserves the unit section $e: S \rightarrow G$, so the derived morphism $L(n_G)$ does indeed send $\text{Lie}(G/S) = L_{X/S}^e$ into itself. If one denotes by i the inclusion $\text{Lie}(G/S) \hookrightarrow T_{G/S}$, then $L(n_G)$ is defined by the equality $i(L(n_G)(x)) = i(x)^n$, for every $S' \rightarrow S$ and every $x \in \text{Lie}(G/S)(S')$. Now, by 3.9, the two group laws on $\text{Lie}(G/S)$ (the one coming from condition (E) and the one coming from the law of G) coincide, i.e. one has $i(x)^n = i(nx)$, whence $L(n_G)(x) = nx$.

Before drawing further consequences from Proposition 3.9, let us prove another functoriality result:

Proposition 3.11. *In the situation of Section 1, one has an isomorphism, functorial in $\mathcal{M}$,*

$$T_{\text{Hom}_{Z/S}(X,Y)/S}(\mathcal{M}) \xrightarrow{\sim} \text{Hom}_{Z/S}(X, T_{Y/Z}(\mathcal{M})).$$

Indeed, one has by definition (cf. 3.1):

$$T_{\text{Hom}_{Z/S}(X,Y)/S}(\mathcal{M}) = \text{Hom}_S(I_S(\mathcal{M}), \text{Hom}_{Z/S}(X, Y)).$$

By isomorphism (4) of 1.1, applied to $T = I_S(\mathcal{M})$, one has:

$$\text{Hom}_S(I_S(\mathcal{M}), \text{Hom}_{Z/S}(X, Y)) \simeq \text{Hom}_{Z/S}(X, \text{Hom}_Z(Z \times_S I_S(\mathcal{M}), Y)).$$

Taking account of the isomorphism $Z \times_S I_S(\mathcal{M}) \simeq I_Z(\mathcal{M})$, this gives

$$\begin{aligned} T_{\text{Hom}_{Z/S}(X,Y)/S}(\mathcal{M}) &\simeq \text{Hom}_{Z/S}(X, \text{Hom}_Z(I_Z(\mathcal{M}), Y)) \\ &= \text{Hom}_{Z/S}(X, T_{Y/Z}(\mathcal{M})). \end{aligned}$$

Corollary 3.11.1. *If Y/Z satisfies (E), then $\text{Hom}_{Z/S}(X, Y)/S$ satisfies (E), and the isomorphism of 3.11 respects the structures of $\mathcal{O}$ -modules over $\text{Hom}_{Z/S}(X, Y)$.⁽³⁶⁾*

PROOF. Let $\mathcal{M}, \mathcal{N}$ be two free $\mathcal{O}_S$ -modules of finite type. If Y/Z satisfies (E), then

$$T_{Y/Z}(\mathcal{M} \oplus \mathcal{N}) \simeq T_{Y/Z}(\mathcal{M}) \times_Y T_{Y/Z}(\mathcal{N}).$$

The right-hand term is a subfunctor of $T_{Y/Z}(\mathcal{M}) \times_S T_{Y/Z}(\mathcal{N})$, and via the isomorphism of 1.2 (*), one obtains an isomorphism

$$\begin{aligned} \text{Hom}_{Z/S}(X, T_{Y/Z}(\mathcal{M} \oplus \mathcal{N})) \\ \simeq \text{Hom}_{Z/S}(X, T_{Y/Z}(\mathcal{M})) \times_{\text{Hom}_{Z/S}(X,Y)} \text{Hom}_{Z/S}(X, T_{Y/Z}(\mathcal{N})). \end{aligned}$$

Combined with 3.11, this yields:

$$\begin{aligned} T_{\text{Hom}_{Z/S}(X,Y)/S}(\mathcal{M} \oplus \mathcal{N}) \\ \simeq T_{\text{Hom}_{Z/S}(X,Y)/S}(\mathcal{M}) \times_{\text{Hom}_{Z/S}(X,Y)} T_{\text{Hom}_{Z/S}(X,Y)/S}(\mathcal{N}), \end{aligned}$$

so $\text{Hom}_{Z/S}(X, Y)$ satisfies (E) with respect to S . This proves the first assertion of the corollary.

Let us turn to the second. Write $H = \text{Hom}_{Z/S}(X, Y)$, and let us take an S -morphism $\Delta: H' \rightarrow \text{Hom}_{Z/S}(X, Y)$, i.e. a Z -morphism $\delta: H' \times_S X \rightarrow Y$, which thus makes $H' \times_S X$ into a Y -object.

⁽³⁶⁾**N.D.E.** Setting $H = \text{Hom}_{Z/S}(X, Y)$, this implies, in particular, that $T_{H/S}(\mathcal{M})$ is equipped with a natural structure of module over $\prod_{X \times_S H/H} \mathcal{O}_H$. This result seemed a little surprising to the editors; for this reason its proof has been spelled out.

On the one hand, one has a commutative diagram:

$$\begin{array}{ccc}
\mathrm{Hom}_H(H', \mathrm{Hom}_{Z/S}(X, \mathrm{T}_{Y/Z}(\mathcal{M}))) & \hookrightarrow & \mathrm{Hom}_S(H', \mathrm{Hom}_{Z/S}(X, \mathrm{T}_{Y/Z}(\mathcal{M}))) \\
\downarrow & & \downarrow \\
\mathrm{Hom}_Y(H' \times_S X, \mathrm{T}_{Y/Z}(\mathcal{M})) & \hookrightarrow & \mathrm{Hom}_Z(H' \times_S X, \mathrm{T}_{Y/Z}(\mathcal{M})) \\
\downarrow & & \downarrow \\
\{\psi \in \mathrm{Hom}_Z(\mathrm{I}_{H' \times_S X}(\mathcal{M}), Y) \mid \psi \circ \varepsilon_{\mathcal{M}} = \delta\} & \hookrightarrow & \mathrm{Hom}_Z(\mathrm{I}_{H' \times_S X}(\mathcal{M}), Y).
\end{array}$$

By 1.3, the action of $\alpha \in \mathcal{O}(H' \times_S X)$ on $\Psi \in \mathrm{Hom}_Y(H' \times_S X, \mathrm{T}_{Y/Z}(\mathcal{M}))$ is given by: for every $U \rightarrow S$ and every $(h, x) \in \mathrm{Hom}_S(U, H' \times_S X)$ (with U then lying over Y via $\delta \circ (h, x)$), one has:

$$(\alpha\Psi)(h, x) = \alpha(h, x) \Psi(h, x),$$

where $\alpha(h, x) \in \mathcal{O}(U)$ acts on $\Psi(h, x) \in \mathrm{T}_{Y/Z}(\mathcal{M})(U)$ via the structure of $\mathcal{O}_Y$ -module on $\mathrm{T}_{Y/Z}(\mathcal{M})$. By 3.4.2, the latter is given, via the identification

$$\begin{aligned}
& \mathrm{Hom}_Y(H' \times_S X, \mathrm{T}_{Y/Z}(\mathcal{M})) \\
&= \{\psi \in \mathrm{Hom}_Z(\mathrm{I}_{H' \times_S X}(\mathcal{M}), Y) \mid \psi \circ \varepsilon_{\mathcal{M}} = \delta\},
\end{aligned}$$

by: for every $(m, h, x) \in \mathrm{Hom}_S(U, \mathrm{I}_S(\mathcal{M}) \times_S H' \times_S X)$,

$$(1) \quad (\alpha\psi)(m, h, x) = \psi(m \cdot \alpha(h, x), h, x).$$

On the other hand, let us consider the tangent space $\mathrm{T}_{H/S}(\mathcal{M}) = \mathrm{Hom}_S(\mathrm{I}_S(\mathcal{M}), H)$; one has a commutative diagram:

$$\begin{array}{ccc}
\mathrm{Hom}_H(H', \mathrm{T}_{H/S}(\mathcal{M})) & \hookrightarrow & \mathrm{Hom}_S(H', \mathrm{T}_{H/S}(\mathcal{M})) \\
\downarrow & & \downarrow \\
\{\Phi \in \mathrm{Hom}_S(\mathrm{I}_{H'}(\mathcal{M}), H) \mid \Phi \circ \varepsilon_{\mathcal{M}} = \Delta\} & \hookrightarrow & \mathrm{Hom}_S(\mathrm{I}_{H'}(\mathcal{M}), H) \\
(*) \downarrow & & \downarrow \\
\{\phi \in \mathrm{Hom}_Z(\mathrm{I}_{H' \times_S X}(\mathcal{M}), Y) \mid \phi \circ \varepsilon_{\mathcal{M}} = \delta\} & \hookrightarrow & \mathrm{Hom}_Z(\mathrm{I}_{H' \times_S X}(\mathcal{M}), Y),
\end{array}$$

where the bijection $(*)$ is given as follows (cf. 1.1 (2) and I 1.7.1): for every $U \rightarrow S$ and every $(m, h, x) \in \mathrm{Hom}_S(U, \mathrm{I}_S(\mathcal{M}) \times_S H' \times_S X)$ (so that U lies over Z via $U \xrightarrow{x} X \rightarrow Z$), one has $\Phi(m, h) \in \mathrm{Hom}_Z(X \times_S U, Y)$ and

$$(\dagger) \quad \phi(m, h, x) = \Phi(m, h) \circ (x \times \mathrm{id}_U) \in \mathrm{Hom}_Z(U, Y).$$

By 3.4.2 (where one replaces X by $\mathrm{Hom}_{Z/S}(X, Y)$ and X' by H'), the action of $a \in \mathcal{O}(H')$ on $\Phi \in \mathrm{Hom}_S(\mathrm{I}_{H'}(\mathcal{M}), H)$ is given by: for every $U \rightarrow S$ and every $(m, h) \in \mathrm{Hom}_S(U, \mathrm{I}_S(\mathcal{M}) \times_S H')$,

$$(a\Phi)(m, h) = \Phi(m \cdot a(h), h).$$

Consequently, if ϕ (resp. $a\phi$) is the element of $\mathrm{Hom}_Z(\mathrm{I}_{H' \times_S X}(\mathcal{M}), Y)$ associated with Φ (resp. $a\Phi$), one has, by $(\dagger)$,

$$\begin{aligned}
(2) \quad (a\phi)(m, h, x) &= \Phi(m \cdot a(h), h) \circ (x \times \mathrm{id}_U) \\
&= \phi(m \cdot a(h), h, x).
\end{aligned}$$

Together with (1), this shows that the isomorphism $\mathrm{T}_{H/S}(\mathcal{M}) \xrightarrow{\sim} \mathrm{Hom}_{Z/S}(X, \mathrm{T}_{Y/Z}(\mathcal{M}))$ of 3.11.1 is an isomorphism of $\mathcal{O}(H)$ -modules; moreover, for every $H' \rightarrow H$, the structure of $\mathcal{O}(H')$ -module on $\mathrm{Hom}_H(H', \mathrm{T}_{H/S}(\mathcal{M}))$ extends, in a manner functorial in H' , to a structure of $\mathcal{O}(H' \times_S X)$ -module. $\square$

In particular, for $Z = S$, one obtains the following corollary.

Corollary 3.11.2. *One has an isomorphism, functorial in $\mathcal{M}$,*

$$T_{\mathrm{Hom}_S(X,Y)/S}(\mathcal{M}) \xrightarrow{\sim} \mathrm{Hom}_S(X, T_{Y/S}(\mathcal{M})).$$

Moreover, if Y/S satisfies (E), then $\mathrm{Hom}_S(X, Y)/S$ satisfies (E), and the preceding isomorphism respects the structures of $\mathcal{O}$ -modules over $\mathrm{Hom}_S(X, Y)$.

⁽³⁷⁾ Let $u: X \rightarrow Y$ be an S -morphism; one identifies it with the constant morphism $u: S \rightarrow \mathrm{Hom}_S(X, Y)$ such that $u(f) = u$ for every $f: S' \rightarrow S$. One then sees at once that the fibered product of u and of $\mathrm{Hom}_S(X, T_{Y/S}(\mathcal{M})) \rightarrow \mathrm{Hom}_S(X, Y)$ is identified with $\mathrm{Hom}_{Y/S}(X, T_{Y/S}(\mathcal{M}))$, where X lies over Y via u . Consequently, from the definition of $L_{\mathrm{Hom}_S(X,Y)/S}^u(\mathcal{M})$ and from the preceding corollary, one deduces the following:

Corollary 3.11.3. *Let $u: X \rightarrow Y$ be an S -morphism. One has an isomorphism functorial in $\mathcal{M}$ (where, in the right-hand term, X lies over Y via u):*

$$L_{\mathrm{Hom}_S(X,Y)/S}^u(\mathcal{M}) \xrightarrow{\sim} \mathrm{Hom}_{Y/S}(X, T_{Y/S}(\mathcal{M})).$$

⁽³⁸⁾ It is an isomorphism of $\mathbf{O}_S$ -modules if Y/S satisfies (E).

In particular, for $Y = X$, $\mathrm{End}_S(X)$ is an S -functor in monoids, hence *a fortiori* an S -H-functor; recalling that $\mathrm{Lie}(\mathrm{End}_S(X)/S, \mathcal{M})$ denotes $L_{\mathrm{End}_S(X)/S}^e(\mathcal{M})$, where e is the unit section (cf. 3.9.0.1), one obtains:

Corollary 3.11.4. *One has an isomorphism functorial in $\mathcal{M}$*

$$\mathrm{Lie}(\mathrm{End}_S(X)/S, \mathcal{M}) \xrightarrow{\sim} \prod_{X/S} T_{X/S}(\mathcal{M});$$

⁽³⁸⁾ it is an isomorphism of $\mathbf{O}_S$ -modules if Y/S satisfies (E).

Remark 3.11.5. ⁽³⁹⁾ Suppose that X/S satisfies (E). Then

$$\prod_{X/S} T_{X/S}(\mathcal{M}) = \mathrm{Hom}_{X/S}(X, T_{X/S}(\mathcal{M}))$$

is endowed with a structure of $(\prod_{X/S} \mathbf{O}_X)$ -module, i.e. for every $S' \rightarrow S$,

$$\begin{aligned} & \mathrm{Hom}_{X/S}(X, T_{X/S}(\mathcal{M}))(S') \\ &= \{\psi \in \mathrm{Hom}_X(I_{S'}(\mathcal{M}) \times_S X, X) \mid \psi \circ (\varepsilon_{\mathcal{M}} \times \mathrm{id}_X) = \mathrm{pr}_X\} \end{aligned}$$

is endowed with a structure of $\mathbf{O}(X \times_S S')$ -module, functorial in S' . This follows, at one's choice, from 3.6 and from the properties of the functor $\prod_{X/S}$ (cf. 1.2), or else from the proof of 3.11.1.

We shall now interpret geometrically the definition of the tangent bundle. ⁽⁴⁰⁾ Let U be an S -functor; by I 1.7.2, one has isomorphisms functorial in $\mathcal{M}$

$$\begin{aligned} T_{X/S}(\mathcal{M})(U) &= \mathrm{Hom}_S(U, \mathrm{Hom}_S(I_S(\mathcal{M}), X)) \\ &\simeq \mathrm{Hom}_S(I_S(\mathcal{M}), \mathrm{Hom}_S(U, X)) \\ &\simeq \mathrm{Hom}_{I_S(\mathcal{M})}(U_{I_S(\mathcal{M})}, X_{I_S(\mathcal{M})}). \end{aligned}$$

⁽³⁷⁾ **N.D.E.** The following sentences have been added.

⁽³⁸⁾ **N.D.E.** The following sentence has been added.

⁽³⁹⁾ **N.D.E.** This remark has been added.

⁽⁴⁰⁾ **N.D.E.** The original has been simplified in what follows.

In particular, the morphism $\mathcal{M} \rightarrow 0$ gives a commutative diagram:

$$\begin{array}{ccc} \mathrm{Hom}_S(U, T_{X/S}(\mathcal{M})) & \xrightarrow{\sim} & \mathrm{Hom}_{I_S(\mathcal{M})}(U_{I_S(\mathcal{M})}, X_{I_S(\mathcal{M})}) \\ \pi_{\mathcal{M}} \downarrow & & \downarrow \\ \mathrm{Hom}_S(U, X) & \xrightarrow{\text{identity}} & \mathrm{Hom}_S(U, X), \end{array}$$

where the second vertical arrow is obtained by the base change $\varepsilon_{\mathcal{M}}: S \rightarrow I_S(\mathcal{M})$. ⁽⁴¹⁾

Consequently:

Proposition 3.12. *Let $h_0: U \rightarrow X$ be an S -morphism. Then $\mathrm{Hom}_X(U, T_{X/S}(\mathcal{M}))$ is identified with the set of $I_S(\mathcal{M})$ -morphisms from $U_{I_S(\mathcal{M})}$ into $X_{I_S(\mathcal{M})}$ which restrict to h_0 on U (viewed as a subobject of $U \times_S I_S(\mathcal{M})$ via $\mathrm{id}_U \times_S \varepsilon_{\mathcal{M}}$).*

⁽⁴²⁾ In particular, for $U = X$ and $h_0 = \mathrm{id}_X$, one obtains:

Corollary 3.12.1. *The set $\Gamma(T_{X/S}(\mathcal{M})/X)$ is identified with the set of $I_S(\mathcal{M})$ -endomorphisms φ of $X_{I_S(\mathcal{M})}$ which induce the identity on X , i.e. such that the diagram below is commutative.⁽⁴³⁾*

$$\begin{array}{ccc} I_X(\mathcal{M}) & \xrightarrow{\varphi} & I_X(\mathcal{M}) \\ \varepsilon_{\mathcal{M}} \uparrow & & \uparrow \varepsilon_{\mathcal{M}} \\ X & \xrightarrow{\mathrm{id}} & X \end{array}$$

⁽⁴⁴⁾ On the other hand, by 3.11.2, $\Gamma(T_{X/S}(\mathcal{M})/X) \simeq \mathrm{Lie}(\mathrm{End}_S(X)/S, \mathcal{M})(S)$; if moreover X/S satisfies (E), then $\mathrm{End}_S(X)/S$ satisfies (E), so $\mathrm{Lie}(\mathrm{End}_S(X)/S, \mathcal{M})$ is an $\mathbf{O}_S$ -module, by 3.6 (and in fact a $(\prod_{X/S} \mathbf{O}_X)$ -module, by 3.11.5). Applying 3.9, one deduces from this:

Proposition 3.13. *If X/S satisfies (E), the abelian group $\Gamma(T_{X/S}(\mathcal{M})/X)$ is identified with the set of $I_S(\mathcal{M})$ -endomorphisms of $X_{I_S(\mathcal{M})}$ which induce the identity on X . Consequently, every $I_S(\mathcal{M})$ -endomorphism of $X_{I_S(\mathcal{M})}$ which induces the identity on X is an automorphism.*

Corollary 3.13.1. *Let $u: X \rightarrow Y$ be an S -isomorphism, with Y/S satisfying (E). Every $I_S(\mathcal{M})$ -morphism from $X_{I_S(\mathcal{M})}$ into $Y_{I_S(\mathcal{M})}$ which extends u is an isomorphism.*

Corollary 3.13.2. *If Y/S satisfies (E), then the monomorphism*

$$\mathrm{Isom}_S(X, Y) \rightarrow \mathrm{Hom}_S(X, Y)$$

induces, for every $u \in \mathrm{Isom}_S(X, Y)$, an isomorphism

$$L_{\mathrm{Isom}_S(X, Y)/S}^u(\mathcal{M}) \xrightarrow{\sim} L_{\mathrm{Hom}_S(X, Y)/S}^u(\mathcal{M}).$$

PROOF. ⁽⁴⁵⁾ One must see that $L_{\mathrm{Isom}_S(X, Y)/S}^u(\mathcal{M})(S') \rightarrow L_{\mathrm{Hom}_S(X, Y)/S}^u(\mathcal{M})(S')$ is a bijection, for every $S' \rightarrow S$. By base change (cf. 3.4), it suffices to do this for $S' = S$. In this case, $L_{\mathrm{Hom}_S(X, Y)/S}^u(\mathcal{M})(S)$ (resp. $L_{\mathrm{Isom}_S(X, Y)/S}^u(\mathcal{M})(S)$) is the set of $I_S(\mathcal{M})$ -morphisms (resp. automorphisms) $X_{I_S(\mathcal{M})} \rightarrow Y_{I_S(\mathcal{M})}$ which extend u , and one applies the preceding corollary. $\square$

⁽⁴¹⁾**N.D.E.** Via the isomorphism $\mathrm{Hom}_{I_S(\mathcal{M})}(U_{I_S(\mathcal{M})}, X_{I_S(\mathcal{M})}) \simeq \mathrm{Hom}_S(U \times_S I_S(\mathcal{M}), X)$, this is equivalent to Remark 3.1.1. Likewise, 3.12 is equivalent to the first part of Remark 3.4.2.

⁽⁴²⁾**N.D.E.** The original has been spelled out in what follows.

⁽⁴³⁾**N.D.E.** When $\mathcal{M} = \mathcal{O}_X$, these are called “infinitesimal endomorphisms” of X ; see 6.2 at the end of this exposé.

⁽⁴⁴⁾**N.D.E.** The original has been spelled out in what follows.

⁽⁴⁵⁾**N.D.E.** The following proof has been added.

Corollary 3.13.3. *If X/S satisfies (E), the monomorphism $\text{Aut}_S(X) \rightarrow \text{End}_S(X)$ induces, for every $u \in \text{Aut}_S(X)$, an isomorphism $L_{\text{Aut}_S(X)/S}^u(\mathcal{M}) \xrightarrow{\sim} L_{\text{End}_S(X)/S}^u(\mathcal{M})$. In particular, one has*

$$\text{Lie}(\text{Aut}_S(X)/S, \mathcal{M}) \xrightarrow{\sim} \text{Lie}(\text{End}_S(X)/S, \mathcal{M}) \xrightarrow{\sim} \prod_{X/S} T_{X/S}(\mathcal{M}),$$

⁽⁴⁶⁾ so that $\text{Lie}(\text{Aut}_S(X)/S, \mathcal{M})$ is endowed with a structure of $(\prod_{X/S} \mathbf{O}_X)$ -module.

3.14. — Suppose, to conclude, that X is representable.⁽⁴⁷⁾ In this case, one has seen in 2.2.2 that the X -functor $T_{X/S}$ is representable by $\mathbf{V}(\Omega_{X/S}^1)$, whence the bijections:

$$\Gamma(T_{X/S}/X) \simeq \text{Hom}_X(\Omega_{X/S}^1, \mathcal{O}_X) \simeq \text{Dér}_{\mathcal{O}_S}(\mathcal{O}_X).$$

This can also be deduced from what precedes, as follows. By 3.13, $\Gamma(T_{X/S}/X)$ is identified with the set of infinitesimal endomorphisms of X (i.e. the I_S -endomorphisms of X_{I_S} inducing the identity on X). Now X and X_{I_S} have the same underlying topological space, the corresponding sheaves of rings being $\mathcal{O}_X$ and $\mathcal{D}\mathcal{O}_X = \mathcal{O}_X \oplus \mathcal{M}$, where $\mathcal{M} = \mathcal{O}_X$ is regarded as a square-zero ideal. Denoting by $\pi: \mathcal{D}\mathcal{O}_X \rightarrow \mathcal{O}_X$ the morphism of $\mathcal{O}_X$ -algebras which vanishes on $\mathcal{M}$, one deduces from this that giving an infinitesimal endomorphism of X is equivalent to giving a morphism of $\mathcal{O}_S$ -algebras $\varphi: \mathcal{O}_X \rightarrow \mathcal{D}\mathcal{O}_X$ such that $\pi \circ \varphi = \text{id}_{\mathcal{O}_X}$, which is equivalent to giving an $\mathcal{O}_S$ -derivation of the sheaf of rings $\mathcal{O}_X$.

Moreover, one easily sees that if $D, D' \in \text{Dér}_{\mathcal{O}_S}(\mathcal{O}_X)$ and if one denotes by φ_D the infinitesimal endomorphism corresponding to D , then

$$\varphi_{D+D'} = \varphi_D \circ \varphi_{D'}.$$

This shows that the identification

$$\{\text{infinitesimal endomorphisms of } X\} \simeq \text{Dér}_{\mathcal{O}_S}(\mathcal{O}_X)$$

is an isomorphism of abelian groups. Taking 3.13 (and 3.11.5) into account, one has therefore constructed an isomorphism of abelian groups (and even of $\mathbf{O}(X)$ -modules)

$$\Gamma(T_{X/S}/X) \xrightarrow{\sim} \text{Dér}_{\mathcal{O}_S}(\mathcal{O}_X)$$

which recovers the classical interpretation of tangent vector fields in terms of derivations of the structure sheaf.⁽⁴⁸⁾ Let us moreover observe that $\Gamma(T_{X/S}/X)$ is equal to $H^0(X, \mathfrak{g}_{X/S})$, where $\mathfrak{g}_{X/S}$ is the dual of $\Omega_{X/S}^1$.

4. Tangent space to a group – Lie algebras

4.1. — Let G be a group functor over S . By 3.9.0.2, $T_{G/S}(\mathcal{M})$ and $\text{Lie}(G/S, \mathcal{M})$ are endowed with structures of groups over S , and one has group morphisms

$$\text{Lie}(G/S, \mathcal{M}) \xrightarrow{i} T_{G/S}(\mathcal{M}) \xrightarrow{\overset{s}{\varphi}} G;$$

by definition i is an isomorphism of $\text{Lie}(G/S)(\mathcal{M})$ onto the kernel of p , and s is a section of p . It then follows from I 2.3.7 that this sequence of morphisms allows one to identify $T_{G/S}(\mathcal{M})$ with the semidirect product of G by $\text{Lie}(G/S, \mathcal{M})$.

Definition 4.1.A. ⁽⁴⁹⁾ The corresponding operation of G on $\text{Lie}(G/S, \mathcal{M})$ is denoted

$$\text{Ad}: G \rightarrow \text{Aut}_{S\text{-gr.}}(\text{Lie}(G/S, \mathcal{M}))$$

⁽⁴⁶⁾**N.D.E.** The following has been added.

⁽⁴⁷⁾**N.D.E.** The reminder of 2.2.2 has been added, and the sequel has been spelled out.

⁽⁴⁸⁾**N.D.E.** Moreover, this isomorphism is functorial in S : if $S' \rightarrow S$, setting $X' = X_{S'}$, one has $\text{Lie}(\text{Aut}_S(X)/S)(S') \simeq \Gamma(T_{X'/S'}/X') \simeq \text{Dér}_{\mathcal{O}_{S'}}(\mathcal{O}_{X'})$.

⁽⁴⁹⁾**N.D.E.** The numbering 4.1.A and 4.1.B has been added in order to set off these definitions.

and called the *adjoint representation* (relative to $\mathcal{M}$) of G ; one therefore has by definition, for $x \in G(S')$ and $X \in \text{Lie}(G/S, \mathcal{M})(S')$:

$$\text{Ad}(x)X = i^{-1}(s(x) i(X) s(x)^{-1}).$$

If G is commutative, then so is $T_{G/S}(\mathcal{M})$, and $\text{Ad}(x)X = X$.

Definition 4.1.B. ⁽⁴⁹⁾ If G and H are two group functors over S and if $f: G \rightarrow H$ is a group morphism, one deduces from it by functoriality a morphism of exact sequences compatible with the canonical sections:

$$\begin{array}{ccccccc} 1 & \longrightarrow & \text{Lie}(G/S, \mathcal{M}) & \longrightarrow & T_{G/S}(\mathcal{M}) & \longrightarrow & G \longrightarrow 1 \\ & & \downarrow L(f) & & \downarrow T(f) & & \downarrow f \\ 1 & \longrightarrow & \text{Lie}(H/S, \mathcal{M}) & \longrightarrow & T_{H/S}(\mathcal{M}) & \longrightarrow & H \longrightarrow 1 \end{array}$$

$L(f)$, which one will also denote $\text{Lie}(f)$, is the *derived morphism* of f .⁽⁵⁰⁾

Remark 4.1.C. ⁽⁵¹⁾ If G/S and H/S satisfy (E), then $L(f)$ respects the structures of $\mathbf{O}_S$ -modules deduced from the “functoriality in $\mathcal{M}$ ” (cf. 3.6).

Proposition 4.1.1. *Let $g \in G(S)$. Then $\text{Ad}(g): \text{Lie}(G/S, \mathcal{M}) \rightarrow \text{Lie}(G/S, \mathcal{M})$ is the derived morphism of $\text{Int}(g): G \rightarrow G$.*

Indeed $\text{Ad}(g)X = i^{-1}(\text{Int}(g) i(X))$, which is none other than $L(\text{Int}(g))(X)$ by the very definition of the derived morphism.

Suppose that G/S satisfies (E). Then, by proposition 3.9, the group structure of $\text{Lie}(G/S, \mathcal{M})$ defined as above is none other than the structure induced by its structure of $\mathbf{O}_S$ -module (defined by means of (E)). One then deduces from the preceding proposition and from the functoriality of the operation of $\mathbf{O}_S$ (cf. 3.6) the corollary:

Corollary 4.1.1.1. *Suppose that G/S satisfies (E). Then Ad sends G into the subgroup*

$$\text{Aut}_{\mathbf{O}_S\text{-mod.}}(\text{Lie}(G/S, \mathcal{M}))$$

of $\text{Aut}_{S\text{-gr.}}(\text{Lie}(G/S, \mathcal{M}))$, i.e. for every $g \in G(S')$, $\text{Ad}(g)$ respects the structure of $\mathbf{O}_{S'}$ -module of $\text{Lie}(G'/S', \mathcal{M})$ (where $G' = G \times_S S'$). In other words, Ad is a linear representation (cf. I, 3.2) of G in the $\mathbf{O}_S$ -module $\text{Lie}(G/S, \mathcal{M})$.

Remarks 4.1.1.2. a) In order that G/S satisfy (E), it is necessary and sufficient that, for every pair $(\mathcal{M}, \mathcal{N})$ of free $\mathbf{O}_S$ -modules of finite type, the diagram

$$\begin{array}{ccc} & \text{Lie}(G/S, \mathcal{M} \oplus \mathcal{N}) & \\ \swarrow & & \searrow \\ \text{Lie}(G/S, \mathcal{M}) & & \text{Lie}(G/S, \mathcal{N}) \\ \searrow & & \swarrow \\ & \text{Lie}(G/S, 0) = S & \end{array},$$

obtained by applying the functor $\text{Lie}(G/S, \)$ to the diagram (*) of 2.1, be cartesian.

b) Suppose that G/S satisfies (E). Then the derived morphism of the group law $\pi: G \times_S G \rightarrow G$ is none other than the addition law in $\text{Lie}(G/S, \mathcal{M})$. (N.B. π is not a group morphism, but $\pi(e, e) = e$, so the derived morphism $L(\pi)$ sends $T_{(G \times_S G)/S}^{(e, e)}(\mathcal{M}) = \text{Lie}(G/S, \mathcal{M}) \times_S \text{Lie}(G/S, \mathcal{M})$ into $\text{Lie}(G/S, \mathcal{M})$, cf. 3.7 and 3.8.) For every $n \in \mathbf{Z}$, one shows likewise that if

⁽⁵⁰⁾**N.D.E.** If f is a monomorphism, then so are $L(f)$ and $T(f)$, cf. 3.7.

⁽⁵¹⁾**N.D.E.** This remark has been added.

one denotes by $n_G: G \rightarrow G$ the morphism of S -functors defined by $g \mapsto g^n$, then the derived morphism $L(n_G)$ is multiplication by n on $\text{Lie}(G/S)$, cf. 3.9.4.

4.1.2.0. — ⁽⁵²⁾ Let us now consider the S -functor $\text{Hom}_{G/S}(G, T_{G/S}(\mathcal{M}))$; for every $S' \rightarrow S$, one has $T_{G/S}(\mathcal{M})_{S'} \simeq T_{G_{S'}/S'}(\mathcal{M})$ (cf. 3.4) and therefore:

$$\text{Hom}_{G/S}(G, T_{G/S}(\mathcal{M}))(S') \simeq \text{Hom}_{G_{S'}}(G_{S'}, T_{G_{S'}/S'}(\mathcal{M})) = \Gamma(T_{G_{S'}/S'}(\mathcal{M})/G_{S'}).$$

Let us first note that one has an isomorphism, functorial in S' ,

$$(*) \quad \text{Hom}_{S'}(G_{S'}, \text{Lie}(G_{S'}/S', \mathcal{M})) \xrightarrow{\sim} \Gamma(T_{G_{S'}/S'}(\mathcal{M})/G_{S'})$$

which associates to every $f: G_{S'} \rightarrow \text{Lie}(G_{S'}/S', \mathcal{M})$ the section $s_f: G_{S'} \rightarrow T_{G_{S'}/S'}(\mathcal{M})$ such that, for every $S'' \rightarrow S'$ and $g \in G(S'')$:

$$s_f(g) = i(f(g)) s(g).$$

Let h be an automorphism of the functor $G_{S'}$ over S' , not necessarily respecting the group structure. To every section τ of $T_{G_{S'}/S'}(\mathcal{M})$, one may associate $h(\tau)$ defined by transport of structure: it is for instance the only section of $T_{G_{S'}/S'}(\mathcal{M})$ making the diagram

$$\begin{array}{ccc} G_{S'} & \xrightarrow{\tau} & T_{G_{S'}/S'}(\mathcal{M}) \\ \downarrow h & & \downarrow T(h) \\ G_{S'} & \xrightarrow{h(\tau)} & T_{G_{S'}/S'}(\mathcal{M}) \end{array}$$

commutative.

In particular, let us take for h the right translation t_x by an element x of $G(S')$, i.e. $h(g) = t_x(g) = g \cdot x$, for every $g \in G(S'')$, $S'' \rightarrow S'$. Then one immediately has

$$t_x(s_f) = s_{t_x(f)}$$

where $t_x(f)$ denotes the morphism from $G_{S'}$ into $\text{Lie}(G_{S'}/S', \mathcal{M})$ defined by

$$t_x(f)(g) = f(g \cdot x^{-1})$$

for every $g \in G(S'')$, $S'' \rightarrow S'$.

It follows that if one lets G operate by right translations on

$$\text{Hom}_{G/S}(G, T_{G/S}(\mathcal{M})) \quad \text{and} \quad \text{Hom}_S(G, \text{Lie}(G/S, \mathcal{M}))$$

in the following way: for every $S' \rightarrow S$, $x \in G(S')$, $\sigma \in \Gamma(T_{G_{S'}/S'}(\mathcal{M})/G_{S'})$ and $f \in \text{Hom}_{S'}(G_{S'}, \text{Lie}(G_{S'}/S', \mathcal{M}))$,

$$(\sigma \cdot x)(g) = \sigma(g \cdot x^{-1}) \cdot s(x) \quad \text{and} \quad (f \cdot x)(g) = f(g \cdot x^{-1}),$$

for every $g \in G(S'')$, $S'' \rightarrow S'$, then the isomorphism $(*)$ above respects the operations of G .

In particular, via this isomorphism, the elements of $\text{Hom}_{G/S}(G, T_{G/S}(\mathcal{M}))^G(S')$ correspond to the *constant* morphisms from $G_{S'}$ into $\text{Lie}(G_{S'}/S', \mathcal{M})$ (i.e. factoring through the projection $G_{S'} \rightarrow S'$) or else to the elements of $\text{Lie}(G_{S'}/S', \mathcal{M})(S') = \text{Lie}(G/S, \mathcal{M})(S')$.

Terminology. — The elements of $\text{Hom}_{G/S}(G, T_{G/S}(\mathcal{M}))^G(S')$ will be called “sections of $T_{G_{S'}/S'}(\mathcal{M})$ invariant under right translation”.

One then obtains:

⁽⁵²⁾**N.D.E.** The original has been spelled out in what follows, in order to exhibit the action of G on the functors $\text{Hom}_{G/S}(G, T_{G/S}(\mathcal{M}))$ and $\text{Hom}_S(G, \text{Lie}(G/S, \mathcal{M}))$.

Proposition 4.1.2. ^{*} *The map $\text{Lie}(G/S, \mathcal{M})(S) \longrightarrow \Gamma(T_{G/S}(\mathcal{M})/G)$ which associates to $X \in \text{Lie}(G/S, \mathcal{M})(S)$ the section $x \mapsto X \cdot x$ is a bijection of $\text{Lie}(G/S, \mathcal{M})(S)$ onto the part of $\Gamma(T_{G/S}(\mathcal{M})/G)$ consisting of the sections invariant under right translation.*

Likewise, one lets G act on the right on $\text{End}_{I_S(\mathcal{M})/S}(G_{I_S(\mathcal{M})})$ as follows: for every $S' \longrightarrow S$, $x \in G(S')$ and $u \in \text{End}_{I_S(\mathcal{M})/S}(G_{I_S(\mathcal{M})})(S') = \text{End}_{I_{S'}(\mathcal{M})}(G_{I_{S'}(\mathcal{M})})$,

$$(u \cdot x)(g) = u(g \cdot x^{-1}) \cdot x,$$

for every $g \in G(S'')$, $S'' \longrightarrow I_{S'}(\mathcal{M})$. Then the morphism of 3.12.1

$$\text{Hom}_{G/S}(G, T_{G/S}(\mathcal{M})) \longrightarrow \text{End}_{I_S(\mathcal{M})/S}(G_{I_S(\mathcal{M})})$$

respects the operations of G and therefore induces, for every $S' \longrightarrow S$, a bijection between $\Gamma(T_{G_{S'}/S'}(\mathcal{M})/G_{S'})$ and the set of $I_{S'}(\mathcal{M})$ -endomorphisms u of $G_{I_{S'}(\mathcal{M})}$ which induce the identity on G and which “commute with the right translations”, i.e. which satisfy $u_{S''} \cdot x = u_{S''}$ for every $S'' \longrightarrow S'$ and $x \in G(S'')$. One therefore obtains:

Proposition 4.1.3. ^(*) *There exists a bijection, functorial in G , between the set*

$$\text{Lie}(G/S, \mathcal{M})(S)$$

and the set of $I_S(\mathcal{M})$ -endomorphisms of $G_{I_S(\mathcal{M})}$ inducing the identity on G and commuting with the right translations of G (in the sense indicated above).

Now taking 3.13 into account:

Theorem 4.1.4. ^(*) *Let G be an S -group functor; suppose that G/S satisfies (E). Then the group $\text{Lie}(G/S, \mathcal{M})(S)$ is identified, functorially in G , with the group of $I_S(\mathcal{M})$ -automorphisms of $G_{I_S(\mathcal{M})}$ inducing the identity on G and commuting with the right translations of G (in the sense indicated above).*

One thus recovers (in the case $\mathcal{M} = \mathbf{O}_S$) one of the classical definitions of the Lie algebra of a group.

4.2.0. — ⁽⁵⁴⁾ Before going further, let us establish new corollaries to 3.11. Let X, Y be over Z , and Z over S , as in section 1. As one saw in 3.11, the isomorphisms 1.1 (4):

$$(1) \quad \begin{array}{ccc} \text{Hom}_S(I_S(\mathcal{M}), \text{Hom}_{Z/S}(X, Y)) & \xrightarrow{\sim} & \text{Hom}_{Z/S}(X, \text{Hom}_Z(I_Z(\mathcal{M}), Y)) \\ & \searrow \simeq \quad \swarrow \simeq & \\ & \text{Hom}_{Z/S}(X \times_S I_S(\mathcal{M}), Y) & \end{array}$$

induce the isomorphism θ below

$$(2) \quad \begin{array}{ccc} T_{\text{Hom}_{Z/S}(X, Y)}(\mathcal{M}) & \xrightarrow[\sim]{\theta} & \text{Hom}_{Z/S}(X, T_{Y/Z}(\mathcal{M})) \\ & \searrow \simeq \quad \swarrow \simeq & \\ & \text{Hom}_{Z/S}(X \times_S I_S(\mathcal{M}), Y). & \end{array}$$

By 1.3, if Y is a Z -group, the same is true of $\text{Hom}_Z(V, Y)$ for every $V \longrightarrow Z$ (in particular for $V = I_Z(\mathcal{M})$); explicitly, if $Z'' \longrightarrow Z' \longrightarrow Z$ and $\varphi, \psi \in \text{Hom}_Z(V_{Z'}, Y)$, then $\varphi \cdot \psi$ is defined by $(\varphi \cdot \psi)(v) = \varphi(v)\psi(v)$, for every $v \in V_{Z'}(Z'')$.

Suppose now that X and Y are Z -groups, and let us set the following definition.

^{*}The statements 4.1.2, 4.1.3 and 4.1.4 are obtained more simply by observing that the automorphisms of G invariant under right translations are the left translations. ⁽⁵³⁾

⁽⁵³⁾**N.D.E.** See for instance the proof of propositions 4.6 and 2.5 of [DG70], § II.4.

⁽⁵⁴⁾**N.D.E.** One has added paragraph 4.2.0, whose results are used implicitly in 4.2 and 4.7 of the original.

Definition 4.2.0.1. Let $\text{Hom}_{(Z/S)\text{-gr.}}(X, Y)$ be the subfunctor of $\text{Hom}_{Z/S}(X, Y)$ defined by: for every $S' \rightarrow S$,

$$(3) \quad \text{Hom}_{(Z/S)\text{-gr.}}(X, Y)(S') = \text{Hom}_{Z_{S'}\text{-gr.}}(X_{S'}, Y_{S'}).$$

This definition applies equally when one replaces Y by the Z -group $T_{Y/Z}(\mathcal{M})$.

One then sees easily that $T_{\text{Hom}_{(Z/S)\text{-gr.}}(X, Y)}(\mathcal{M})(S')$ corresponds, in the preceding isomorphisms (2), to the $Z_{S'}$ -morphisms

$$\varphi: X_{S'} \times_{S'} I_{S'}(\mathcal{M}) \rightarrow Y_{S'}$$

which are “multiplicative in X ”, i.e. which satisfy $\varphi(x_1 x_2, m) = \varphi(x_1, m) \varphi(x_2, m)$, and that these correspond to the morphisms of $Z_{S'}$ -groups $X_{S'} \rightarrow T_{Y/Z}(\mathcal{M})_{S'}$. One has thus obtained:

Proposition 4.2.0.2. *Let X, Y be Z -groups, with Z over S . One has isomorphisms of S -functors, functorial in $\mathcal{M}$:*

$$T_{\text{Hom}_{(Z/S)\text{-gr.}}(X, Y)}(\mathcal{M}) \xrightarrow{\sim} \text{Hom}_{(Z/S)\text{-gr.}}(X, T_{Y/Z}(\mathcal{M})).$$

In particular, for $Z = S$, one obtains the following corollary. Before stating it, let us remark that if Y is a commutative S -group, the same is true of $T_{Y/S}(\mathcal{M})$, then of $H = \text{Hom}_{S\text{-gr.}}(X, Y)$ and $\text{Hom}_{S\text{-gr.}}(X, T_{Y/S}(\mathcal{M}))$, and finally of $T_{H/S}(\mathcal{M})$.

Corollary 4.2.0.3. *Let X, Y be S -groups. One has isomorphisms of S -functors, functorial in $\mathcal{M}$:*

$$T_{\text{Hom}_{S\text{-gr.}}(X, Y)/S}(\mathcal{M}) \xrightarrow{\sim} \text{Hom}_{S\text{-gr.}}(X, T_{Y/S}(\mathcal{M})).$$

If Y is commutative, these are moreover isomorphisms of abelian S -groups.

Definition 4.2.0.4. ⁽⁵⁵⁾ If Y is an $\mathbf{O}_S$ -module, the functor $T_{Y/S}(\mathcal{M})$ (resp. $\text{Lie}(Y/S, \mathcal{M})$) is equipped with an $\mathbf{O}_S$ -module structure deduced from that of Y . Equipped with this structure, one will denote it by $T'_{Y/S}(\mathcal{M})$ (resp. $\text{Lie}'(Y/S, \mathcal{M})$).

Consequently, if X, Y are $\mathbf{O}_S$ -modules, then $T'_{Y/S}(\mathcal{M}) = \text{Hom}_S(I_S(\mathcal{M}), Y)$ and $H = \text{Hom}_{\mathbf{O}_{S\text{-mod.}}}(X, Y)$, and then $\text{Hom}_{\mathbf{O}_{S\text{-mod.}}}(X, T'_{Y/S}(\mathcal{M}))$ and $T'_{H/S}(\mathcal{M})$, are equipped with an $\mathbf{O}_S$ -module structure, and one obtains the following:

Corollary 4.2.0.5. *If X, Y are $\mathbf{O}_S$ -modules, one has isomorphisms of $\mathbf{O}_S$ -modules, functorial in $\mathcal{M}$:*

$$T'_{\text{Hom}_{\mathbf{O}_{S\text{-mod.}}}(X, Y)/S}(\mathcal{M}) \xrightarrow{\sim} \text{Hom}_{\mathbf{O}_{S\text{-mod.}}}(X, T'_{Y/S}(\mathcal{M})).$$

Definition 4.2.A. ⁽⁵⁶⁾ Let X, L be S -groups, with X operating on L by group automorphisms (cf. I 2.3.5). One defines the subfunctor $Z_S^1(X, L)$ of $\text{Hom}_S(X, L)$ as follows: for every $S' \rightarrow S$,

$$Z_S^1(X, L)(S') = \left\{ \varphi \in \text{Hom}_{S'}(X_{S'}, L_{S'}) \left| \begin{array}{l} \varphi(x_1 x_2) = \varphi(x_1) (x_1 \cdot \varphi(x_2)), \\ \text{for every } x_1, x_2 \in X(S''), S'' \rightarrow S' \end{array} \right. \right\}.$$

One calls it the “functor of crossed homomorphisms from X to L ”.

Remark 4.2.B. ⁽⁵⁶⁾ a) If L' is a second S -group on which X operates by group automorphisms, one has

$$Z_S^1(X, L \times_S L') \simeq Z_S^1(X, L) \times_S Z_S^1(X, L').$$

b) If L is a $G\text{-}\mathbf{O}_S$ -module, $Z_S^1(X, L)$ coincides with the kernel of the differential

$$\partial: \text{Hom}_S(X, L) \rightarrow \text{Hom}_S(X^2, L)$$

⁽⁵⁵⁾**N.D.E.** One has inserted here this definition, which in the original appeared in 4.3.

⁽⁵⁶⁾**N.D.E.** One has expanded the original in what follows; in particular, one has added definition 4.2.A and remark 4.2.B.

defined in I 5.1; in particular, $Z_S^1(X, L)$ is in this case an $\mathbf{O}_S$ -module.

4.2. — Let $u: X \rightarrow Y$ be a morphism of S -groups; then $L_{\text{Hom}_{S\text{-gr.}}(X, Y)/S}^u(\mathcal{M})$ is described as follows. First, one saw in 3.11.3 that one has an isomorphism of S -functors, functorial in $\mathcal{M}$:

$$(\dagger) \quad L_{\text{Hom}_S(X, Y)/S}^u(\mathcal{M}) \xrightarrow{\sim} \text{Hom}_{Y/S}(X, T_{Y/S}(\mathcal{M})).$$

On the other hand, as Y is an S -group; one then has

$$T_{Y/S}(\mathcal{M}) = \text{Lie}(Y/S, \mathcal{M}) \cdot Y = \text{Lie}(Y/S, \mathcal{M})_Y;$$

there results an isomorphism of S -functors, functorial in $\mathcal{M}$:

$$L_{\text{Hom}_S(X, Y)/S}^u(\mathcal{M}) \xrightarrow{\sim} \text{Hom}_S(X, \text{Lie}(Y/S, \mathcal{M})).$$

For every $S' \rightarrow S$, let us denote by $u': X' \rightarrow Y'$ the morphism deduced from u by base change. Let us consider the S -functor defined as follows: ⁽⁵⁷⁾ for every $S' \rightarrow S$,

$$\begin{aligned} \text{Hom}_{(Y/S)\text{-gr.}}(X, \text{Lie}(Y/S, \mathcal{M}) \cdot Y)(S') &= \text{Hom}_{Y'\text{-gr.}}(X', (\text{Lie}(Y/S, \mathcal{M}) \cdot Y)_{S'}) \\ &= \text{Hom}_{Y'\text{-gr.}}(X', \text{Lie}(Y'/S', \mathcal{M}) \cdot Y'). \end{aligned}$$

Then one sees easily that the isomorphism $(\dagger)$ induces an isomorphism

$$(\dagger') \quad L_{\text{Hom}_{S\text{-gr.}}(X, Y)/S}^u(\mathcal{M}) \xrightarrow{\sim} \text{Hom}_{(Y/S)\text{-gr.}}(X, \text{Lie}(Y/S, \mathcal{M}) \cdot Y).$$

On the other hand, the morphism

$$X \xrightarrow{u} Y \xrightarrow{\text{Ad}} \text{Aut}_{S\text{-gr.}}(\text{Lie}(Y/S, \mathcal{M}))$$

defines an operation of X on $L = \text{Lie}(Y/S, \mathcal{M})$ by group automorphisms.

If $\Phi \in \text{Hom}_{Y/S}(X, \text{Lie}(Y/S, \mathcal{M}) \cdot Y)$ then, for every $S'' \rightarrow S' \rightarrow S$ and $x \in X(S'')$, one can write in a unique way

$$\Phi(S')(x) = \varphi(S')(x) \cdot u'(x), \quad \text{where} \quad \varphi(S')(x) \in \text{Lie}(Y'/S', \mathcal{M})(S'');$$

this determines an element φ of $\text{Hom}_S(X, \text{Lie}(Y/S, \mathcal{M}))$. Then $\Phi(S')$ is a morphism of groups if, and only if, for every $x_1, x_2 \in X(S'')$ one has:

$$\begin{aligned} \varphi(S')(x_1 x_2) &= \varphi(S')(x_1) (u(x_1) \varphi(S')(x_2) u(x_1)^{-1}) \\ &= \varphi(S')(x_1) (x_1 \cdot \varphi(S')(x_2)), \end{aligned}$$

i.e. if and only if $\varphi \in Z_S^1(X, \text{Lie}(Y/S, \mathcal{M}))$. One has thus obtained the following:

Proposition 4.2. *Let $u: X \rightarrow Y$ be a morphism of S -groups. One has an isomorphism of S -functors, functorial in $\mathcal{M}$:*

$$L_{\text{Hom}_{S\text{-gr.}}(X, Y)/S}^u(\mathcal{M}) \xrightarrow{\sim} Z_S^1(X, \text{Lie}(Y/S, \mathcal{M})).$$

⁽⁵⁸⁾ Suppose moreover that Y/S satisfies (E). Then it follows from 4.2.0.3, exactly as in the proof of 3.11.1, that $\text{Hom}_{S\text{-gr.}}(X, Y)/S$ satisfies (E). Thus one has (cf. 3.5.1):

$$L_{\text{Hom}_{S\text{-gr.}}(X, Y)/S}^u(\mathcal{M} \oplus \mathcal{N}) \simeq L_{\text{Hom}_{S\text{-gr.}}(X, Y)/S}^u(\mathcal{M}) \times_S L_{\text{Hom}_{S\text{-gr.}}(X, Y)/S}^u(\mathcal{N}).$$

(This also follows from 4.2 and 4.2.B a.) Consequently, $L_{\text{Hom}_{S\text{-gr.}}(X, Y)/S}^u(\mathcal{M})$ is equipped, as is $Z_S^1(X, \text{Lie}(Y/S, \mathcal{M}))$ (cf. 4.2.B b)), with an $\mathbf{O}_S$ -module structure, deduced from the functoriality in $\mathcal{M}$. One deduces from this that the isomorphism of 4.2 is, in this case, an isomorphism of $\mathbf{O}_S$ -modules:

Proposition 4.2. bis. ⁽⁵⁸⁾ *Let $u: X \rightarrow Y$ be a morphism of S -groups; suppose that Y/S satisfies (E). One has an isomorphism of $\mathbf{O}_S$ -modules, functorial in $\mathcal{M}$:*

$$L_{\text{Hom}_{S\text{-gr.}}(X, Y)/S}^u(\mathcal{M}) \xrightarrow{\sim} Z_S^1(X, \text{Lie}(Y/S, \mathcal{M})).$$

⁽⁵⁷⁾ **N.D.E.** This is definition 4.2.0.1 applied to $Z = Y$ and to the Y -groups $u: X \rightarrow Y$ and $T_{Y/S}(\mathcal{M})$.

⁽⁵⁸⁾ **N.D.E.** One has added the sentences which follow and proposition 4.2 bis, implicit in the original.

Moreover, when Y/S satisfies (E), one deduces from 3.13.1, exactly as in the proof of 3.13.2, that for every $u \in \text{Isom}_{S\text{-gr.}}(X, Y)$ one has an isomorphism functorial in $\mathcal{M}$:

$$(*) \quad L_{\text{Isom}_{S\text{-gr.}}(X, Y)/S}^u(\mathcal{M}) \xrightarrow{\sim} L_{\text{Hom}_{S\text{-gr.}}(X, Y)/S}^u(\mathcal{M}).$$

One deduces from this the following two corollaries.

Corollary 4.2.1. *Let $u: X \rightarrow Y$ be a morphism of S -groups; if Y/S satisfies (E), one has an isomorphism of $\mathbf{O}_S$ -modules, functorial in $\mathcal{M}$:*

$$L_{\text{Isom}_{S\text{-gr.}}(X, Y)/S}^u(\mathcal{M}) \xrightarrow{\sim} Z_S^1(X, \text{Lie}(Y/S, \mathcal{M})).$$

Corollary 4.2.2. *Let X be an S -group; if X/S satisfies (E), one has an isomorphism of $\mathbf{O}_S$ -modules, functorial in $\mathcal{M}$:*

$$\text{Lie}(\text{Aut}_{S\text{-gr.}}(X)/S, \mathcal{M}) \xrightarrow{\sim} Z_S^1(X, \text{Lie}(X/S, \mathcal{M})).$$

⁽⁵⁹⁾ Furthermore, if Y is commutative, the adjoint action of Y on $L = \text{Lie}(X/S, \mathcal{M})$ is trivial, whence $Z_S^1(X, L) = \text{Hom}_{S\text{-gr.}}(X, L)$. Thus:

Corollary 4.2.3. *Let Y be a commutative S -group; one has an isomorphism of S -functors, functorial in $\mathcal{M}$:*

$$L_{\text{Hom}_{S\text{-gr.}}(X, Y)/S}^u(\mathcal{M}) \xrightarrow{\sim} \text{Hom}_{S\text{-gr.}}(X, \text{Lie}(Y/S, \mathcal{M})).$$

4.3. — Let us now consider the case in which X and Y are $\mathbf{O}_S$ -modules. Let us recall (cf. 4.2.0.4) that one denotes by $T'_{Y/S}(\mathcal{M})$ (resp. $\text{Lie}'(Y/S, \mathcal{M})$) the functor $T_{Y/S}(\mathcal{M})$ (resp. $\text{Lie}(Y/S, \mathcal{M})$) equipped with the $\mathbf{O}_S$ -module structure deduced from that of Y .

When Y/S satisfies (E), one will always denote by $\text{Lie}(Y/S, \mathcal{M})$ the functor $\text{Lie}(Y/S, \mathcal{M})$ equipped with the $\mathbf{O}_S$ -module structure defined for every functor satisfying (E). In this case, we know (cf. 3.9) that the abelian group structures of $\text{Lie}(Y/S, \mathcal{M})$ and $\text{Lie}'(Y/S, \mathcal{M})$ coincide, but the same is not a priori true of the module structures (see a counter-example in paragraph 6.3). For every $S' \rightarrow S$ and $a \in \mathbf{O}(S')$, one will denote by $a \cdot' m$ (resp. $a \cdot m$) the action of a on $m \in \text{Lie}'(Y/S, \mathcal{M})(S')$ (resp. on $m \in \text{Lie}(Y/S, \mathcal{M})(S')$), and likewise for the action of a on $T'_{Y/S}(\mathcal{M})$ and $T_{Y/S}(\mathcal{M})$.

One has $T'_{Y/S}(\mathcal{M}) \simeq \text{Lie}'(Y/S, \mathcal{M}) \oplus Y$ as $\mathbf{O}_S$ -modules; consequently, one obtains, exactly as for propositions 4.2 and 4.2 bis, the following:

Proposition 4.3. *Let $u: X \rightarrow Y$ be a morphism of $\mathbf{O}_S$ -modules. One has an isomorphism of S -functors, functorial in $\mathcal{M}$:*

$$(*) \quad L_{\text{Hom}_{\mathbf{O}_S\text{-mod.}}(X, Y)/S}^u(\mathcal{M}) \xrightarrow{\sim} \text{Hom}_{\mathbf{O}_S\text{-mod.}}(X, \text{Lie}'(Y/S, \mathcal{M})).$$

⁽⁶⁰⁾ If Y/S satisfies (E), then $\text{Hom}_{\mathbf{O}_S\text{-mod.}}(X, Y)/S$ satisfies (E) and $(*)$ is an isomorphism of $\mathbf{O}_S$ -modules when one equips the two members with the $\mathbf{O}_S$ -module structure deduced from the functoriality in $\mathcal{M}$. ⁽⁶¹⁾

Remark 4.3. bis. ⁽⁶²⁾ Let $u: X \rightarrow Y$ be a morphism of $\mathbf{O}_S$ -modules; let us denote by τ_u the map which to every morphism of $\mathbf{O}_S$ -modules $\varphi: X \rightarrow \text{Lie}'(Y/S, \mathcal{M})$ associates the morphism

$$u \oplus \varphi: X \rightarrow T'_{Y/S}(\mathcal{M}) = Y \oplus \text{Lie}'(Y/S, \mathcal{M}).$$

⁽⁵⁹⁾**N.D.E.** One has added the sentence which follows.

⁽⁶⁰⁾**N.D.E.** One has added the sentence which follows.

⁽⁶¹⁾**N.D.E.** Note that on the right-hand term, it is the structure defined by $(a\varphi)(x) = a \cdot \varphi(x)$, where $a \cdot \varphi(x)$ denotes the action of a on $\varphi(x) \in \text{Lie}(Y/S, \mathcal{M})$. This differs from the action $(a \cdot' \varphi)(x) = a \cdot' \varphi(x) = \varphi(ax)$, but coincides with the latter if $\text{Lie}(Y/S, \mathcal{M}) = \text{Lie}'(Y/S, \mathcal{M})$.

⁽⁶²⁾**N.D.E.** One has added this remark, which will be useful in 4.5.1.

Then the isomorphism of 4.3 fits into the following commutative diagram, functorial in $\mathcal{M}$:

$$\begin{array}{ccc} L_{\text{Hom}_{\mathbf{O}_S\text{-mod.}}(X,Y)/S}^u(\mathcal{M}) & \xrightarrow{\sim} & \text{Hom}_{\mathbf{O}_S\text{-mod.}}(X, \text{Lie}'(Y/S, \mathcal{M})) \\ \downarrow & & \downarrow \tau_u \\ T_{\text{Hom}_{\mathbf{O}_S\text{-mod.}}(X,Y)/S}(\mathcal{M}) & \xrightarrow{\sim} & \text{Hom}_{\mathbf{O}_S\text{-mod.}}(X, T'_{Y/S}(\mathcal{M})). \end{array}$$

Moreover, when Y/S satisfies (E), one deduces from 3.13.1, exactly as in the proof of 3.13.2, that for every $u \in \text{Isom}_{\mathbf{O}_S\text{-mod.}}(X, Y)$, one has

$$(*) \quad L_{\text{Isom}_{\mathbf{O}_S\text{-mod.}}(X,Y)/S}^u(\mathcal{M}) = L_{\text{Hom}_{\mathbf{O}_S\text{-mod.}}(X,Y)/S}^u(\mathcal{M}).$$

Corollary 4.3.1. *Let X be an $\mathbf{O}_S$ -module satisfying (E) with respect to S . One has a functorial isomorphism in $\mathcal{M}$:*

$$\text{Lie}(\text{Aut}_{\mathbf{O}_S\text{-mod.}}(X)/S, \mathcal{M}) \xrightarrow{\sim} \text{Hom}_{\mathbf{O}_S\text{-mod.}}(X, \text{Lie}'(X/S, \mathcal{M}))$$

which respects the $\mathbf{O}_S$ -module structures deduced from the functoriality in $\mathcal{M}$. ⁽⁶³⁾ In particular, $\text{Aut}_{\mathbf{O}_S\text{-mod.}}(X)/S$ satisfies (E).

PROOF. The first assertion follows from $(*)$ and 4.3; let us prove the second. Since X/S satisfies (E), one has isomorphisms of $\mathbf{O}_S$ -modules $\text{Lie}'(X/S, \mathcal{M} \oplus \mathcal{N}) \simeq \text{Lie}'(X/S, \mathcal{M}) \times_S \text{Lie}'(X/S, \mathcal{N})$, and hence:

$$\begin{aligned} \text{Lie}(\text{Aut}_{\mathbf{O}_S\text{-mod.}}(X)/S, \mathcal{M} \oplus \mathcal{N}) &\simeq \text{Lie}(\text{Aut}_{\mathbf{O}_S\text{-mod.}}(X)/S, \mathcal{M}) \\ &\quad \times_S \text{Lie}(\text{Aut}_{\mathbf{O}_S\text{-mod.}}(X)/S, \mathcal{N}). \end{aligned}$$

In view of 4.1.1.2 a), this proves that $\text{Aut}_{\mathbf{O}_S\text{-mod.}}(X)/S$ satisfies (E). $\square$

4.3.2. — Before continuing in this direction, let us examine more closely the relations between Y , $\text{Lie}(Y/S)$ and $\text{Lie}'(Y/S)$. Let us first remark that

$$(1) \quad \text{Lie}(\mathbf{O}_S/S, \mathcal{M}) = \text{Lie}'(\mathbf{O}_S/S, \mathcal{M}) = W(\mathcal{M})$$

(where $W(\mathcal{M})$ is defined in I 4.6) and that one therefore has a canonical isomorphism

$$(2) \quad d: \mathbf{O}_S \xrightarrow{\sim} \text{Lie}(\mathbf{O}_S/S).$$

Now let F be an $\mathbf{O}_S$ -module. For every $S_2 \rightarrow S_1 \rightarrow S$ ⁽⁶⁴⁾, one has a dihomomorphism

$$(3) \quad \begin{cases} F(S_1) \rightarrow F(S_2) \\ \mathbf{O}(S_1) \rightarrow \mathbf{O}(S_2), \end{cases}$$

whence a morphism of $\mathbf{O}(S_2)$ -modules

$$F(S_1) \otimes_{\mathbf{O}(S_1)} \mathbf{O}(S_2) \rightarrow F(S_2).$$

In particular, setting $S_1 = S'$ and $S_2 = I_{S'}(\mathcal{M})$, one deduces from this morphisms of $\mathbf{O}(S')$ -modules, functorial in $\mathcal{M}$,

$$F(S') \otimes_{\mathbf{O}(S')} T_{\mathbf{O}_S/S}(\mathcal{M})(S') \rightarrow T'_{F/S}(\mathcal{M})(S');$$

letting S' vary, one obtains morphisms of $\mathbf{O}_S$ -modules, functorial in $\mathcal{M}$,

$$(4) \quad F \otimes_{\mathbf{O}_S} T_{\mathbf{O}_S/S}(\mathcal{M}) \rightarrow T'_{F/S}(\mathcal{M}).$$

These morphisms are functorial in $\mathcal{M}$, hence compatible with the projections of the tangent bundles onto their bases; they therefore define morphisms of $\mathbf{O}_S$ -modules

$$(5) \quad F \otimes_{\mathbf{O}_S} \text{Lie}(\mathbf{O}_S/S, \mathcal{M}) \rightarrow \text{Lie}'(F/S, \mathcal{M})$$

⁽⁶³⁾**N.D.E.** Cf. N.D.E. (61). On the other hand, one has added the sentence which follows, as well as its proof.

⁽⁶⁴⁾**N.D.E.** One has changed $S' \rightarrow S$ into $S_2 \rightarrow S_1 \rightarrow S$, for one must let S_2 and S_1 vary (cf. below). On the other hand, one has expanded the original in what follows.

such that one has the commutative diagram:

$$\begin{array}{ccccccc}
 0 & \longrightarrow & F \otimes_{\mathbf{O}_S} \mathrm{Lie}(\mathbf{O}_S/S, \mathcal{M}) & \longrightarrow & F \otimes_{\mathbf{O}_S} T_{\mathbf{O}_S/S}(\mathcal{M}) & \longrightarrow & F \longrightarrow 0 \\
 & & \downarrow & & \downarrow & & \\
 0 & \longrightarrow & \mathrm{Lie}'(F/S, \mathcal{M}) & \longrightarrow & T'_{F/S}(\mathcal{M}) & \longrightarrow & F \longrightarrow 0.
 \end{array}$$

One may regard the morphisms (5) as morphisms of abelian S -groups

$$(6) \quad F \otimes_{\mathbf{O}_S} \mathrm{Lie}(\mathbf{O}_S/S, \mathcal{M}) \longrightarrow \mathrm{Lie}(F/S, \mathcal{M});$$

by tensoring F with the isomorphism $d: \mathbf{O}_S \xrightarrow{\sim} \mathrm{Lie}(\mathbf{O}_S/S)$, one deduces from this (for $\mathcal{M} = \mathbf{O}_S$) a morphism of abelian S -groups

$$(7) \quad F \xrightarrow{\sim} F \otimes_{\mathbf{O}_S} \mathrm{Lie}(\mathbf{O}_S/S) \longrightarrow \mathrm{Lie}(F/S)$$

also denoted $d: F \longrightarrow \mathrm{Lie}(F/S)$.

Remark 4.3.3. ⁽⁶⁵⁾ When F/S satisfies (E), the morphisms (6) and (7) are not necessarily morphisms of $\mathbf{O}_S$ -modules, when one equips the two members with the module structures deduced from that of $\mathcal{M}$ by means of condition (E).

For example, let k be a field of characteristic $p > 0$, $S = \mathrm{Spec}(k)$, and F the $\mathbf{O}_S$ -module which to every S -scheme T associates $F(T) = \Gamma(T, \mathbf{O}_T)$ equipped with the $\mathbf{O}(T)$ -module structure obtained by letting the scalars operate via the p -th power, i.e. $r \cdot f = r^p f$, for $r \in \mathbf{O}(T)$, $f \in F(T)$. As an S -group functor, F is isomorphic to $\mathbf{G}_{a,S}$. Thus F satisfies (E) and $\mathrm{Lie}(F/S)$ is identified with $\mathrm{Lie}(\mathbf{G}_{a,S}/S) \simeq \mathbf{O}_S$. Then the canonical morphism $d: F \longrightarrow \mathrm{Lie}(F/S)$ is, for every $T \longrightarrow S$, the identity map $F(T) \longrightarrow \mathbf{O}(T)$: it does respect the abelian group structure but not the $\mathbf{O}_S$ -module structure.

Remark 4.3.4. ⁽⁶⁶⁾ One can make the morphisms (4) and (5) explicit as follows. The morphism $\Theta: F \otimes_{\mathbf{O}_S} T_{\mathbf{O}_S/S}(\mathcal{M}) \longrightarrow T'_{F/S}(\mathcal{M}) = \mathrm{Hom}_S(I_S(\mathcal{M}), F)$ is defined by: for every $S' \longrightarrow S$, $\alpha \in \mathbf{O}(I_{S'}(\mathcal{M}))$, and $f: S' \longrightarrow F$,

$$\Theta(f \otimes \alpha) = \alpha \cdot (\tau_0 \circ f) = \alpha \cdot (f \circ \rho),$$

where τ_0 is the zero section $F \longrightarrow T'_{F/S}(\mathcal{M})$ and ρ the structural morphism $I_{S'}(\mathcal{M}) \longrightarrow S'$. Then Θ induces a morphism $\theta: F \otimes_{\mathbf{O}_S} \mathrm{Lie}(\mathbf{O}_S/S, \mathcal{M}) \longrightarrow \mathrm{Lie}'(F/S, \mathcal{M})$; this results from the “functoriality in $\mathcal{M}$ ” already mentioned after (4), and can be seen explicitly as follows. On the one hand, one has

$$\mathrm{Lie}'(F/S, \mathcal{M})(S') = \{\varphi \in \mathrm{Hom}_S(I_{S'}(\mathcal{M}), F) \mid \varphi \circ \varepsilon_{\mathcal{M}} = e\},$$

where e denotes the unit section $S' \longrightarrow F$. On the other hand, $\mathrm{Lie}(\mathbf{O}_S/S, \mathcal{M})(S') = \Gamma(S', \mathcal{M})$ is the kernel of the augmentation $\eta: \mathbf{O}(I_{S'}(\mathcal{M})) \longrightarrow \mathbf{O}(S')$, and it is therefore a matter of seeing that if $f \in F(S')$ and $\alpha \in \Gamma(S', \mathcal{M})$, then $\alpha \cdot (f \circ \rho) \circ \varepsilon_{\mathcal{M}} = e$.

Let us consider the dihomomorphism (3), in the case where $S_2 \longrightarrow S_1$ is the S -morphism $\varepsilon_{\mathcal{M}}: S' \longrightarrow I_{S'}(\mathcal{M})$; one then has a commutative diagram

$$\begin{array}{ccc}
 F(I_{S'}(\mathcal{M})) & \xrightarrow{F(\varepsilon_{\mathcal{M}})} & F(S') \\
 \alpha \downarrow & & \downarrow \eta(\alpha) \\
 F(I_{S'}(\mathcal{M})) & \xrightarrow{F(\varepsilon_{\mathcal{M}})} & F(S').
 \end{array}$$

⁽⁶⁵⁾**N.D.E.** One found in the original the assertion that when F/S satisfies (E), the morphisms (6) (and hence (7)) are morphisms of $\mathbf{O}_S$ -modules, an assertion contradicted by a counter-example given in 6.3. One has deleted this assertion, inserted here the aforementioned example, and modified definition 4.4 accordingly. This is of no consequence for what follows.

⁽⁶⁶⁾**N.D.E.** One has added this remark, which will be useful in 4.6.2.

For every $\varphi: I_{S'}(\mathcal{M}) \rightarrow F$, one therefore has $(\alpha \cdot \varphi) \circ \varepsilon_{\mathcal{M}} = \eta(\alpha) \cdot (\varphi \circ \varepsilon_{\mathcal{M}})$, whence $(\alpha \cdot \varphi) \circ \varepsilon_{\mathcal{M}} = e$ if $\eta(\alpha) = 0$.

In particular, taking $\mathcal{M} = t\mathbf{O}_S$, one has $\text{Lie}(\mathbf{O}_S/S) = t\mathbf{O}_S$ and the morphism $F \rightarrow \text{Lie}'(F/S)$ is given by $f \mapsto t \cdot (f \circ \rho)$.

Remark 4.3.5. ⁽⁶⁷⁾ Let F be an $\mathbf{O}_S$ -module, set $E = \text{End}_{\mathbf{O}_S\text{-mod.}}(F)$ and denote by d_F and d_E the morphisms of $\mathbf{O}_S$ -modules given by 4.3.2 (5):

$$\begin{aligned} d_F: F \otimes_{\mathbf{O}_S} \text{Lie}(\mathbf{O}_S/S, \mathcal{M}) &\rightarrow \text{Lie}'(F/S, \mathcal{M}), \\ d_E: E \otimes_{\mathbf{O}_S} \text{Lie}(\mathbf{O}_S/S, \mathcal{M}) &\rightarrow \text{Lie}'(E/S, \mathcal{M}). \end{aligned}$$

One deduces from 4.3.4 the following commutative diagram of morphisms of $\mathbf{O}_S$ -modules:

$$\begin{array}{ccc} E \otimes_{\mathbf{O}_S} \text{Lie}(\mathbf{O}_S/S, \mathcal{M}) & \xrightarrow{d_E} & \text{Lie}'(E/S, \mathcal{M}) \\ & & \uparrow \simeq (*) \\ \text{Hom}_{\mathbf{O}_S}(F, F \otimes_{\mathbf{O}_S} \text{Lie}(\mathbf{O}_S/S, \mathcal{M})) & \xrightarrow{d_F} & \text{Hom}_{\mathbf{O}_S}(F, \text{Lie}'(F/S, \mathcal{M})) \end{array}$$

where the right-hand vertical arrow is the isomorphism $(*)$ of 4.3. Thus (*loc. cit.*), if F/S satisfies (E), then E/S satisfies (E) and $(*)$ is also an isomorphism of $\mathbf{O}_S$ -modules when one equips the right-hand terms with the $\mathbf{O}_S$ -module structure deduced from (E).

Remark 4.4.0. ⁽⁶⁸⁾ In 4.3.2, the morphisms (4) are isomorphisms if and only if the morphisms (5) are. Moreover, if these conditions are satisfied, then F/S satisfies (E). Indeed, it suffices to check that $\text{Lie}'(F/S, \mathcal{M} \oplus \mathcal{N}) \simeq \text{Lie}'(F/S, \mathcal{M}) \times_S \text{Lie}'(F/S, \mathcal{N})$. Now one has the commutative diagram below, in which by hypothesis the horizontal arrows are isomorphisms:

$$\begin{array}{ccc} F \otimes_{\mathbf{O}_S} \text{Lie}(\mathbf{O}_S/S, \mathcal{M} \oplus \mathcal{N}) & \xrightarrow{\sim} & \text{Lie}'(F/S, \mathcal{M} \oplus \mathcal{N}) \\ \downarrow \wr & & \downarrow \\ F \otimes_{\mathbf{O}_S} (\text{Lie}(\mathbf{O}_S/S, \mathcal{M}) \times_S \text{Lie}(\mathbf{O}_S/S, \mathcal{N})) & \xrightarrow{\sim} & \text{Lie}'(F/S, \mathcal{M}) \times_S \text{Lie}'(F/S, \mathcal{N}); \end{array}$$

the second vertical arrow is therefore also an isomorphism, i.e. F/S satisfies (E).

Definition 4.4. One says that F is a *good* $\mathbf{O}_S$ -module if the morphisms

$$F \otimes_{\mathbf{O}_S} T_{\mathbf{O}_S/S}(\mathcal{M}) \rightarrow T_{F/S}(\mathcal{M})$$

or, equivalently,

$$F \otimes_{\mathbf{O}_S} \text{Lie}(\mathbf{O}_S/S, \mathcal{M}) \rightarrow \text{Lie}(F/S, \mathcal{M}),$$

are isomorphisms of abelian S -groups (so that F/S satisfies (E)) and if moreover they respect the $\mathbf{O}_S$ -module structures deduced from condition (E).

Corollary 4.4.1. ⁽⁶⁹⁾ Let F be an $\mathbf{O}_S$ -module. Consider the following conditions:

- (i) F is a good $\mathbf{O}_S$ -module.
- (ii) F/S satisfies (E) and $d: F \rightarrow \text{Lie}(F/S)$ is an isomorphism of $\mathbf{O}_S$ -modules.
- (iii) $\text{Lie}(F/S, \mathcal{M}) = \text{Lie}'(F/S, \mathcal{M})$.

Then one has (i) $\Leftrightarrow$ (ii) $\Rightarrow$ (iii).

PROOF. (i) $\Rightarrow$ (ii) follows from the definition. To prove (ii) $\Rightarrow$ (i), one must show that the morphisms of abelian S -groups (functorial in $\mathcal{M}$)

$$F \otimes_{\mathbf{O}_S} \text{Lie}(\mathbf{O}_S/S, \mathcal{M}) \xrightarrow{\sim} \text{Lie}(F/S, \mathcal{M})$$

⁽⁶⁷⁾**N.D.E.** One has added this remark, which will be useful in 4.5 and 4.7.

⁽⁶⁸⁾**N.D.E.** In view of the modifications in definition 4.4 (cf. N.D.E. (65)), one has inserted here this remark, which in the original appeared in the proof of 4.4.1.

⁽⁶⁹⁾**N.D.E.** The original showed that (i) implies (ii) and (iii); one has added the implication (ii) $\Rightarrow$ (i).

are isomorphisms of $\mathbf{O}_S$ -modules. Since F/S satisfies (E), the two members transform finite direct sums of copies of $\mathbf{O}_S$ into finite products of abelian S -groups. This reduces us to the case $\mathcal{M} = \mathbf{O}_S$, which follows from the hypothesis.

Finally (i) $\Rightarrow$ (iii) follows from the definition and from the fact that the isomorphisms

$$F \otimes_{\mathbf{O}_S} \text{Lie}(\mathbf{O}_S/S, \mathcal{M}) \xrightarrow{\sim} \text{Lie}'(F/S, \mathcal{M})$$

of 4.3.2 (5) are morphisms of $\mathbf{O}_S$ -modules. $\square$

Examples 4.4.2. For every quasi-coherent $\mathbf{O}_S$ -Module $\mathcal{E}$, the $\mathbf{O}_S$ -modules $V(\mathcal{E})$ and $W(\mathcal{E})$ defined in I 4.6 are good.

(70) Indeed, for every $f: S' \rightarrow S$, the morphisms

$$\begin{aligned} V(\mathcal{E})(S') \otimes_{\mathbf{O}(S')} \mathbf{O}(I_{S'}(\mathcal{M})) &\rightarrow T_{V(\mathcal{E})/S}(\mathcal{M})(S') \\ W(\mathcal{E})(S') \otimes_{\mathbf{O}(S')} \mathbf{O}(I_{S'}(\mathcal{M})) &\rightarrow T_{W(\mathcal{E})/S}(\mathcal{M})(S') \end{aligned}$$

correspond, respectively, to the morphisms:

$$\begin{aligned} \text{Hom}_{\mathbf{O}_{S'}\text{-mod.}}(f^*(\mathcal{E}), \mathbf{O}_{S'}) \otimes_{\mathbf{O}(S')} \Gamma(S', \mathcal{D}_{\mathbf{O}_{S'}}(\mathcal{M})) \\ \rightarrow \text{Hom}_{\mathbf{O}_{S'}\text{-mod.}}(f^*(\mathcal{E}), \mathcal{D}_{\mathbf{O}_{S'}}(\mathcal{M})) \\ \Gamma(S', f^*(\mathcal{E})) \otimes_{\mathbf{O}(S')} \Gamma(S', \mathcal{D}_{\mathbf{O}_{S'}}(\mathcal{M})) \\ \rightarrow \Gamma(S', f^*(\mathcal{E}) \otimes_{\mathbf{O}_{S'}} \mathcal{D}_{\mathbf{O}_{S'}}(\mathcal{M})); \end{aligned}$$

these are isomorphisms since $\mathcal{D}_{\mathbf{O}_{S'}}(\mathcal{M})$ is isomorphic, as an $\mathbf{O}_{S'}$ -module, to a finite direct sum of copies of $\mathbf{O}_{S'}$.

Proposition 4.5. *Let F be a good $\mathbf{O}_S$ -module. Then:*

(i) $\text{Aut}_{\mathbf{O}_S\text{-mod.}}(F)/S$ satisfies (E) and one has a functorial isomorphism

$$\text{Lie}(\text{Aut}_{\mathbf{O}_S\text{-mod.}}(F)/S, \mathcal{M}) \xrightarrow{\sim} \text{Hom}_{\mathbf{O}_S\text{-mod.}}(F, \text{Lie}(F/S, \mathcal{M}))$$

which respects the $\mathbf{O}_S$ -module structures deduced from condition (E). In particular, one has an isomorphism of $\mathbf{O}_S$ -modules

$$\text{Lie}(\text{Aut}_{\mathbf{O}_S\text{-mod.}}(F)/S) \xrightarrow{\sim} \text{End}_{\mathbf{O}_S\text{-mod.}}(F).$$

(ii) (71) Moreover, $\text{End}_{\mathbf{O}_S\text{-mod.}}(F)$ is a good $\mathbf{O}_S$ -module.

Indeed, by 4.4.1, F/S satisfies (E) and

$$(1) \quad \text{Lie}(F/S, \mathcal{M}) = \text{Lie}'(F/S, \mathcal{M}) \simeq F \otimes_{\mathbf{O}_S} \text{Lie}(\mathbf{O}_S/S, \mathcal{M}).$$

Assertion (i) then follows from 4.3.1. Set $E = \text{End}_{\mathbf{O}_S\text{-mod.}}(F)$. By (1) and remark 4.3.5, one has the following commutative diagram of morphisms of abelian S -groups:

$$\begin{array}{ccc} \text{End}_{\mathbf{O}_S\text{-mod.}}(F) \otimes_{\mathbf{O}_S} \text{Lie}(\mathbf{O}_S/S, \mathcal{M}) & \xrightarrow{d_E} & \text{Lie}(\text{End}_{\mathbf{O}_S\text{-mod.}}(F)/S, \mathcal{M}) \\ & & \uparrow \simeq (*) \\ \text{Hom}_{\mathbf{O}_S}(F, F \otimes_{\mathbf{O}_S} \text{Lie}(\mathbf{O}_S/S, \mathcal{M})) & \xrightarrow{d_F} & \text{Hom}_{\mathbf{O}_S}(F, \text{Lie}(F/S, \mathcal{M})) \end{array}$$

where d_F and $(*)$ are isomorphisms of $\mathbf{O}_S$ -modules; consequently, the same is true of d_E . This proves (ii).

(70) **N.D.E.** One has added what follows.

(71) **N.D.E.** One has added point (ii), an immediate consequence of what precedes.

Scholium 4.5.1. ⁽⁷²⁾ Set (cf. 2.1) $\mathbf{O}_{I_S} = \mathbf{O}_S \oplus t\mathbf{O}_S$ (with $t^2 = 0$), and let F be a good $\mathbf{O}_S$ -module. Then, for every $S' \rightarrow S$, the morphism

$$F(S') \oplus tF(S') = F(S') \otimes_{\mathbf{O}(S')} \mathbf{O}(I_{S'}) \rightarrow F(I_{S'}) = F(S') \oplus \text{Lie}(F/S)(S')$$

(which is the identity on $F(S')$) induces an isomorphism of $\mathbf{O}(S')$ -modules $tF(S') \simeq \text{Lie}(F/S)(S')$.

Letting S' vary, one obtains an isomorphism which one may denote $\text{Lie}(\mathbf{F}/S) \simeq t\mathbf{F}$. For every $S' \rightarrow S$, one therefore has, by 4.5, a commutative diagram:

$$\begin{array}{ccc} \text{End}_{\mathbf{O}_{S'}\text{-mod.}}(\mathbf{F}_{S'}) & \xrightarrow{\sim} \text{Hom}_{\mathbf{O}_{S'}\text{-mod.}}(\mathbf{F}_{S'}, t\mathbf{F}_{S'}) & \xrightarrow{\sim} \text{Lie}(\text{Aut}_{\mathbf{O}_S\text{-mod.}}(\mathbf{F})/S)(S') \\ \downarrow & & \downarrow \\ \text{Aut}_{\mathbf{O}_{I_{S'}}\text{-mod.}}(\mathbf{F}_{I_{S'}}) & & T_{\text{Aut}_{\mathbf{O}_S\text{-mod.}}(\mathbf{F})/S}(S') \end{array}$$

and one deduces from 4.3 bis that every $X \in \text{End}_{\mathbf{O}_{S'}\text{-mod.}}(\mathbf{F}_{S'})$ corresponds to the element $\text{id} + tX$ of $\text{Aut}_{\mathbf{O}_{I_{S'}}\text{-mod.}}(\mathbf{F}_{I_{S'}})$.

Definition 4.6. One says that the group S -functor $\mathbf{G}$ is *good* if $\mathbf{G}/S$ satisfies condition (E) and if $\text{Lie}(\mathbf{G}/S)$ is a good $\mathbf{O}_S$ -module. ⁽⁷³⁾ Note that if $\mathbf{F}$ is a good $\mathbf{O}_S$ -module, it is a good S -group; indeed $\mathbf{F}/S$ satisfies (E) and $\text{Lie}(\mathbf{F}/S) \simeq \mathbf{F}$ is a good $\mathbf{O}_S$ -module.

Example 4.6.1. If $\mathbf{G}$ is representable, it is good. Indeed, $\mathbf{G}/S$ satisfies (E) and $\text{Lie}(\mathbf{G}/S)$ is of the form $\mathbf{V}(\mathcal{E})$, hence good, by 4.4.2.

Lemma 4.6.2. ⁽⁷⁴⁾ Let $\mathbf{G}$ be a group S -functor such that $\mathbf{G}/S$ satisfies (E), and let $\mathbf{F} = \text{Lie}(\mathbf{G}/S)$. Then $\mathbf{F}$ satisfies (E) and the morphism of abelian groups $d: \mathbf{F} \rightarrow \text{Lie}(\mathbf{F}/S)$ respects the $\mathbf{O}_S$ -module structures.

Consequently, $\mathbf{G}$ is good if and only if $\mathbf{F} \rightarrow \text{Lie}(\mathbf{F}/S)$ is bijective.

PROOF. Suppose that $\mathbf{G}/S$ satisfies (E). Let $\mathcal{M}, \mathcal{N}$ be free $\mathcal{O}_S$ -modules of finite rank. Write $\mathbf{F}(\mathcal{N}) = \text{Lie}(\mathbf{G}/S, \mathcal{N})$ and let e be the unit section of $\mathbf{G}$.

For every $S' \rightarrow S$, one has $\mathbf{F}(\mathcal{N})(S') = \{g \in \text{Hom}_S(I_{S'}(\mathcal{N}), \mathbf{G}) \mid g \circ \varepsilon_{\mathcal{N}} = e\}$ and $\text{Lie}(\mathbf{F}(\mathcal{N})/S, \mathcal{M})(S') = \text{Hom}_S(I_{S'}(\mathcal{M}), \mathbf{F}(\mathcal{N}))$ is identified with

$$\left\{ \Phi \in \text{Hom}_S(I_{S'}(\mathcal{N}) \times_{S'} I_{S'}(\mathcal{M}), \mathbf{G}) \left| \begin{array}{l} \Phi \circ (\varepsilon_{\mathcal{N}} \times \text{id}_{I_{S'}(\mathcal{M})}) = e \in \mathbf{G}(I_{S'}(\mathcal{M})) \\ \Phi \circ (\text{id}_{I_{S'}(\mathcal{N})} \times \varepsilon_{\mathcal{M}}) = e \in \mathbf{G}(I_{S'}(\mathcal{N})) \end{array} \right. \right\}.$$

This shows that

$$(1) \quad \text{Lie}(\mathbf{F}(\mathcal{N})/S, \mathcal{M}) \simeq \text{Lie}(\mathbf{F}(\mathcal{M})/S, \mathcal{N}).$$

As $\mathbf{G}/S$ satisfies (E), one deduces from this:

$$\begin{aligned} \text{Lie}(\mathbf{F}(\mathcal{N})/S, \mathcal{M}_1 \oplus \mathcal{M}_2) &\simeq \text{Lie}(\mathbf{F}(\mathcal{M}_1 \oplus \mathcal{M}_2)/S, \mathcal{N}) \\ &\simeq \text{Lie}((\mathbf{F}(\mathcal{M}_1) \times_S \mathbf{F}(\mathcal{M}_2))/S, \mathcal{N}). \end{aligned}$$

By 3.8, the right-hand term is isomorphic to

$$\text{Lie}(\mathbf{F}(\mathcal{M}_1)/S, \mathcal{N}) \times_S \text{Lie}(\mathbf{F}(\mathcal{M}_2)/S, \mathcal{N}) \simeq \text{Lie}(\mathbf{F}(\mathcal{N})/S, \mathcal{M}_1) \times_S \text{Lie}(\mathbf{F}(\mathcal{N})/S, \mathcal{M}_2).$$

It follows that

$$(2) \quad \text{Lie}(\mathbf{F}(\mathcal{N})/S, \mathcal{M}_1 \oplus \mathcal{M}_2) \simeq \text{Lie}(\mathbf{F}(\mathcal{N})/S, \mathcal{M}_1) \times_S \text{Lie}(\mathbf{F}(\mathcal{N})/S, \mathcal{M}_2),$$

hence $\mathbf{F}(\mathcal{N})/S$ satisfies (E), by remark 4.1.1.2 a).

⁽⁷²⁾**N.D.E.** One has added this scholium, implicit in the original, which will be useful in 4.7.1. (Here and in what follows, one denotes by t, t' , etc., variables of square zero, the letter ε being reserved for the unit section of groups.)

⁽⁷³⁾**N.D.E.** One has added the following sentence.

⁽⁷⁴⁾**N.D.E.** One has added this lemma.

Let us now show that the morphism of abelian groups $d: \mathbf{F}(\mathcal{N}) \rightarrow \text{Lie}(\mathbf{F}(\mathcal{N})/S)$ respects the $\mathbf{O}_S$ -module structures. Let us consider the free $\mathcal{O}_S$ -module $\mathcal{M} = t\mathcal{O}_S$, so that

$$\mathbf{O}(\text{I}_{S'}(\mathcal{M})) = \mathbf{O}(S')[t]/(t^2) = \mathbf{O}(S') \oplus t\mathbf{O}(S')$$

and $\text{Lie}(\mathbf{O}_S/S) \simeq t\mathbf{O}_S$. One will denote by ρ_t the structural morphism $\text{I}_S(\mathcal{M}) \rightarrow S$, which corresponds to the injection $u_t: \mathcal{O}_S \hookrightarrow \mathcal{D}_{\mathcal{O}_S}(\mathcal{M}) = \mathcal{O}_S \oplus t\mathcal{O}_S$. Recall (cf. 3.4.2) that, for every S -functor X , $\mathbf{F}(\mathcal{N})(X) = \text{Lie}(\mathbf{G}/S, \mathcal{N})(X)$ is the set of S -morphisms $\phi: \text{I}_X(\mathcal{N}) \rightarrow \mathbf{G}$ such that $\phi \circ \varepsilon_{\mathcal{N}} = e$, and that the operation of $a \in \mathbf{O}(X)$ is given by $a \cdot \phi = \phi \circ a^*$, where a^* is the endomorphism of $\text{I}_X(\mathcal{N})$ associated with a , cf. 2.1.3.

Consequently, by 4.3.4, the morphism $d: \mathbf{F}(\mathcal{N}) \xrightarrow{\sim} \mathbf{F}(\mathcal{N}) \otimes_{\mathbf{O}_S} t\mathbf{O}_S \rightarrow \text{Lie}(\mathbf{F}(\mathcal{N})/S)$ is given by: for every $S' \rightarrow S$ and $f \in \text{Hom}_S(S', \mathbf{F}(\mathcal{N}))$,

$$f \mapsto f \otimes t \mapsto t \cdot (f \circ \rho_t) = f \circ \rho_t \circ t^*,$$

where, in the last term,

$$f \circ \rho_t \in \mathbf{F}(\mathcal{N})(\text{I}_{S'}(\mathcal{M}))$$

is regarded as an S -morphism $\phi: \text{I}_{S'}(\mathcal{M})(\mathcal{N}) \rightarrow \mathbf{G}$ (such that $\phi \circ \varepsilon_{\mathcal{N}} = e$). It remains to see that d is a morphism of $\mathbf{O}_S$ -modules, i.e. that $d(a \cdot f) = u_t(a) \cdot d(f)$ for every $a \in \mathbf{O}(S')$.

Regarding $f \in \mathbf{F}(\mathcal{N})(S')$ as a morphism $\text{I}_{S'}(\mathcal{N}) \rightarrow \mathbf{G}$, one likewise has $a \cdot f = f \circ a^*$. On the other hand, by the functoriality in X of the operation of $\mathbf{O}(X)$ on $\text{I}_X(\mathcal{N})$ (cf. 2.1.3), one has the commutative diagram below:

$$\begin{array}{ccc} \text{I}_{S'}(\mathcal{N}) & \xrightarrow{a^*} & \text{I}_{S'}(\mathcal{N}) \\ \uparrow \rho_t & & \uparrow \rho_t \\ \text{I}_{\text{I}_{S'}(\mathcal{M})}(\mathcal{N}) & \xrightarrow{u_t(a)^*} & \text{I}_{\text{I}_{S'}(\mathcal{M})}(\mathcal{N}) \end{array}$$

One therefore has

$$\begin{aligned} d(a \cdot f) &= f \circ a^* \circ \rho_t \circ t^* = f \circ \rho_t \circ u_t(a)^* \circ t^* \\ &= f \circ \rho_t \circ t^* \circ u_t(a)^* = u_t(a) \cdot d(f) \end{aligned}$$

(the penultimate equality resulting from the fact that $\mathbf{O}$ is commutative). This completes the proof of lemma 4.6.2. $\square$

Theorem 4.7. *If $\mathbf{F}$ is a good $\mathbf{O}_S$ -module, the S -group $\text{Aut}_{\mathbf{O}_S\text{-mod.}}(\mathbf{F})$ is good.*

(75) Indeed, by 4.5, $\text{Aut}_{\mathbf{O}_S\text{-mod.}}(\mathbf{F})/S$ satisfies (E) and

$$\text{Lie}(\text{Aut}_{\mathbf{O}_S\text{-mod.}}(\mathbf{F})/S) \simeq \text{End}_{\mathbf{O}_S\text{-mod.}}(\mathbf{F})$$

is a good $\mathbf{O}_S$ -module.

4.7.1. — (76) Let now $\mathbf{G}$ be an S -group and $\mathbf{F}$ a good $\mathbf{O}_S$ -module. Suppose given a *linear representation* of $\mathbf{G}$ in $\mathbf{F}$, that is (I 4.7.1), a morphism of S -groups

$$\rho: \mathbf{G} \rightarrow \text{Aut}_{\mathbf{O}_S\text{-mod.}}(\mathbf{F}).$$

If $\mathbf{G}/S$ satisfies (E), one deduces from this, by 4.1.C and 4.5, a morphism of $\mathbf{O}_S$ -modules, denoted ρ' or $d\rho$:

$$\text{Lie}(\mathbf{G}/S) \rightarrow \text{Lie}(\text{Aut}_{\mathbf{O}_S\text{-mod.}}(\mathbf{F})/S) \simeq \text{End}_{\mathbf{O}_S\text{-mod.}}(\mathbf{F}).$$

(77) Moreover, setting $\mathcal{O}_{\text{I}_S} = \mathcal{O}_S \oplus t\mathcal{O}_S$ (with $t^2 = 0$), one deduces from 4.5.1 that, if $S' \rightarrow S$ and $X \in \text{Lie}(\mathbf{G}/S)(S') \subset \mathbf{G}(\text{I}_{S'})$, then one has in $\text{Aut}_{\text{I}_{S'}\text{-mod.}}(\mathbf{F}_{\text{I}_{S'}})$ the following equality:

$$(*) \quad \rho(X) = \text{id} + t\rho'(X),$$

(75) **N.D.E.** One has simplified the original, using the additions made in 4.3.1 and 4.5.

(76) **N.D.E.** One has added the numbering 4.7.1, 4.7.2, 4.7.3.

(77) **N.D.E.** One has added the following sentence.

i.e. for every $S'' \rightarrow I_{S'}$ and $f \in \mathbf{F}(S'')$, one has in $\mathbf{F}(S'')$ the equality $\rho(X)(f) = f + t\rho'(X)(f)$.

Definition 4.7.2. Let $\mathbf{G}$ be a *good* S -group. Then $\mathrm{Lie}(\mathbf{G}/S)$ is a good $\mathbf{O}_S$ -module, and one has a morphism of S -groups $\mathrm{Ad}: \mathbf{G} \rightarrow \mathrm{Aut}_{\mathbf{O}_S\text{-mod.}}(\mathrm{Lie}(\mathbf{G}/S))$. One deduces from this, by 4.7.1, a morphism of $\mathbf{O}_S$ -modules

$$\mathrm{ad}: \mathrm{Lie}(\mathbf{G}/S) \rightarrow \mathrm{End}_{\mathbf{O}_S\text{-mod.}}(\mathrm{Lie}(\mathbf{G}/S)),$$

or, what amounts to the same thing, an $\mathbf{O}_S$ -bilinear morphism:

$$\mathrm{Lie}(\mathbf{G}/S) \times_S \mathrm{Lie}(\mathbf{G}/S) \rightarrow \mathrm{Lie}(\mathbf{G}/S), \quad (x, y) \mapsto [x, y] = \mathrm{ad}(x) \cdot y$$

(where x and y denote two arbitrary elements of $\mathrm{Lie}(\mathbf{G}/S)(S') = \mathrm{Lie}(\mathbf{G}_{S'}/S')(S')$). If $\mathbf{G}$ is commutative, then $[x, y] = 0$.

4.7.3. — One may give an equivalent definition of the bracket, as follows: let us first observe that it suffices to do so for $x, y \in \mathrm{Lie}(\mathbf{G}/S)(S)$. Let us next observe that there is a canonical isomorphism $I_S \times_S I_S \simeq I_{I_S}$; to avoid confusion, let us denote by I and I' two copies of I_S and set $\mathcal{O}_I = \mathcal{O}_S[t]$, $\mathcal{O}_{I'} = \mathcal{O}_S[t']$, where $t^2 = 0 = t'^2$. One then has a commutative diagram

$$\begin{array}{ccc} I \times I' & \longrightarrow & I' \\ \downarrow & & \downarrow \\ I & \longrightarrow & S, \end{array}$$

the two arrows issuing from $I \times I'$ identifying the latter with the scheme of dual numbers on I or on I' . There results a commutative diagram of groups (where one writes $L = \mathrm{Lie}(\mathbf{G}/S)$):

$$(1) \quad \begin{array}{ccccccc} & & 1 & & 1 & & \\ & & \downarrow & & \downarrow & & \\ & & L(I) & \longrightarrow & L(S) & \longrightarrow & 1 \\ & & \downarrow & & \downarrow & & \\ 1 & \longrightarrow & L(I') & \longrightarrow & \mathbf{G}(I \times I') & \longrightarrow & \mathbf{G}(I') \longrightarrow 1 \\ & & \downarrow & & \downarrow & & \downarrow \\ 1 & \longrightarrow & L(S) & \longrightarrow & \mathbf{G}(I) & \longrightarrow & \mathbf{G}(S) \longrightarrow 1 \\ & & \downarrow & & \downarrow & & \downarrow \\ & & 1 & & 1 & & 1 \end{array}$$

The ninth piece of the puzzle is none other than $\mathrm{Lie}(L/S)(S)$. If $\mathbf{G}$ is *good*, this is $L(S)$, and one therefore has the following commutative diagram, in which the rows and the columns are exact sequences of groups, the five $L(\)$ are commutative,⁽⁷⁸⁾ and in which, taking into account the identification $L(I) = L(S) \oplus tL(S)$ (resp. $L(I') = L(S) \oplus t'L(S)$), the injection $L(S) \hookrightarrow L(I)$ (resp. $L(S) \hookrightarrow L(I')$) is given by $u \mapsto tu$ (resp. $u \mapsto t'u$):

$$(2) \quad \begin{array}{ccccc} z \in L(S) & \xrightarrow{t} & L(I) & \longrightarrow & L(S) \ni x \\ t' \downarrow & & \downarrow & & \downarrow \\ L(I') & \longrightarrow & \mathbf{G}(I \times I') & \longrightarrow & \mathbf{G}(I') \\ \downarrow & & \downarrow & & \downarrow \\ y \in L(S) & \longrightarrow & \mathbf{G}(I) & \longrightarrow & \mathbf{G}(S). \end{array}$$

Now in such a diagram, if one takes two elements x and y as marked, and if one lifts x resp. y arbitrarily to an element $\tilde{x} \in L(I)$ resp. $y' \in L(I')$, the commutator $\tilde{x}y'\tilde{x}^{-1}y'^{-1}$ in

⁽⁷⁸⁾**N.D.E.** One has expanded the original in what follows.

$\mathbf{G}(\mathbf{I} \times \mathbf{I}')$ does not depend on the chosen lifts and is the image of an element z as marked. The reader will verify that one has $z = [x, y]$.⁽⁷⁹⁾

Indeed, if one still denotes by x the image of x under the canonical section $L(S) \rightarrow L(\mathbf{I})$ (and likewise for y), then $\tilde{x} = xu$ and $y' = yv$, with $u, v \in L(S) = L(\mathbf{I}) \cap L(\mathbf{I}')$, and since $L(\mathbf{I})$ and $L(\mathbf{I}')$ are commutative one has

$$\tilde{x} y' \tilde{x}^{-1} y'^{-1} = x u y v u^{-1} x^{-1} v^{-1} y^{-1} = x u y u^{-1} v x^{-1} v^{-1} y^{-1} = x y x^{-1} y^{-1}.$$

Moreover, this element is sent to the unit element of $\mathbf{G}(\mathbf{I})$ and of $\mathbf{G}(\mathbf{I}')$, hence comes from a (unique) z as indicated. Finally, regarding y (resp. x) as an element of $L_{\mathbf{I}'}(\mathbf{I}')$ (resp. of $L(S) \subset \mathbf{G}(\mathbf{I}')$), one has by 4.7.1 (*):

$$xyx^{-1} = \text{Ad}(x)(y) = (\text{id} + t' \text{ad}(x))(y) = y + t'[x, y],$$

hence the element $xyx^{-1}y^{-1}$ of $L(\mathbf{I}')$ is the image of the element $z = [x, y]$ of $L(S)$.

In this construction the following two properties appear:

- (i) the bracket is “functorial in $\mathbf{G}$ ”: precisely, $\mathbf{G} \mapsto \text{Lie}(\mathbf{G}/S)$ is a functor from the category of good S -groups to the category of good $\mathbf{O}_S$ -modules equipped with an $\mathbf{O}_S$ -bilinear composition law.
- (ii) One has $[x, y] + [y, x] = 0$: indeed the diagram is symmetric with respect to the main diagonal.⁽⁸⁰⁾

Proposition 4.8. *Let $\mathbf{F}$ be a good $\mathbf{O}_S$ -module. Via the identification*

$$\text{Lie}(\text{Aut}_{\mathbf{O}_{S\text{-mod}}}(\mathbf{F})/S) = \text{End}_{\mathbf{O}_{S\text{-mod}}}(\mathbf{F})$$

one has

$$\text{Ad}(g) \cdot Y = g \circ Y \circ g^{-1} \quad \text{and} \quad [X, Y] = X \circ Y - Y \circ X,$$

for every $S' \rightarrow S$, $g \in \text{Aut}_{\mathbf{O}_{S'\text{-mod}}}(\mathbf{F}_{S'})$, and

$$X, Y \in \text{Lie}(\text{Aut}_{\mathbf{O}_{S\text{-mod}}}(\mathbf{F})/S)(S') = \text{End}_{\mathbf{O}_{S'\text{-mod}}}(\mathbf{F}_{S'}).$$

PROOF. ⁽⁸¹⁾ By base change, one reduces to $S' = S$, which makes it possible to lighten the notation. Set $\mathbf{I} = \mathbf{I}_S$ and $\mathcal{O}_{\mathbf{I}} = \mathcal{O}_S[t]$ (with $t^2 = 0$). Recall (cf. 4.5.1) that the inclusion $i: \text{End}_{\mathbf{O}_{S\text{-mod}}}(\mathbf{F}) \hookrightarrow \text{Aut}_{\mathbf{O}_{\mathbf{I}\text{-mod}}}(\mathbf{F}_{\mathbf{I}})$ sends Y to $\text{id} + tY$. Then, by definition of $\text{Ad}(g)$ (cf. 4.1.A), one has:

$$\text{id} + t \text{Ad}(g)(Y) = g \circ (\text{id} + tY) \circ g^{-1} = \text{id} + t(g \circ Y \circ g^{-1}),$$

whence $\text{Ad}(g)(Y) = g \circ Y \circ g^{-1}$.

Let $\mathbf{I}'$ be a second copy of $\mathbf{I}_S$, and set $\mathcal{O}_{\mathbf{I}'} = \mathcal{O}_S[t']$ (with $t'^2 = 0$). Let us apply the results of 4.7.3 to $\mathbf{G} = \text{Aut}_{\mathbf{O}_{S\text{-mod}}}(\mathbf{F})$ and $L = \text{Lie}(\mathbf{G}/S) = \text{Aut}_{\mathbf{O}_{S\text{-mod}}}(\mathbf{F})$. One identifies X with its image under the canonical section $L(S) \hookrightarrow L(\mathbf{I})$; its image in $\mathbf{G}(\mathbf{I} \times \mathbf{I}')$ is then $\text{id} + t'X$, whose inverse is $\text{id} - t'X$. Likewise, Y lifts to $\text{id} + tY$, whose inverse is $\text{id} - tY$. Then the commutator

$$(\text{id} + t'X) \circ (\text{id} + tY) \circ (\text{id} - t'X) \circ (\text{id} - tY) = \text{id} + tt'(X \circ Y - Y \circ X)$$

is the image in $\mathbf{G}(\mathbf{I} \times \mathbf{I}')$ of the element $Z = X \circ Y - Y \circ X$ of $L(S)$ (indeed, Z is sent to $tZ \in L(\mathbf{I})$, then to $\text{id} + t'tZ \in \mathbf{G}(\mathbf{I} \times \mathbf{I}')$). By 4.7.3, this shows that $[X, Y] = X \circ Y - Y \circ X$. $\square$

⁽⁷⁹⁾**N.D.E.** The original indicated in a footnote: “(*) The writer acknowledges, at Gabriel’s request, that the exercise is not immediate; this is moreover the reason why it is not in the text”. One has given the details, taking into account the addition made in 4.7.1; see also [DG70], § II.4, 4.2.

⁽⁸⁰⁾**N.D.E.** I.e., x and y play symmetric roles: the image in $\mathbf{G}(\mathbf{I} \times \mathbf{I}')$ of $[y, x]$ equals the commutator $y' \tilde{x} y'^{-1} \tilde{x}^{-1}$, which is the inverse of $\tilde{x} y' \tilde{x}^{-1} y'^{-1} = [x, y]$. On the other hand (cf. 4.10 below), one does not know whether one necessarily has $[x, x] = 0$: if one regards x as an element $\tilde{x} \in L(\mathbf{I})$ (resp. $x' \in L(\mathbf{I}')$) it is not a priori clear that $\tilde{x}$ and x' commute...

⁽⁸¹⁾**N.D.E.** One has expanded and simplified the original in what follows.

Corollary 4.8.1. *Let $\mathbf{G}$ be a good S -group and $x, y, z \in \text{Lie}(\mathbf{G}/S)(S')$. One has:*

$$[x, [y, z]] + [y, [z, x]] + [z, [x, y]] = 0.$$

⁽⁸²⁾ Indeed, since $\mathbf{G}$ is good, $\text{Lie}(\mathbf{G}/S)$ is a good $\mathbf{O}_S$ -module and hence, by 4.7,

$$\text{Aut}_{\mathbf{O}_S\text{-mod.}}(\text{Lie}(\mathbf{G}/S))$$

is a good S -group. Then the morphism of S -groups

$$\text{Ad}: \mathbf{G} \longrightarrow \text{Aut}_{\mathbf{O}_S\text{-mod.}}(\text{Lie}(\mathbf{G}/S))$$

gives, by the functoriality 4.7.3 (i): $\text{ad}[x, y] = [\text{ad } x, \text{ad } y]$. Combined with 4.8, this gives:

$$\text{ad}[x, y] = [\text{ad } x, \text{ad } y] = \text{ad } x \circ \text{ad } y - \text{ad } y \circ \text{ad } x,$$

which, applied to an element z , gives the Jacobi relation.

Corollary 4.8.2. *Let $\mathbf{G}$ be a good S -group operating linearly on a good $\mathbf{O}_S$ -module $\mathbf{F}$ (i.e. let $\mathbf{F}$ be a $\mathbf{G}$ - $\mathbf{O}_S$ -module, $\mathbf{G}$ and $\mathbf{F}$ good). Then the linear map $\rho': \text{Lie}(\mathbf{G}/S) \longrightarrow \text{End}_{\mathbf{O}_S\text{-mod.}}(\mathbf{F})$ is a representation, that is, one has*

$$\rho'([x, y]) = \rho'(x) \circ \rho'(y) - \rho'(y) \circ \rho'(x).$$

Scholium 4.9. *To every good S -group (for example a representable one), one has associated a good $\mathbf{O}_S$ -module $\text{Lie}(\mathbf{G}/S)$ equipped functorially with a bilinear map satisfying*

$$[x, y] + [y, x] = 0, \quad [x, [y, z]] + [y, [z, x]] + [z, [x, y]] = 0.$$

We shall call $\text{Lie}(\mathbf{G}/S)$ equipped with this structure the “Lie algebra” of $\mathbf{G}$ over S (the quotation marks being justified by the fact that one does not know whether $\text{Lie}(\mathbf{G}/S)$ is strictly speaking an $\mathbf{O}_S$ -Lie algebra⁽⁸³⁾). To every linear representation of $\mathbf{G}$ in a good $\mathbf{O}_S$ -module $\mathbf{F}$ is associated a representation of its “Lie algebra”. In particular, to the adjoint representation of $\mathbf{G}$ is associated the adjoint representation of its “Lie algebra”.

Definition 4.10. A group functor $\mathbf{G}$ over S is said to be *very good* if it is good and if $\text{Lie}(\mathbf{G}/S)$ is an $\mathbf{O}_S$ -Lie algebra (that is, if one has identically $[x, x] = 0$).

Examples 4.10.1. The following S -groups are very good: $\text{Aut}_{\mathbf{O}_S\text{-mod.}}(\mathbf{F})$ for every good $\mathbf{O}_S$ -module $\mathbf{F}$ (cf. 4.7 and 4.8), every representable group (see below), every good S -group admitting a monomorphism into a very good S -group, for example every good group subfunctor of a representable group, or every good S -group admitting a faithful linear representation in a good $\mathbf{O}_S$ -module, for example every good S -group such that Ad is a monomorphism. . .

4.11. — Suppose now that G is a group scheme over S . By 4.1.4, $\text{Lie}(G/S)(S)$ is identified with the group of right-invariant infinitesimal automorphisms of G/S , that is, by 3.14, with the group of derivations of $\mathcal{O}_G$ over $\mathcal{O}_S$ invariant under right translation. Moreover this identification respects the module structure and is an⁽⁸⁴⁾ anti-isomorphism of Lie algebras, as one sees by arguing as in 4.7.3: let us set $\mathcal{O}_I = \mathcal{O}_S[t]$ and $\mathcal{O}_{I'} = \mathcal{O}_S[t']$, and let $x \in L(I)$ and $y \in L(I')$. The left translation λ_x (resp. λ_y) is an S -automorphism of $G_{I \times I'}$ which induces the identity on $G_{I'}$ (resp. G_I) and which corresponds to an $\mathcal{O}_S$ -automorphism

$$u = \text{id} + t d_x \quad \text{resp.} \quad v = \text{id} + t' d_y$$

of $\mathcal{O}_{G_{I \times I'}} = \mathcal{O}_G[t, t']/(t^2, t'^2)$, where d_x, d_y are $\mathcal{O}_S$ -derivations of $\mathcal{O}_G$ invariant under right translation. As the correspondence between S -automorphisms of $G_{I \times I'}$ and $\mathcal{O}_S$ -automorphisms of $\mathcal{O}_{G_{I \times I'}}$ is contravariant, $\lambda_x \lambda_y \lambda_x^{-1} \lambda_y^{-1}$ corresponds to $v^{-1} u^{-1} v u = \text{id} + t t' (d_y d_x - d_x d_y)$. One deduces from this, by 4.7.3, that the map $x \mapsto -d_x$ is an isomorphism of Lie algebras (for more details, see [DG70], § II.4, 4.4 and 4.6). The foregoing is valid

⁽⁸²⁾N.D.E. One has expanded the original in what follows.

⁽⁸³⁾N.D.E. For one does not know whether $[x, x] = 0$; see 4.10 below.

⁽⁸⁴⁾N.D.E. One has corrected a sign error in the original, cf. [DG70], § II.4, 4.4 and 4.6.

for $\mathrm{Lie}(G/S)(S') = \mathrm{Lie}(G_{S'}/S')(S')$ for every $S' \rightarrow S$. One then recovers the classical definition.⁽⁸⁵⁾

Scholium 4.11.1. *Via the isomorphism $x \mapsto -d_x$, $\mathrm{Lie}(G/S)$ is identified with the functor which, to every S' over S , associates the $\mathcal{O}(S')$ -Lie algebra of the derivations of $G_{S'}$ with respect to S' invariant under right translation.*

As one already knows, by 4.6.1, that every representable group is good, it follows from the foregoing:

Corollary 4.11.2. *Every representable group is very good.*

Let $\varepsilon: S \rightarrow G$ be the unit section of G . Set $\omega_{G/S}^1 = \varepsilon^*(\Omega_{G/S}^1)$ and recall (cf. 3.3) that $\mathrm{Lie}(G/S)$ is representable by the vector bundle $\mathbb{V}(\omega_{G/S}^1)$.

Scholium 4.11.3. *One has therefore associated functorially, to every S -group scheme G , a vector bundle $\mathbb{L}\mathrm{ie}(G/S) = \mathbb{V}(\omega_{G/S}^1)$ over S , which represents the functor $\mathrm{Lie}(G/S)$, and hence is equipped with a structure of S -scheme in $\mathcal{O}_S$ -Lie algebras. Moreover (cf. 3.4 and 3.8), this construction commutes with base change and with finite products.*

Remarks 4.11.4. ⁽⁸⁶⁾ Let π denote the morphism $G \rightarrow S$.

a) The $\mathcal{O}_G$ -module $\Omega_{G/S}^1$ is evidently $(G \times_S G)$ -equivariant (cf. I, § 6) and hence, by I 6.8.1, one has $\Omega_{G/S}^1 \simeq \pi^*(\omega_{G/S}^1)$. It follows for example that $\Omega_{G/S}^1$ is locally free (resp. locally free of finite type) if $\omega_{G/S}^1$ is, which is in particular the case if S is the spectrum of a field (resp. if S is the spectrum of a field and G is locally of finite type over S).

b) Moreover, by I 6.8.2, $\omega_{G/S}^1$ is equipped with a canonical structure of G - $\mathcal{O}_S$ -module, which induces on $\mathbb{V}(\omega_{G/S}^1) = \mathbb{L}\mathrm{ie}(G/S)$ the adjoint operation.⁽⁸⁷⁾

c) On the other hand (cf. EGA I, 5.3.11 and 5.4.6), ε is an immersion, and is a closed immersion if G is separated over S . Thus $\omega_{G/S}^1$ is identified with $\mathcal{I}/\mathcal{I}^2$, where $\mathcal{I}$ is the quasi-coherent ideal defining $\varepsilon(S)$ in an open U of G in which $\varepsilon(S)$ is closed (if $G \rightarrow S$ is separated, one may take $U = G$, and if $G = \mathrm{Spec} \mathcal{A}(G)$ is affine over S , $\mathcal{I}$ is none other than the augmentation ideal of $\mathcal{A}(G)$, i.e. the kernel of $\varepsilon^\sharp: \mathcal{A}(G) \rightarrow \mathcal{O}_S$, cf. I 4.2).⁽⁸⁸⁾

Remark 4.11.5. One can deduce from the isomorphism $\Omega_{G/S}^1 \simeq \pi^*(\omega_{G/S}^1)$ that the $\mathcal{O}_S$ -module $\omega_{G/S}^1$ is identified with the sheaf $\pi_*^G(\Omega_{G/S}^1)$ of “differentials of G with respect to S invariant on the right”, that is, the sheaf whose sections over an open U of S are the sections of $\Omega_{G/S}^1$ over $\pi^{-1}(U)$, invariant under right translation (cf. I, 6.8.3, compare also with VII_A, 2.4).⁽⁸⁹⁾

⁽⁸⁵⁾**N.D.E.** One has set out the original in more detail in what follows; in particular, one has added, for later references, the numbering 4.11.1 to 4.11.8.

⁽⁸⁶⁾**N.D.E.** One has set out what follows in more detail, taking into account the additions made in Exp. I, § 6.8.

⁽⁸⁷⁾**N.D.E.** For this reason, the linear operation of G on $\omega_{G/S}^1$ could be called the “pre-adjoint” operation; in fact, by abuse of language, one will still speak of “the adjoint operation” on $\omega_{G/S}^1$. Let us point out here a somewhat more general construction, which will be used in Exp. III (cf. in particular III 0.8). Suppose given, for every $Y \rightarrow S$, an $\mathcal{O}_Y$ -module $\mathcal{N}(Y)$, and for every S -morphism $\varphi: Z \rightarrow Y$, a morphism of $\mathcal{O}_Z$ -modules $\mathcal{N}(\varphi): \varphi^*\mathcal{N}(Y) \rightarrow \mathcal{N}(Z)$, functorial in φ (this is the case, for example, if $\mathcal{J}$ is a quasi-coherent ideal of $\mathcal{O}_S$ and if one sets $\mathcal{N}(Y) = \mathcal{J}\mathcal{O}_Y$, cf. Exp. III); then the operation of G on $\omega_{G/S}^1$ induces a (generalized) “adjoint operation” of G on the S -functor in abelian groups which, to every $f: Y \rightarrow S$, associates $\mathrm{Hom}_{\mathcal{O}_Y}(f^*(\omega_{G/S}^1), \mathcal{N}(Y))$.

⁽⁸⁸⁾**N.D.E.** One has added the preceding description of $\omega_{G/S}^1$, which will be useful later (for example, in VII_A, 5.5).

⁽⁸⁹⁾**N.D.E.** This description of $\omega_{G/S}^1$ in terms of invariant differentials will not be used in the sequel.

Notation 4.11.6. One denotes by $\mathcal{L}ie(G/S)$ the sheaf of sections of the vector bundle $\mathbb{L}ie(G/S) \rightarrow S$; it is the $\mathcal{O}_S$ -module $(\omega_{G/S}^1)^\vee = \mathcal{H}om_{\mathcal{O}_S}(\omega_{G/S}^1, \mathcal{O}_S)$ dual to $\omega_{G/S}^1$ (cf. EGA II 1.7.9). It is equipped with a structure of $\mathcal{O}_S$ -Lie algebra.

As this construction does not commute with base change (in general), the Lie algebra structure on this module does not make it possible to reconstruct the structure of S -scheme in $\mathcal{O}_S$ -Lie algebras on $\mathbb{L}ie(G/S)$.⁽⁹⁰⁾

However one has:

Lemma 4.11.7. *One supposes $\omega_{G/S}^1$ locally free of finite type (which occurs in particular if G is smooth over S (cf. EGA IV₄, 17.2.4), or if S is the spectrum of a field and G is locally of finite type over S). Then $\mathcal{L}ie(G/S)^\vee \simeq (\omega_{G/S}^1)^{\vee\vee} \simeq \omega_{G/S}^1$ and hence*

$$\mathbb{L}ie(G/S) = \mathbb{V}(\omega_{G/S}^1) = \mathbb{V}(\mathcal{L}ie(G/S)^\vee) = \mathbb{W}(\mathcal{L}ie(G/S))$$

(the last equality resulting from I 4.6.5.1).

Finally, let $G \rightarrow H$ be a monomorphism of group functors. Then $\mathbb{L}ie(G/S) \rightarrow \mathbb{L}ie(H/S)$ is likewise a monomorphism (cf. 3.7). As every vector monomorphism of vector bundles is a closed immersion⁽⁹¹⁾ one obtains:

Corollary 4.11.8. *Let $G \hookrightarrow H$ be a monomorphism of S -group schemes.*

- (i) $\mathbb{L}ie(G/S) \rightarrow \mathbb{L}ie(H/S)$ is a closed immersion, and hence $\omega_{H/S}^1 \rightarrow \omega_{G/S}^1$ is an epimorphism.
- (ii) If $\omega_{G/S}^1$ is locally free of finite type, then the corresponding morphism $\mathcal{L}ie(G/S) \rightarrow \mathcal{L}ie(H/S)$ is an isomorphism of $\mathcal{L}ie(G/S)$ onto a submodule of $\mathcal{L}ie(H/S)$ which is locally a direct factor.

5. Computation of some Lie algebras

5.1. Examples of Lie algebras: the diagonalizable groups. — Let $G = D_S(M)$ be a diagonalizable group over S (I 4.4). As the formation of $\mathbb{L}ie(G/S)$ commutes with base change, it suffices to carry out the construction for $G = D(M)$. One then has:

$$G(I_S) = \mathrm{Hom}_{\mathrm{gr.}}(M, \Gamma(I_S, \mathcal{O}_{I_S})^\times) = \mathrm{Hom}_{\mathrm{gr.}}(M, \Gamma(S, D\mathcal{O}_S)^\times).$$

Now one has a split exact sequence

$$1 \rightarrow \Gamma(S, \mathcal{O}_S) \rightarrow \Gamma(S, D\mathcal{O}_S)^\times \rightarrow \Gamma(S, \mathcal{O}_S)^\times \rightarrow 1,$$

which gives that $\mathbb{L}ie(G)(S)$ is identified with $\mathrm{Hom}_{\mathrm{gr.}}(M, \mathcal{O}(S))$ equipped with its evident structure of $\mathcal{O}(S)$ -module. One therefore obtains, after base change:

Proposition 5.1. *One has isomorphisms*

$$\mathrm{Hom}_{S\text{-gr.}}(M_S, \mathcal{O}_S) \xrightarrow{\sim} \mathcal{L}ie(D_S(M)/S) \quad \text{and} \quad \mathcal{H}om_{\mathrm{gr.}}(\widetilde{M}_S, \mathcal{O}_S) \xrightarrow{\sim} \mathcal{L}ie(D_S(M)/S),$$

(where, in the second isomorphism, $\widetilde{M}_S$ denotes the constant sheaf of groups on S defined by M , and $\mathcal{H}om_{\mathrm{gr.}}$ the sheaf of homomorphisms of sheaves of groups).

⁽⁹⁰⁾**N.D.E.** For $\mathcal{L}ie(G/S) \simeq (\omega_{G/S}^1)^\vee$ does not necessarily determine $\omega_{G/S}^1$. For example, if $S = \mathbf{A}_k^1$ (k a field) and if G is the S -group whose fiber at $s = 0$ is $\mathbf{G}_{a,k}$ and whose other fibers are the trivial group, then $\mathcal{L}ie(G/S) = 0$ but $\mathcal{L}ie(G_s/k)(k) = k$.

⁽⁹¹⁾**N.D.E.** Let $f: \mathcal{M} \rightarrow \mathcal{N}$ be a morphism of $\mathcal{O}_S$ -modules and $\mathcal{P} = \mathrm{Coker}(f)$. If $\mathbb{V}(\mathcal{N}) \rightarrow \mathbb{V}(\mathcal{M})$ is a monomorphism, the surjective morphism $\mathrm{Sym}(\mathcal{N}) \rightarrow \mathrm{Sym}(\mathcal{P})$ factors through $\mathcal{O}_S$, hence $\mathcal{P} = 0$.

Corollary 5.1.1. *If M is free of finite type (or, as we shall say later, if $D_S(M)$ is a split torus) then (see I, 4.6.5 for the definition of $\mathbb{W}$)*

$$\begin{aligned}\mathbb{W}(\mathrm{Lie}(D_S(M)/S)) &\xrightarrow{\sim} \mathrm{Lie}(D_S(M)/S), \\ M^\vee \otimes_{\mathbf{Z}} \mathcal{O}_S &\xrightarrow{\sim} \mathrm{Lie}(D_S(M)/S),\end{aligned}$$

where $M^\vee$ denotes the dual of the abelian group M . In particular

$$\mathcal{O}_S \xrightarrow{\sim} \mathrm{Lie}(\mathbf{G}_{m,S}/S) \quad \text{and} \quad \mathcal{O}_S \xrightarrow{\sim} \mathrm{Lie}(\mathbf{G}_{m,S}/S).$$

5.2. Normalizers and centralizers. — Let us first prove a few lemmas. Recall (cf. I 3.1.1) that a sequence $0 \rightarrow F' \rightarrow F \rightarrow F'' \rightarrow 0$ of $\mathcal{O}_S$ -modules is said to be *exact* if for every $S' \rightarrow S$ the sequence $0 \rightarrow F'(S') \rightarrow F(S') \rightarrow F''(S') \rightarrow 0$ of $\mathcal{O}(S')$ -modules is exact.

Likewise, a sequence $1 \rightarrow G' \rightarrow G \rightarrow G'' \rightarrow 1$ of S -groups is said to be exact if for every $S' \rightarrow S$ the sequence of groups $1 \rightarrow G'(S') \rightarrow G(S') \rightarrow G''(S') \rightarrow 1$ is exact.

Lemma 5.2.1. ⁽⁹²⁾ *Let $1 \rightarrow G' \rightarrow G \rightarrow G'' \rightarrow 1$ be an exact sequence of S -groups.*

(i) *The sequences $1 \rightarrow T_{G'/S}(\mathcal{M}) \rightarrow T_{G/S}(\mathcal{M}) \rightarrow T_{G''/S}(\mathcal{M}) \rightarrow 1$ and*

$$1 \rightarrow \mathrm{Lie}(G'/S, \mathcal{M}) \rightarrow \mathrm{Lie}(G/S, \mathcal{M}) \rightarrow \mathrm{Lie}(G''/S, \mathcal{M}) \rightarrow 1$$

are then exact.

(ii) *Let $1 \rightarrow H' \rightarrow H \rightarrow H'' \rightarrow 1$ be a second sequence of groups; it is exact if and only if the sequence below is exact:*

$$1 \rightarrow G' \times_S H' \rightarrow G \times_S H \rightarrow G'' \times_S H'' \rightarrow 1.$$

(iii) *If two of the S -groups G', G, G'' satisfy (E), the third also satisfies (E).*

(iv) *If $0 \rightarrow F' \rightarrow F \rightarrow F'' \rightarrow 0$ is an exact sequence of $\mathcal{O}_S$ -modules and if two of the modules F', F, F'' are good, the third is also good.*

(v) *If two of the S -groups G', G, G'' are good, the third is also good.*

PROOF. ⁽⁹³⁾ The first part of (i) is immediate, and the second part follows from it. Likewise, (ii) is immediate. Let us prove (iii). For brevity, let us write $L(\mathcal{M}) = \mathrm{Lie}(G, \mathcal{M})$, $L'(\mathcal{M}) = \mathrm{Lie}(G', \mathcal{M})$, etc. Then one has the following commutative diagram

$$\begin{array}{ccccccc} 1 & \longrightarrow & L'(\mathcal{M} \oplus \mathcal{N}) & \longrightarrow & L(\mathcal{M} \oplus \mathcal{N}) & \longrightarrow & L''(\mathcal{M} \oplus \mathcal{N}) \longrightarrow 1 \\ & & \downarrow & & \downarrow & & \downarrow \\ 1 & \longrightarrow & L'(\mathcal{M}) \times_S L'(\mathcal{N}) & \longrightarrow & L(\mathcal{M}) \times_S L(\mathcal{N}) & \longrightarrow & L''(\mathcal{M}) \times_S L''(\mathcal{N}) \longrightarrow 1 \end{array}$$

in which the first (resp. the second) row is exact by (i) (resp. by (i) and (ii)). Assertion (iii) follows from this.

Let us prove (iv). One has a commutative diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & F' \otimes_{\mathcal{O}_S} T_{\mathcal{O}_S/S}(\mathcal{M}) & \longrightarrow & F \otimes_{\mathcal{O}_S} T_{\mathcal{O}_S/S}(\mathcal{M}) & \longrightarrow & F'' \otimes_{\mathcal{O}_S} T_{\mathcal{O}_S/S}(\mathcal{M}) \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & T_{F'/S}(\mathcal{M}) & \longrightarrow & T_{F/S}(\mathcal{M}) & \longrightarrow & T_{F''/S}(\mathcal{M}) \longrightarrow 0 \end{array}$$

with exact rows (the first, because $T_{\mathcal{O}_S/S}(\mathcal{M})$ is a free $\mathcal{O}_S$ -module, hence flat, the second by (i)). It follows that if two of the modules F', F, F'' are good, the third is also good. Finally, (v) follows from (iii) and (iv).

Lemma 5.2.2. *Let G be an S -group, E, H two G -objects, and F a G - $\mathcal{O}_S$ -module.*

⁽⁹²⁾**N.D.E.** One has set out the statement of the lemma (and its proof) in more detail; this will be useful in 5.3.1.

⁽⁹³⁾**N.D.E.** One has added the proof of points (iii) and (v).

- (i) The canonical homomorphism $E^G \times_S H^G \longrightarrow (E \times_S H)^G$ is an isomorphism.
- (ii) If F is a good $\mathcal{O}_S$ -module, then so is F^G .

PROOF. ⁽⁹⁴⁾ The first assertion is immediate; let us prove the second. For every $S' \longrightarrow S$, one has a commutative diagram

$$\begin{array}{ccc} F^G(I_{S'}(\mathcal{M})) & \xhookrightarrow{\quad} & F(I_{S'}(\mathcal{M})) \\ \uparrow \varphi & & \simeq \uparrow \varphi_1 \\ F^G(S') \otimes_{\mathcal{O}(S')} \mathcal{O}(I_{S'}(\mathcal{M})) & \xhookrightarrow{\quad} & F(S') \otimes_{\mathcal{O}(S')} \mathcal{O}(I_{S'}(\mathcal{M})) \end{array}$$

and one must prove that φ is bijective; now, it is evidently injective; let us show that it is surjective. Let $(t_0, \dots, t_n)$ be a basis of the free $\mathcal{O}(S')$ -module $\mathcal{O}(I_{S'}(\mathcal{M}))$ and let

$$u = \sum_{i=0}^n f_i \otimes t_i$$

be an element of $F(S') \otimes_{\mathcal{O}(S')} \mathcal{O}(I_{S'}(\mathcal{M}))$ such that $\varphi_1(u)$ belongs to $F^G(I_{S'}(\mathcal{M}))$. Let us show that the f_i belong to $F^G(S')$.

Let $S'' \longrightarrow S'$; one may regard S'' as lying over $I_{S'}(\mathcal{M})$ via the zero section $\varepsilon_{\mathcal{M}}$. Then, for every $g \in G(S'')$, one has:

$$u_{S''} = g \cdot u_{S''} = \sum_{i=0}^n g \cdot (f_i)_{S''} \otimes (t_i)_{S''}.$$

As the $(t_i)_{S''}$ form a basis of $\mathcal{O}(I_{S''}(\mathcal{M})) \simeq \mathcal{O}(I_{S'}(\mathcal{M})) \otimes_{\mathcal{O}(S')} \mathcal{O}(S'')$ over $\mathcal{O}(S'')$, it follows that $g \cdot (f_i)_{S''} = (f_i)_{S''}$, whence $f_i \in F^G(S')$ for every i .

Notations. — ⁽⁹⁵⁾ If E is an S -group and F a sub- S -group of E , one denotes by E/F the S -functor which to every $S' \longrightarrow S$ associates the set $E(S')/F(S')$ of classes $x = x F(S')$, $x \in E(S')$. If E is a commutative S -group then E/F is a commutative S -group.

Let now G be an S -group and K a sub- S -group of G ; set $E = \text{Lie}(G/S, \mathcal{M})$ and $F = \text{Lie}(K/S, \mathcal{M})$. The adjoint action of K on E stabilizes F , hence induces an action of K on the S -functor E/F . Then, for every $S' \longrightarrow S$, one has:

$$(E/F)^K(S') = \left\{ x \in E(S')/F(S') \left| \begin{array}{l} \text{for every } f: S'' \longrightarrow S' \text{ and } k \in K(S''), \\ f^*(x^{-1}) \text{ Ad}(k)(f^*(x)) \in F(S'') \end{array} \right. \right\},$$

where $f^*(x)$ denotes the image of x in $E(S'')$.

Theorem 5.2.3. *Let G be an S -group, and K a sub- S -group of G . Let us denote (I 2.3.3)*

$$N = \text{Norm}_G(K), \quad Z = \text{Centr}_G(K).$$

Let us make K operate on $\text{Lie}(G/S, \mathcal{M})$ by means of the adjoint representation of G .

- (i) *If the group law of $\text{Lie}(G/S, \mathcal{M})$ is commutative^(*)⁽⁹⁶⁾, then*

$$\text{Lie}(N/S, \mathcal{M})/\text{Lie}(K/S, \mathcal{M}) = (\text{Lie}(G/S, \mathcal{M})/\text{Lie}(K/S, \mathcal{M}))^K.$$

- (ii) *If the group law of $\text{Lie}(G/S, \mathcal{M})$ is commutative $(*)$, then*

$$\text{Lie}(Z/S, \mathcal{M}) = \text{Lie}(G/S, \mathcal{M})^K.$$

- (iii) *If G satisfies (E) (resp. if G and K satisfy (E)), then Z satisfies (E) (resp. N satisfies (E)).*

⁽⁹⁴⁾**N.D.E.** One has set out the original in more detail in what follows.

⁽⁹⁵⁾**N.D.E.** One has set out these notations in more detail.

^(*)Condition automatically satisfied if G satisfies (E) (cf. 3.9), for example if G is representable.

⁽⁹⁶⁾**N.D.E.** For an example where the S -group $\text{Lie}(G/S)$ is not commutative, see 6.3.

- (iv) Suppose G is good; then Z is good; if moreover G is very good, then Z is very good.
- (v) Suppose G and K are good; then N is good; if moreover G is very good, then N is very good.

To prove (i) and (ii)⁽⁹⁷⁾ we shall use the following lemma, which results from the diagram of exact sequences considered in 4.1.B (with G and H interchanged).

Lemma 5.2.3.0. *Let G be an S -group, H a sub- S -group of G , and $\mathcal{M}$ a free $\mathcal{O}_S$ -module of finite type. Then $T_{H/S}(\mathcal{M})$ and $\text{Lie}(G/S, \mathcal{M})$ are sub- S -groups of $T_{G/S}(\mathcal{M})$ and one has:*

$$T_{H/S}(\mathcal{M}) \cap \text{Lie}(G/S, \mathcal{M}) = \text{Lie}(H/S, \mathcal{M}),$$

where one has set $T_{H/S}(\mathcal{M}) \cap \text{Lie}(G/S, \mathcal{M}) \stackrel{\text{def}}{=} T_{H/S}(\mathcal{M}) \times_{T_{G/S}(\mathcal{M})} \text{Lie}(G/S, \mathcal{M})$.

As the functors considered in (i) and (ii) commute with base change, it suffices to show the equalities of S -points.

Set $H = N$ (resp. $= Z$) and let $\alpha = \pm 1$. By the preceding lemma and the definition of N and Z (cf. I 2.3.3), one has: $\text{Lie}(H/S, \mathcal{M})(S) =$

$$\left\{ \begin{array}{l} X \in \text{Lie}(G/S, \mathcal{M})(S) \subset G(I_S(\mathcal{M})) \mid \\ \text{for every } f: S' \longrightarrow I_S(\mathcal{M}) \text{ and } u \in K(S'), \\ f^*(X^\alpha) \cdot u \cdot f^*(X^{-\alpha}) \cdot u^{-1} \in K(S') \\ \text{resp. } f^*(X) \cdot u \cdot f^*(X^{-1}) \cdot u^{-1} = 1 \end{array} \right\} \quad (*),$$

where $f^*(X)$ denotes the image of X in $G(S')$.

To simplify the writing, let us set

$$\mathfrak{g} = \text{Lie}(G/S, \mathcal{M}), \quad \mathfrak{k} = \text{Lie}(K/S, \mathcal{M}), \quad \mathfrak{n} = \text{Lie}(N/S, \mathcal{M}), \quad \mathfrak{z} = \text{Lie}(Z/S, \mathcal{M}).$$

If $X \in \text{Lie}(H/S, \mathcal{M})(S)$, the equalities $(*)$ are valid for every $f: S' \longrightarrow S$, since $f = \rho \circ \varepsilon_{\mathcal{M}} \circ f$ factors through $I_S(\mathcal{M})$ (where $\rho: I_S(\mathcal{M}) \longrightarrow S$ is the structural morphism and $\varepsilon_{\mathcal{M}}: S \longrightarrow I_S(\mathcal{M})$ the zero section). One deduces from this that

$$\mathfrak{n}(S)/\mathfrak{k}(S) \subset (\mathfrak{g}/\mathfrak{k})^K(S) \quad \text{and} \quad \mathfrak{z}(S) \subset \mathfrak{g}^K(S).$$

To prove the reverse inclusions, let us henceforth suppose $\mathfrak{g} = \text{Lie}(G/S, \mathcal{M})$ commutative; then $\mathfrak{g}^K$ and $(\mathfrak{g}/\mathfrak{k})^K$ are commutative S -groups. Let $X \in \mathfrak{g}(S)$ and X its image in $(\mathfrak{g}/\mathfrak{k})(S)$, suppose that $X \in (\mathfrak{g}/\mathfrak{k})^K(S)$ (resp. $X \in \mathfrak{g}^K(S)$) and let us show that $X \in \mathfrak{n}(S)$ (resp. $X \in \mathfrak{z}(S)$).

Let $f: S' \longrightarrow I_S(\mathcal{M})$; let us show that the preceding condition $(*)$ is satisfied for every $u \in K(S')$. Consider the cartesian square

$$\begin{array}{ccc} I_{S'}(\mathcal{M}) & \xrightarrow{p} & I_S(\mathcal{M}) \\ \rho' \downarrow & & \downarrow \rho \\ S' & \xrightarrow{\rho \circ f} & S \end{array}$$

and let h be the section of ρ' defined by f . It suffices to show that, for every $v \in K(I_{S'}(\mathcal{M}))$, one has

$$\left. \begin{array}{l} p^*(X^\alpha) \cdot v \cdot p^*(X^{-\alpha}) \cdot v^{-1} \in K(I_{S'}(\mathcal{M})) \\ \text{resp. } p^*(X) \cdot v \cdot p^*(X^{-1}) \cdot v^{-1} = 1 \end{array} \right\} \quad (**).$$

Indeed, taking $v = \rho'^*(u)$ and applying h^* to $(**)$, one obtains $(*)$, since $\rho' \circ h = \text{id}_{S'}$ and $p \circ h = f$.

Let us now show $(**)$; to simplify, one will write X in place of $p^*(X)$. Every $v \in K(I_{S'}(\mathcal{M}))$ may be written uniquely $Y \cdot k$ where $Y \in \text{Lie}(K/S, \mathcal{M})(S')$ and $k \in K(S')$. The expression $X^\alpha \cdot u \cdot X^{-\alpha} \cdot u^{-1}$ then becomes $X^\alpha \cdot Y \cdot (k \cdot X^{-\alpha} \cdot k^{-1}) \cdot Y^{-1}$ which, as $\text{Lie}(G/S, \mathcal{M})$ is

⁽⁹⁷⁾**N.D.E.** One has set out the proof in more detail, adding in particular Lemma 5.2.3.0, implicit in the original.

commutative, is written $X^\alpha \text{Ad}(k)(X^{-\alpha})$. Now the latter lies a priori in $\text{Lie}(G/S, \mathcal{M})(S')$; taking Lemma 5.2.3.0 into account, the condition $(**)$ for X therefore becomes: for every $k \in K(S')$,

$$\begin{cases} \text{Ad}(k)(X) = X & \text{if } H = Z; \\ X^\alpha \text{Ad}(k)(X^{-\alpha}) \in \text{Lie}(K/S, \mathcal{M})(S') & \text{if } H = N. \end{cases}$$

When $H = Z$, this condition is indeed a consequence of the hypothesis $X \in \mathfrak{g}^K(S)$. When $H = N$, the condition may also be written:

$$(*)' \quad \text{Ad}(k)(X) = X \quad \text{and} \quad \text{Ad}(k)(X^{-1}) = X^{-1},$$

now the second condition of $(*)'$ is a consequence of the first, since the operation of K on $\mathfrak{g}/\mathfrak{k}$ respects the group structure of the latter. Thus, when $H = N$, the condition $(**)$ for X is indeed a consequence of the hypothesis $X \in (\mathfrak{g}/\mathfrak{k})^K(S)$. This proves (i) and (ii).

To prove (iii)–(v), let us write $\mathfrak{g}(\mathcal{M}) = \text{Lie}(G/S, \mathcal{M})$ and let us define likewise $\mathfrak{k}(\mathcal{M})$, $\mathfrak{z}(\mathcal{M})$ and $\mathfrak{n}(\mathcal{M})$. If G/S satisfies (E), then $\mathfrak{g}(\mathcal{M} \oplus \mathcal{N}) \simeq \mathfrak{g}(\mathcal{M}) \times_S \mathfrak{g}(\mathcal{N})$ and hence, by 5.2.2 (i),

$$\mathfrak{g}(\mathcal{M} \oplus \mathcal{N})^K \simeq \mathfrak{g}(\mathcal{M})^K \times_S \mathfrak{g}(\mathcal{N})^K$$

whence $\mathfrak{z}(\mathcal{M} \oplus \mathcal{N}) \simeq \mathfrak{z}(\mathcal{M}) \times_S \mathfrak{z}(\mathcal{N})$, hence Z satisfies (E). If moreover K/S satisfies (E), one obtains successively isomorphisms:

$$\begin{aligned} (\mathfrak{g}/\mathfrak{k})(\mathcal{M} \oplus \mathcal{N}) &\simeq (\mathfrak{g}/\mathfrak{k})(\mathcal{M}) \times_S (\mathfrak{g}/\mathfrak{k})(\mathcal{N}) \\ (\mathfrak{g}/\mathfrak{k})^K(\mathcal{M} \oplus \mathcal{N}) &\simeq (\mathfrak{g}/\mathfrak{k})(\mathcal{M})^K \times_S (\mathfrak{g}/\mathfrak{k})(\mathcal{N})^K, \end{aligned}$$

then a commutative diagram:

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathfrak{k}(\mathcal{M} \oplus \mathcal{N}) & \longrightarrow & \mathfrak{n}(\mathcal{M} \oplus \mathcal{N}) & \longrightarrow & (\mathfrak{n}/\mathfrak{k})(\mathcal{M} \oplus \mathcal{N}) \longrightarrow 0 \\ & & \downarrow \wr & & \downarrow & & \downarrow \wr \\ 0 & \longrightarrow & \mathfrak{k}(\mathcal{M}) \times_S \mathfrak{k}(\mathcal{N}) & \longrightarrow & \mathfrak{n}(\mathcal{M}) \times_S \mathfrak{n}(\mathcal{N}) & \longrightarrow & (\mathfrak{n}/\mathfrak{k})(\mathcal{M}) \times_S (\mathfrak{n}/\mathfrak{k})(\mathcal{N}) \longrightarrow 0 \end{array}$$

whence it follows that $\mathfrak{n}(\mathcal{M} \oplus \mathcal{N}) \simeq \mathfrak{n}(\mathcal{M}) \times_S \mathfrak{n}(\mathcal{N})$, hence N satisfies (E).

Henceforth, let us write $\mathfrak{g} = \mathfrak{g}(\mathcal{O}_S) = \text{Lie}(G/S)$ and let us define likewise $\mathfrak{k}$, $\mathfrak{z}$, $\mathfrak{n}$. If G is *good*, $\mathfrak{g}$ is a *good* $\mathcal{O}_S$ -module, hence, by 5.2.2 (ii), so is $\mathfrak{z} \simeq \mathfrak{g}^K$, hence Z (which satisfies (E) by (iii)) is *good*. If moreover K is *good*, then $\mathfrak{k}$ is a *good* $\mathcal{O}_S$ -module, hence, by 5.2.1 (iv) and 5.2.2 (ii), the same is true of $\mathfrak{g}/\mathfrak{k}$ and $(\mathfrak{g}/\mathfrak{k})^K$. In view of the exact sequence

$$0 \longrightarrow \mathfrak{k} \longrightarrow \mathfrak{n} \longrightarrow (\mathfrak{g}/\mathfrak{k})^K \longrightarrow 0,$$

one obtains, by 5.2.1 (iv) again, that $\mathfrak{n}$ is *good*. Finally, if in addition to the preceding conditions G is *very good*, that is, if one has identically $[x, x] = 0$ for every $x \in \mathfrak{g}$, it is clear that Z and N are *very good*. This proves (iii), (iv) and (v).

Corollary 5.2.3.1. *One has $\text{Lie}(\text{Centr}(G)/S) = \text{Lie}(G/S)^G$ if the group law of $\text{Lie}(G/S)$ is commutative.*

Corollary 5.2.3.2. *If the group law of $\text{Lie}(G/S)$ is commutative and if K is an invariant subgroup of G , then*

$$(\text{Lie}(G/S)/\text{Lie}(K/S))^K = \text{Lie}(G/S)/\text{Lie}(K/S).$$

5.3. Linear representations. — Let G be a *good* S -group operating linearly on a *good* $\mathcal{O}_S$ -module F via

$$\rho: G \longrightarrow \text{Aut}_{\mathcal{O}_S\text{-mod.}}(F).$$

One has defined (4.7.1 and 4.8.2) a corresponding linear representation

$$\rho': \text{Lie}(G/S) \longrightarrow \text{End}_{\mathcal{O}_S\text{-mod.}}(F).$$

The sub- S -groups $\text{Norm}_G(E)$ and $\text{Centr}_G(E)$ are defined for every subset E of F , for example for every sub- $\mathcal{O}_S$ -module E of F .

Definition 5.3.0. One will set, analogously: for every $S' \rightarrow S$,

$$\text{Norm}_{\text{Lie}(G/S)}(E)(S') = \{X \in \text{Lie}(G/S)(S') \mid \rho'(X)E_{S'} \subset E_{S'}\};$$

$$\text{Centr}_{\text{Lie}(G/S)}(E)(S') = \{X \in \text{Lie}(G/S)(S') \mid \rho'(X)E_{S'} = 0\}.$$

(Let us note that this construction is carried out for every linear representation of an $\mathcal{O}_S$ -“Lie algebra” (in the sense of 4.9) and that the two subobjects constructed are sub- $\mathcal{O}_S$ -modules stable under the bracket).

Theorem 5.3.1. *Let G be a good S -group operating linearly on a good $\mathcal{O}_S$ -module F and let E be a sub- $\mathcal{O}_S$ -module of F .*

- (i) *One has $\text{Lie}(\text{Centr}_G(E)/S) = \text{Centr}_{\text{Lie}(G/S)}(E)$ and $\text{Centr}_G(E)$ is a good S -group; it is very good if G is.*
- (ii) *Suppose that E is a good $\mathcal{O}_S$ -module. Then⁽⁹⁸⁾ $\text{Lie}(\text{Norm}_G(E)/S) = \text{Norm}_{\text{Lie}(G/S)}(E)$ and $\text{Norm}_G(E)$ is a good S -group; it is very good if G is.*

The proof is left to the reader.

5.3.2. — Let G be a good S -group; the foregoing applies in particular to the case where one takes for ρ the adjoint representation of G . Let E be a *good*⁽⁹⁹⁾ submodule of $\text{Lie}(G/S)$; one therefore associates with it two subgroups of G , its centralizer and its normalizer. By 5.3.1, their Lie algebras are respectively the centralizer and the normalizer of E in $\text{Lie}(G/S)$, computed as usual by means of the bracket:

$$\text{Centr}_{\text{Lie}(G/S)}(E)(S') = \{X \in \text{Lie}(G/S)(S') \mid [X, E_{S'}] = 0\}.$$

$$\text{Norm}_{\text{Lie}(G/S)}(E)(S') = \{X \in \text{Lie}(G/S)(S') \mid [X, E_{S'}] \subset E_{S'}\}.$$

5.3.3. — Let K be a sub- S -group of G , then $\text{Lie}(K/S)$ is a sub- $\mathcal{O}_S$ -module of $\text{Lie}(G/S)$; suppose that $\text{Lie}(K/S)$ is a *good* $\mathcal{O}_S$ -module⁽⁹⁹⁾ (which is the case if K is a *good* S -group). One evidently has

$$\text{Centr}_G(K) \subset \text{Centr}_G(\text{Lie}(K/S))$$

$$\text{Norm}_G(K) \subset \text{Norm}_G(\text{Lie}(K/S)),$$

whence, by 5.3.1,

$$\text{Lie}(\text{Centr}_G(K)/S) \subset \text{Centr}_{\text{Lie}(G/S)}(\text{Lie}(K/S))$$

$$\text{Lie}(\text{Norm}_G(K)/S) \subset \text{Norm}_{\text{Lie}(G/S)}(\text{Lie}(K/S)),$$

but none of these four inclusions is a priori an identity; we shall see many examples of this later.

It follows in particular from these inclusions that if K is an *invariant subgroup* of G , then $\text{Lie}(K/S)$ is an *ideal* of $\text{Lie}(G/S)$.

6. Miscellaneous remarks

6.1. — One can define the bracket of two infinitesimal automorphisms for an S -functor X which is not necessarily a group. It suffices to apply the results of this exposé to the group $\text{Aut}_S(X)$. In order to arrive at a pleasant formalism, one is led to suppose X *good*, that is, to suppose that the $\mathcal{O}_X$ -module $T_{X/S}$ is *good* (if X is an S -group, this definition evidently coincides with definition 4.6).

⁽⁹⁸⁾**N.D.E.** One has corrected the original, which stated the following equality without hypotheses on E ; see 6.5.1 for counterexamples.

⁽⁹⁹⁾**N.D.E.** One has added this hypothesis; cf. 6.5.1.

6.2. — There exist functors possessing infinitesimal endomorphisms which are not automorphisms, hence *a fortiori* not satisfying condition (E).⁽¹⁰⁰⁾ For every pointed set (E, x_0) , let $\text{MA}(E)$ be the free abelian monoid generated by E and let $\text{MA}_P(E, x_0)$ be the abelian monoid obtained by quotienting $\text{MA}(E)$ by the equivalence relation generated by the relation $m \sim x_0 + m$. Then $(E, x_0) \mapsto \text{MA}_P(E, x_0)$ is the left adjoint of the forgetful functor from the category of abelian monoids to that of pointed sets; one will say that $\text{MA}_P(E, x_0)$ is the “free abelian monoid on the pointed set (E, x_0) ”.

Let us then take for X the functor which to every scheme S associates the free abelian monoid on the set $\text{O}(S)$, pointed by the zero element. Each morphism $f: S \rightarrow I_{\mathbf{Z}} = \text{Spec}(\mathbf{Z}[\varepsilon])$ corresponds to an element u_f of square zero of $\text{O}(S)$, hence defines an endomorphism of $X(S)$ by $x \mapsto x + u_f$ (sum in $\text{MA}_P(\text{O}(S), 0)$). One thus obtains an endomorphism ϕ of $X_{I_{\mathbf{Z}}} = X \times_{\mathbf{Z}} I_{\mathbf{Z}}$, defined as follows. For every $f \in I_{\mathbf{Z}}(S)$ and $x \in X(S)$,

$$\phi((x, f)) = (x + u_f, f).$$

If $f_0: S \rightarrow I_{\mathbf{Z}}$ is the composite of the structural morphism $S \rightarrow \text{Spec}(\mathbf{Z})$ and of the zero section of $I_{\mathbf{Z}}$, the corresponding element is $u_{f_0} = 0$, and hence $\phi((x, f_0)) = (x, f_0)$, since $x + 0 = x$ in $\text{MA}_P(\text{O}(S), 0)$. This shows that ϕ induces the identity on X ; it is therefore an infinitesimal endomorphism of X which is evidently not an automorphism.

6.3. — There exist modules which are not *good*. On the one hand, one can slightly modify the preceding counterexample:⁽¹⁰¹⁾ let us take for $F(S)$ the free $\text{O}(S)$ -module with basis the elements of $\text{O}(S)$; then F does not satisfy condition (E) with respect to $\text{Spec}(\mathbf{Z})$; moreover, let G be the $\mathbf{Z}$ -group defined by $G(S) = \text{GL}_n(R(S))$, where $n \geq 2$ and $R(S)$ is the $\text{O}(S)$ -algebra of the abelian group $\text{O}(S)$; then the $\mathbf{Z}$ -group $\text{Lie}(G/\mathbf{Z})$ is not commutative.

⁽¹⁰²⁾On the other hand, one can give the following counterexamples. Let $S = \text{Spec}(k)$, k a field of characteristic $p > 0$.

a) Let F be the $\mathcal{O}_S$ -module which to every S -scheme T associates $F(T) = \Gamma(T, \mathcal{O}_T)$ equipped with the structure of $\text{O}(T)$ -module obtained by making the scalars operate via the p -th power, that is, $r \cdot f = r^p f$, for $r \in \text{O}(T)$, $f \in F(T)$. As an S -group functor, F is isomorphic to $\mathbf{G}_{a,S}$. Hence F/S satisfies (E) and $\text{Lie}(F/S)$ is identified with $\text{Lie}(\mathbf{G}_{a,S}/S) \simeq \mathcal{O}_S$. Then, the canonical morphism $F \rightarrow \text{Lie}(F/S)$ is, for every $T \rightarrow S$, the identity map $F(T) \rightarrow \text{O}(T)$: it does respect the structure of abelian group but not the structure of module. Hence F is not *good* (cf. 4.4.1).

b) Let $\alpha_{p,k}$ be the k -group functor which to every S -scheme T associates

$$\alpha_{p,k}(T) = \{x \in \text{O}(T) \mid x^p = 0\},$$

it is represented by $\text{Spec}(k[X]/(X^p))$ hence it is a *very good* S -group; it is also equipped with a structure of $\mathcal{O}_S$ -module, but it is not a *good* $\mathcal{O}_S$ -module, for the canonical morphism $\alpha_{p,k} \rightarrow \text{Lie}(\alpha_{p,k}/k) = \mathbf{G}_{a,k}$ is not bijective.

6.4. — Let G be a group functor over S . One has by definition the following implications

$$(G/S \text{ satisfies (E)}) \iff (G \text{ is good}) \iff (G \text{ is very good}).$$

⁽¹⁰³⁾ It would be interesting to prove or to give counterexamples for the implications in the reverse direction.

⁽¹⁰⁰⁾**N.D.E.** One has set out the original in more detail in what follows. In particular, the correct definition of the “free abelian monoid on a pointed set” was pointed out to us by O. Gabber.

⁽¹⁰¹⁾**N.D.E.** One has added what follows.

⁽¹⁰²⁾**N.D.E.** In what follows, one has set out example a) in more detail and added example b).

⁽¹⁰³⁾**N.D.E.** And G is very good if it is representable (4.11). For criteria of representability, see for example [MO67]. On the other hand, for the automorphisms of an affine algebraic group (over an algebraically closed field of characteristic 0), let us cite [HM69].

6.5. — ⁽¹⁰⁴⁾Let $\mathbf{Nil}$ be the $\mathbf{Z}$ -group functor defined as follows: for every scheme S , $\mathbf{Nil}(S)$ is the ideal of $\mathcal{O}(S)$ consisting of the nilpotent elements, i.e.

$$\mathbf{Nil}(S) = \{x \in \mathcal{O}(S) \mid \text{there exists } n \in \mathbf{N} \text{ such that } x^n = 0\}.$$

($\mathbf{Nil}$ is *very good* but is not representable). Let $\mathbf{Nil}^2$, $\mathcal{O}_{\text{red}}$ and F be the $\mathbf{Z}$ -group functors which to every scheme S associate, respectively, the ideal $\mathbf{Nil}(S)^2$ and

$$\mathcal{O}_{\text{red}}(S) = \mathcal{O}(S)/\mathbf{Nil}(S), \quad F(S) = \mathcal{O}(S)/\mathbf{Nil}(S)^2.$$

One sees easily that $\text{Lie}(\mathcal{O}_{\text{red}}/\mathbf{Z}) = 0$, hence the $\mathcal{O}_{\mathbf{Z}}$ -module $\mathcal{O}_{\text{red}}$ is not *good* (although it is a *good* $\mathbf{Z}$ -group). If $\mathcal{M}, \mathcal{N}$ are free $\mathbf{Z}$ -modules of finite rank, one has

$$\mathbf{Nil}^2(I_S(\mathcal{M} \oplus \mathcal{N})) \simeq \mathbf{Nil}^2(S) \oplus \mathbf{Nil}(S) \otimes_{\mathbf{Z}} \mathcal{M} \oplus \mathbf{Nil}(S) \otimes_{\mathbf{Z}} \mathcal{N}$$

and hence

$$F(I_S(\mathcal{M} \oplus \mathcal{N})) \simeq F(S) \oplus \mathcal{O}_{\text{red}}(S) \otimes_{\mathbf{Z}} \mathcal{M} \oplus \mathcal{O}_{\text{red}}(S) \otimes_{\mathbf{Z}} \mathcal{N}.$$

One deduces from this, on the one hand, that the $\mathbf{Z}$ -group functor F satisfies condition (E) and, on the other hand, that $\text{Lie}(F/\mathbf{Z}) = \mathcal{O}_{\text{red}}$; as the latter is not a *good* $\mathcal{O}_{\mathbf{Z}}$ -module, this shows that F is an example of a $\mathbf{Z}$ -group which satisfies (E) but which is not *good*.

6.5.1. — Let us also give the counterexamples indicated in 5.3.1–5.3.3. Let S be a scheme, F the *good* $\mathcal{O}_S$ -module $\mathcal{O}_S^{\oplus 2}$ equipped with the natural action of the *good* S -group $G = \text{GL}_{2,S}$, and E the sub- $\mathcal{O}_S$ -module of F consisting of the pairs (x_1, x_2) such that x_2 is nilpotent. Set $N = \text{Norm}_G(E)$. Then $\text{Lie}(N/S) = \text{Lie}(G/S)$ whereas, for every $S' \rightarrow S$, one has

$$\text{Norm}_{\text{Lie}(G/S)}(E)(S') = \left\{ \begin{pmatrix} a & b \\ x & c \end{pmatrix} \mid a, b, c, x \in \mathcal{O}(S'), x \text{ nilpotent} \right\}$$

hence $\text{Lie}(\text{Norm}_G(E)/S) \neq \text{Norm}_{\text{Lie}(G/S)}(E)$.

By considering the semidirect product $G_1 = F \cdot G$, one obtains an analogous counterexample in which E is a sub- $\mathcal{O}_S$ -module of $\text{Lie}(G_1/S)$; moreover, with the notations introduced above, $E = \text{Lie}(K/S)$ where K is the subgroup $\mathcal{O}_S \oplus \mathbf{Nil}^2$ of F (that is, for every $S' \rightarrow S$, $K(S')$ consists of the pairs (x_1, x_2) such that $x_2 \in \mathbf{Nil}(S')^2$).

Bibliography⁽¹⁰⁵⁾

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- [MO67] H. Matsumura, F. Oort, *Representability of group functors and automorphisms of algebraic schemes*, Invent. math. **4** (1967), 1–25.

⁽¹⁰⁴⁾**N.D.E.** One has added this paragraph.

⁽¹⁰⁵⁾**N.D.E.** Additional references cited in this Exposé.